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Travel Patterns and Characteristics of Millennial Population in New York State



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IN NEW YORK STATE**

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ABBREVIATIONS

ACS	American Community Survey
MOE	margin of error
NHTS	National Household Travel Survey
NYC	New York City
NYMTC	New York Metropolitan Transportation Council
NYS	New York State
NYSDOT	New York State Department of Transportation
ORNL	Oak Ridge National Laboratory
PMT	person miles of travel
POV	privately owned vehicle
PUMS	Public Use Microdata Sample
VMT	vehicle miles traveled

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EXECUTIVE SUMMARY

This study investigates the travel behaviors, demographics, and transportation preferences of millennials (born 1981–1996), a generation that significantly influences urban living and mobility trends. Using data from the National Household Travel Survey (2009, 2017, and 2022) and other sources, this study examines key factors such as trip rates, trip length, travel modes, trip purposes, and travel time while also analyzing the effects of transportation technologies and COVID-19 on millennials’ travel behaviors in New York State (NYS). Additionally, travel patterns of millennials are compared with those of younger (i.e., Gen Z) and older (i.e., Gen X and baby boomer) generations across different geographical regions in NYS (e.g., New York City).

Sociodemographic Characteristics

- In 2017, the millennial population in NYS was 24% of the state’s total population and was stable compared with 2009. Considering the gender breakdown, millennials comprised 25% of males and 23% of females in NYS in 2017, maintaining an equal gender distribution from 2009. In the rest of the United States, the gender distribution was similar, with millennials comprising 24% of US males and 23% of US females in 2017.
- New York City (NYC) had a higher millennial concentration, exceeding 30% in areas like Brooklyn, Queens, and parts of Manhattan, whereas upstate and rural regions’ millennial populations were below 20%.
- In NYC, millennials as heads of household showed a significant skew toward higher income, with 35% earning more than \$100K annually. In nonurban areas, 37% of them earn less than \$25K.
- In 2017, more than half (64%) of the millennial-headed households in NYC had zero vehicles, which is considerably higher than that in other regions as well as among other household types. Approximately 37%–56% of households comprising both millennials and older members owned more than three vehicles, which is higher than that of other households across regions. In 2022, millennials living with older members also had higher vehicle ownership rates in the Middle Atlantic (Mid-Atlantic) Census Division.

Trip Patterns

- In 2009, millennials had significantly lower daily personal trips than those of Gen Xers and baby boomers across regions in NYS and the rest of US. For instance, in NYC, millennials averaged 3.0 trips per day, compared with 4.2 trips for Gen Xers and 3.8 for baby boomers. However, millennials made significantly more trips than those of Gen Zers. Similar trends were also observed in 2017.
- In 2022, a significant decrease occurred in the average daily trips for all generations across the Mid-Atlantic Census Division and the rest of the United States. Millennials in the Mid-Atlantic dropped to 2.3 trips per day, down from 3.3 in 2017. This drop aligns with a broader trend of reduced travel across all generations, likely influenced by societal changes such as the COVID-19 pandemic, increased remote work, and shifting lifestyles.
- NYC millennials made only 16% of their trips as the driver and 9% as a passenger in private vehicles; this represents less than one-third of their daily trips and even less than those of baby boomers. This is the lowest among all urban and rural regions of NYS and the rest of the United

States. In contrast, walking and public transit account for a much larger share of daily trips (e.g., 41% and 28%, respectively, in NYC).

- Millennials across all regions dedicated a significant portion (~28%) of their trips to earning a living, which is the highest across all generations. They made fewer trips for family and personal business compared with older generations (i.e., Gen X and baby boomer).
- Households with zero vehicles are more likely to be in the low-income category. The share of zero-vehicle millennial households with <\$25K income is significantly higher outside NYC. In the NYS nonurban areas, an overwhelming 88% of millennial households do not own a vehicle.
- Across all regions, millennials' person miles traveled are significantly higher when they travel alone. In contrast, Gen Zers with a travel party of four or more people travel farther on average than millennials travel with a similarly sized travel party. Similar trends were observed for vehicle miles traveled.

Views of Transportation Services

- A higher share of households with millennials as heads disagree (23%) or strongly disagree (30%) than other household types in NYC regarding whether gas price affected their travel.
- Nearly half of millennial-headed households in other urban areas of NYS agree or strongly agree that they walked to save money. In NYC, 66% of millennial-headed households agree or strongly agree with this statement.
- In NYC, 78% of millennial-headed households agree or strongly agree that they used public transportation to save money, compared with 62% of nonmillennial-headed households.

Use of Transportation Technologies

- Millennials showed a greater tendency than older generations toward online shopping. In NYC, for instance, only 18% of millennials reported not making online purchases, compared with 55% of baby boomers and 30% of Gen Xers.
- Work-from-home rates across regions were higher among baby boomers and Gen Xers than among millennials in 2017. In NYC, 8% of millennials reported working from home, compared with 12% of Gen Xers. In nonurban regions, 14% of baby boomers worked from home; this was the highest across all groups, possibly reflecting flexible working arrangements.
- Millennials were the leading adopters of rideshare services, especially in urban areas. In NYS, 25% of millennials reported using rideshares 1–5 times and 8% used these services 6–10 times, surpassing rates in the rest of the United States.

Travel Behavior During COVID-19

- In the Mid-Atlantic Census Division, 39% of millennials reported shopping online during COVID-19 “more often than before” the pandemic; that was lower than rates of Gen Xers (45%) and baby boomers (43%) but higher than that of Gen Zers (28%).
- About 47% of millennial females in the Mid-Atlantic reported ordering for delivery more often than before, compared with only 30% of millennial males.

- During COVID-19, 46% of millennials in the Mid-Atlantic reported reduced public transit use. Baby boomers showed the highest reduction at 54%.

E-bike Use

- Millennials have experienced substantial growth in e-bike usage over the years. Male millennials consistently accounted for over 70% of e-bike trips, indicating a gender imbalance in the usage of e-bike trips in NYC.
- Among the four generations, millennials recorded the highest monthly median trip length, with Gen Zers having the shortest. Millennials also exhibited a distinct weekday usage pattern, with trips concentrated during morning (7–9 a.m.) and evening (5–7 p.m.) peaks, likely reflecting commuter trips.
- Millennials' trip origins and destinations were mainly concentrated in areas such as Grand Central Terminal and Union Square, indicating their use of e-bikes for multimodal commutes, possibly as the first- or last-mile connections. In contrast, Gen Zers' hotspots were found near educational institutions.

1. INTRODUCTION

1.1 BACKGROUND

Millennials (also called Generation Y), defined as individuals born between 1981 and 1996, now comprise a significant portion of the population, and their distinct lifestyle preferences, technological affinities, and transportation behaviors have wide-reaching implications for urban planning and infrastructure development. In recent demographic data, millennials represent the largest share of the US population, surpassing the baby boomers (Fry, 2020; Korhonen, 2024). Their consumption choices and their activity participation (both virtual and in-person) have profound implications for public services, private businesses, and infrastructure planning. Previous research has shown that millennials differ from earlier generations in several key ways. They are more likely to prefer urban living, exhibit lower vehicle ownership, and delay major life milestones such as employment, homebuying, marriage, and parenting (The Council of Economic Advisers, 2014; Fry et al., 2018). These trends suggest a need for adjustments in housing markets, business practices, transportation investments, and land-use planning to accommodate the demographic shifts (Moos et al., 2017). Importantly, millennials' evolving preferences and behaviors have the potential to redefine travel and commuting patterns, further distinguishing their transportation needs from those of prior generations.

This study explores these demographic shifts by examining the motivations, travel patterns, and emerging transportation preferences within the millennial population, with a focus on New York State (NYS). Understanding millennials' travel behaviors is essential for several reasons. First, millennials have shown a strong inclination toward sustainable and clean transportation options, such as electric vehicles, public transit, ridesharing, and active modes of travel like biking and walking (Lee et al., 2021; The Millennial Report, 2024). Their reduced reliance on private vehicles, increased use of technology-driven transportation services, and openness to flexible work arrangements challenge traditional transportation models, which have largely been car-centric. As millennials continue to prioritize urban living and embrace digital platforms for mobility, planners and policymakers need to address the unique needs of this generation to advance resilient, adaptive, and sustainable transportation systems.

Millennials' travel behaviors also pose challenges to existing transportation planning and infrastructure. Traditional models often assume stable commuting patterns centered around 9-to-5 work schedules and heavy reliance on personal vehicles. However, recent surveys indicate that millennials are more likely to engage in remote work, vary their travel times, and use multimodal transportation options, including ride-, bike-, and carsharing (Humi, 2024; Lee et al., 2021). These behaviors require a reevaluation of existing models to ensure they accurately capture new travel trends and can adequately plan for infrastructure and service needs. Furthermore, the COVID-19 pandemic has accelerated changes in travel behavior, particularly among millennials, thereby highlighting the importance of adaptable transportation networks capable of responding to evolving public health and lifestyle factors (Patwary & Khattak, 2022).

This study aims to address the critical need to understand and anticipate millennials' travel preferences and to assess how these evolving patterns influence transportation planning in NYS. By examining millennials' travel behaviors in comparison with those of other generations, this study provides insights that will help guide infrastructure development, policy decisions, and service design to better serve this generation's needs. Additionally, this research aligns with broader transportation goals, such as reducing carbon emissions, promoting equitable access to transportation, and improving quality of life within New York's urban and suburban communities. As millennials continue to shape urban living dynamics, understanding their unique transportation needs will become increasingly crucial for creating sustainable, forward-looking transportation solutions that benefit both millennials and the broader population. This report represents the fourth installment in a series of special population analyses, focusing on demographics and mobility through the use of the 2017 National Household Travel Survey (NHTS) data

in NYS. Previous reports explored the travel patterns and demographics of populations living in rural areas and of elderly and low-income population subgroups (Liu et al., 2022; Pan et al., 2024; Uddin et al., 2024).

1.2 OBJECTIVES

This study investigates the travel behaviors, demographics, and transportation preferences of millennials to understand how their distinct characteristics and lifestyle choices shape their mobility patterns. It aims to assess the impact of generational trends, technological advancements, and the COVID-19 pandemic on millennials' transportation behaviors and to evaluate the implications for transportation planning and infrastructure development. By analyzing and comparing millennials' travel behaviors across various regions (e.g., New York City, NYS, Mid-Atlantic region, and the rest of the United States) and time periods (2009, 2017, and 2022, corresponding to NHTS data years), this study seeks to provide a comprehensive understanding of this generation's mobility patterns. Additionally, travel patterns of millennials are compared with those of other generations. The study has the following objectives:

- Analyze sociodemographic characteristics of millennials by gender, region, and household status.
- Evaluate travel behaviors of the millennials in terms of daily trip rates, trip lengths, trip purposes, and travel modes.
- Compare millennials' travel behaviors with those of older and younger generations.
- Investigate millennials' views on transportation services to understand factors such as the influence of gas pricing on travel decisions and access to the internet.
- Examine millennials' behavior in terms of their use of transportation technologies (i.e., ride-, car-, and bikesharing; online shopping; and working from home) to gauge the potential to reduce reliance on traditional personal vehicle travel.
- Explore how the COVID-19 pandemic has affected transportation-related behaviors, including shifts toward making online purchases and changing commuting habits.

1.3 A BRIEF DESCRIPTION OF MAJOR DATA SOURCES

1.3.1 National Household Travel Survey

The NHTS, sponsored by the Federal Highway Administration, is a fundamental source for analyzing the travel behavior of the American population (NHTS, 2017). The survey collects comprehensive daily travel data, linking trip characteristics—such as frequency, distance, duration, mode of transportation, and purpose—with personal, household, and vehicle attributes.

We used three NHTS surveys: 2009, 2017, and 2022. The 2009 NHTS collected data from approximately 150,000 households nationwide, offering insights into travel behaviors across different demographic groups, including millennials, who were in their early adulthood during that time. This dataset provides a baseline for understanding millennials' travel habits, including their use of public transit, driving preferences, and early adoption of emerging transportation technologies. This survey included NYS as an add-on partner.

In the 2017 NHTS, over 129,000 households were surveyed, comprising 26,000 from a national sample and additional households from add-on samples contributed by 13 state or Metropolitan Planning

Organization partners. These included data from over 17,000 households in NYS. This enriched dataset enabled more granular analyses of millennials' travel behaviors in NYS, thus offering insights at a finer geographical scale compared with the national dataset alone. By capturing detailed travel patterns and choices during millennials' prime working years, the 2017 survey provides an opportunity to assess shifts in their transportation preferences, such as reduced reliance on personal vehicles and increased use of ridesharing.

The 2022 NHTS, part of the Next Generation NHTS initiative, continued to build on previous datasets by incorporating insights into the evolving travel behavior of Americans, particularly in light of the COVID-19 pandemic. The 2022 dataset, which captured over 25,000 households across the nation, included updates on telework, e-commerce, and travel preferences during and after the pandemic. Although no NYS-specific add-on data were collected for this cycle, the survey may provide insights into the effects of the pandemic on millennials' travel patterns, such as the shift to remote work, changes in commuting frequency, and the role of transportation technologies like ride- and bikesharing. The 2022 NHTS also helps to contextualize generational trends in millennials' travel behavior as they age into mid-adulthood, offering opportunities to evaluate the long-term effects of the pandemic on transportation planning and infrastructure development.

1.3.2 American Community Survey

The American Community Survey (ACS), conducted by the US Census Bureau, is among the most comprehensive sources of demographic, social, economic, and housing data in the United States. The ACS provides detailed annual estimates for a wide range of variables that are essential for understanding population characteristics and trends. The ACS Public Use Microdata Sample (PUMS) is a subset of the ACS that allows users to access anonymized, individual-level data, offering greater flexibility for detailed analyses. PUMS provides microdata at the person or household level, enabling researchers to explore relationships between variables and conduct custom analyses not possible with pretabulated ACS summary data. However, the data are aggregated into Public Use Microdata Areas, which are statistical geographic areas containing at least 100,000 residents. For this study, we used the 5 year estimates of ACS PUMS 2017 to explore the distribution of the millennial population across NYS. The PUMS data allow us to identify and compare the percentage of millennials in different Public Use Microdata Areas in the state.

1.3.3 Citi Bike

Citi Bike is a bikesharing system that launched in May 2013. It has grown to become the largest bikesharing system in the nation, boasting over 150,000 active annual members and averaging more than one million trips per month. The system operates an open data policy, offering publicly available trip history datasets that provide insights into the use and adoption of bikesharing as a mode of transportation. The Citi Bike dataset includes disaggregated trip information such as trip start and end times, the stations used (start and end points), and user demographics. User data include membership type, categorized into annual members and casual users (those with 24 hour or 3 day passes); gender; and year of birth. Starting in 2020, user demographics are not included in the published dataset. Although Citi Bike primarily operates within New York City, the system expanded to New Jersey in 2016 with a smaller-scale deployment of 50 docking stations.

For this study, Citi Bike data from 2014 to 2019 were analyzed to explore millennials' participation in bikesharing within NYS, focusing on e-bike trips. Specifically, this study examined trip distributions across years, seasonal variations, differences between weekdays and weekends, and hourly trends. The patterns of millennials were then compared with those of younger (e.g., Gen Z) and older generations

(e.g., Gen X and baby boomers) to understand generational differences in adoption and usage of bikesharing services.

1.4 GENERATION TIMELINE

Millennials, born between 1981 and 1996, are positioned between Gen Xers (1965–1980) and Gen Zers (1997–2012). Using the age variable (*R_AGE_IMP*) from the NHTS, this study classifies respondents into these generational cohorts to enable detailed analyses of travel behavior.

As shown in Table 1-1, millennials were 13–28 years old during the 2009 NHTS survey, representing a demographic at the threshold of early adulthood. By the time of the 2017 NHTS survey, millennials were 21–36 years old, transitioning further into adulthood with increasing household and professional responsibilities. In the latest (2022) NHTS survey, millennials were ages 26–41, marking their progression into middle age, a phase often associated with more stable household structures, family travel, and shifts in work patterns.

Table 1-1. Generational age ranges across NHTS survey years (2009, 2017, and 2022).

Generation	Years born	Ages in 2009	Ages in 2017	Ages in 2022
Gen Z	1997–2012	0–12	5–20	10–25
Millennial	1981–1996	13–28	21–36	26–41
Gen X	1965–1980	29–44	37–52	42–57
Baby boomer	1946–1964	45–63	53–71	58–76

1.4.1 Millennials' Household Breakdown

To further understand millennials' travel behavior, we not only analyzed generational trends but also examined household dynamics. For this purpose, we categorized households into three distinct types:

- **No-millennials household:** These households do not have any millennials present. They typically belong to older or younger generations, such as baby boomers, Gen X, or Gen Z, and serve as a baseline for comparison.
- **Millennials living with older household members:** These households include millennials who reside with older generations, such as Gen Xers or baby boomers. This group reflects intergenerational living arrangements, often influenced by economic, cultural, or social factors, such as delayed independence, cost-sharing, or caregiving responsibilities.
- **Millennials as heads of the household:** Millennials are the primary heads of the family. This category represents millennials who have achieved a degree of independence, often correlating with life events like marriage, parenthood, or homeownership. This group provides insights into the evolving responsibilities and travel patterns unique to millennials.

1.5 GEOGRAPHICAL REGION BREAKDOWN BY NHTS YEAR

To capture the diverse travel behaviors and trends across different types of regions, this study uses a geographical breakdown based on urban, suburban, and rural areas, as well as broader comparisons with state and national trends. This breakdown includes specific areas within NYS and a broader regional context for the Middle Atlantic Census Division.

For the 2009 and 2017 NHTS data, the analysis focuses on NYS and the following subregions:

- **New York City (NYC):** Comprising the five boroughs (New York County, Kings County, Queens County, Richmond County, and Bronx County).
- **NYS other urban areas:** These include mainly the Metropolitan Planning Organization regions outside NYC.
- **NYS nonurban areas:** Rural or less dense regions in NYS.
- **NYS:** This aggregates all areas in NYS, offering a statewide perspective.

The 2022 NHTS data lack state-level information, thus limiting direct comparisons for NYS and necessitating the following alternative approach:

- **Middle Atlantic (Mid-Atlantic) Census Division:** This Census Division includes New York, New Jersey, and Pennsylvania.
- **Rest of US:** Used for broader national comparisons outside the Mid-Atlantic region.

1.6 ORGANIZATION OF REPORT

The report is structured to comprehensively analyze millennials' travel patterns, demographics, and evolving transportation behaviors, with a particular focus on NYS. Chapter 2 examines demographic characteristics such as race, household size, and income, along with vehicle-related metrics like ownership and age, comparing generations, NYS regions with the rest of the US and, for 2022 data, the Mid-Atlantic Census Division. Chapter 3 explores mobility patterns, including person-trip rates, person miles traveled, vehicle-trip rates, and vehicle miles traveled, with detailed analyses of trip purposes and transportation modes across generations and regions. Chapter 4 investigates millennials' perceptions of transportation services, highlighting regional variations within NYS. Chapter 5 investigates millennials' adoption of transportation technologies (e.g., online shopping; working from home; and ride-, car-, and bikesharing) and implications for reducing physical travel. Chapter 6 focuses on the effect of COVID-19 on millennials' travel behaviors, examining changes in commuting to physical work locations and in public transit use, with comparisons across generations and regions. Chapter 7 analyzes millennials' participation in bikesharing programs, emphasizing e-bike usage trends by year, season, weekdays versus weekends, and time of day, with generational comparisons. Last, Chapter 8 concludes the report with key findings and future research directions.

2. DEMOGRAPHIC PROFILE OF NYS MILLENNIAL POPULATION

2.1 POPULATION AND GENDER BY GENERATIONS

2.1.1 Population Statistics Based on the NHTS Data

The distribution of population (ages 5 years and older) by generation in NYS and the rest of the US based on NHTS data is shown in Figure 2-1. The Gen Z population has increased in both NYS and the rest of the US between 2009 and 2017, from 10% to 20% in NYS and from 11% to 22% in the rest of the US. The millennial population in NYS was 24% in 2017, remaining relatively stable compared with 2009. In the rest of the US, millennials accounted for 23% and 22% of the population in 2017 and 2009, respectively. This suggests that this generation continues to comprise a significant portion of the population. Table 2-1 complements Figure 2-1 by providing a gender breakdown within these generational categories. In 2017, millennials comprised 25% of males and 23% of females in NYS, maintaining an equal gender distribution from 2009. In the rest of the US, the gender distribution was similar, with millennials comprising 24% of US males and 23% of US females in 2017.

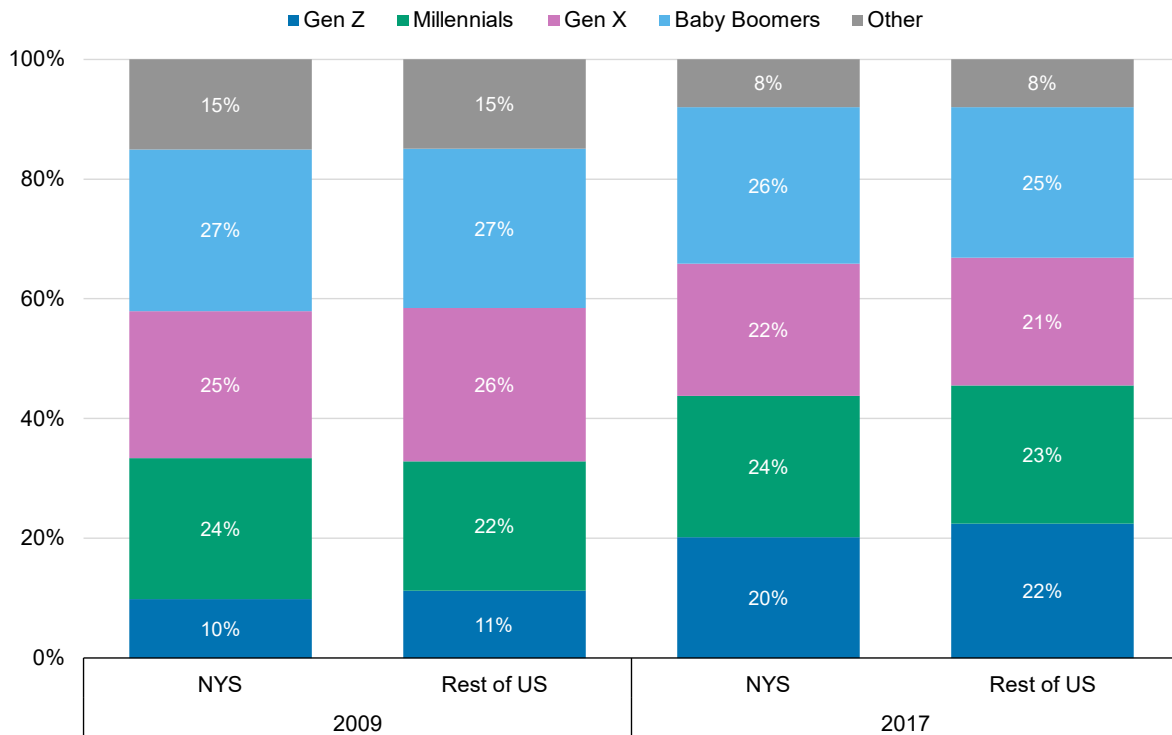


Figure 2-1. Distribution of population by generation in NYS and the rest of the US (NHTS data).

Table 2-1. Distribution of population by generation and gender (NHTS data).

	NYS				Rest of US			
	2017		2009		2017		2009	
	Male	Female	Male	Female	Male	Female	Male	Female
Gen Z	22%	19%	10%	9%	24%	21%	12%	11%
Millennials	25%	23%	25%	23%	24%	23%	23%	21%
Gen X	22%	22%	25%	24%	21%	21%	27%	25%
Baby boomers	25%	27%	27%	27%	25%	26%	26%	27%
Other	7%	9%	13%	17%	7%	9%	13%	17%
Total population	9,071,725	9,652,080	8,845,089	9,436,713	138,962,730	143,912,634	130,411,972	134,360,098

2.1.2 Population Statistics Based on Census Data

We used the 5 year estimates of the 2017 ACS PUMS to show the distribution of the millennial population across NYS by PUMS region (Figure 2-2). The map in Figure 2-2 shows the percentage estimate of the millennial population by PUMS area and indicates that certain areas, particularly in NYC, have a concentration of millennials significantly higher than those of other generations. Areas such as Brooklyn, Queens, and parts of Manhattan exhibit the darkest shades on the map, representing a millennial population exceeding 30%. This concentration likely reflects the urban appeal of NYC, where job opportunities, lifestyle amenities, and cultural attractions draw younger populations, particularly millennials. In contrast, upstate New York, particularly rural and suburban areas, shows much lighter shades on the map, indicating a lower percentage of millennials. These areas have millennial populations below 20%, reflecting the broader national trend of younger generations gravitating toward urban centers rather than rural regions. The margin of error (MOE) of the millennial population distribution was further examined. Figure 2-3 presents an upper bound estimate of the millennial population, incorporating the MOE adjustment. This map reinforces previous observations, showing that urban areas, particularly in NYC, have an even higher percentage of millennials. For the lower bound estimate, Figure 2-4 still shows NYC and its boroughs as key areas with significant millennial populations, but the concentration is less intense compared with the upper bound estimate. These adjustments account for uncertainty of the estimates.

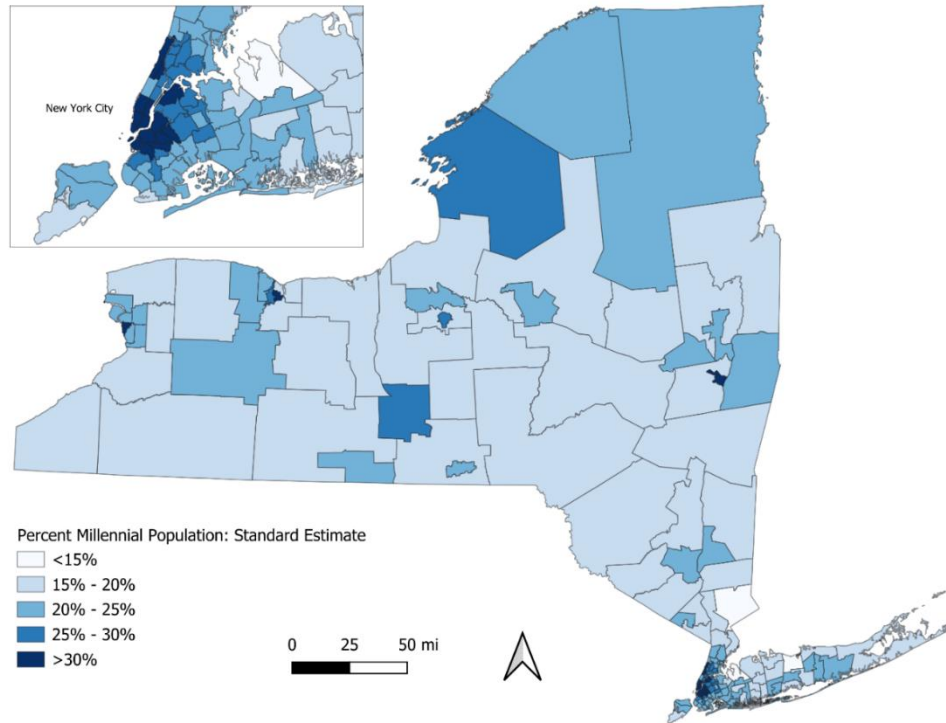


Figure 2-2. Distribution of millennial population across NYS by PUMS region (2017 ACS PUMS data).

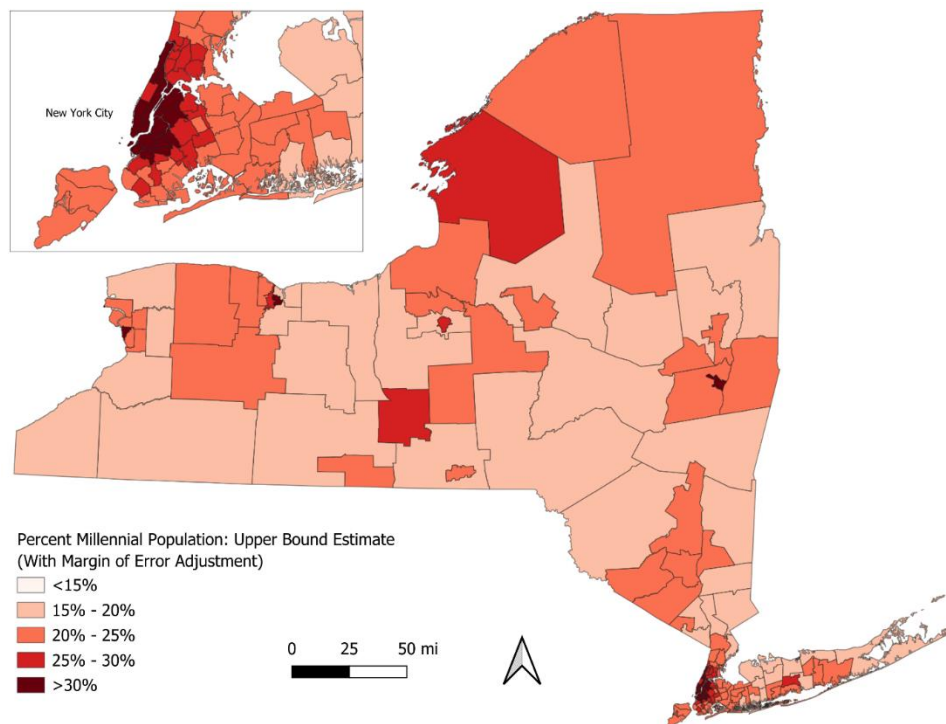


Figure 2-3. Upper bound estimate of millennial population across NYS by PUMS region (2017 ACS PUMS data).

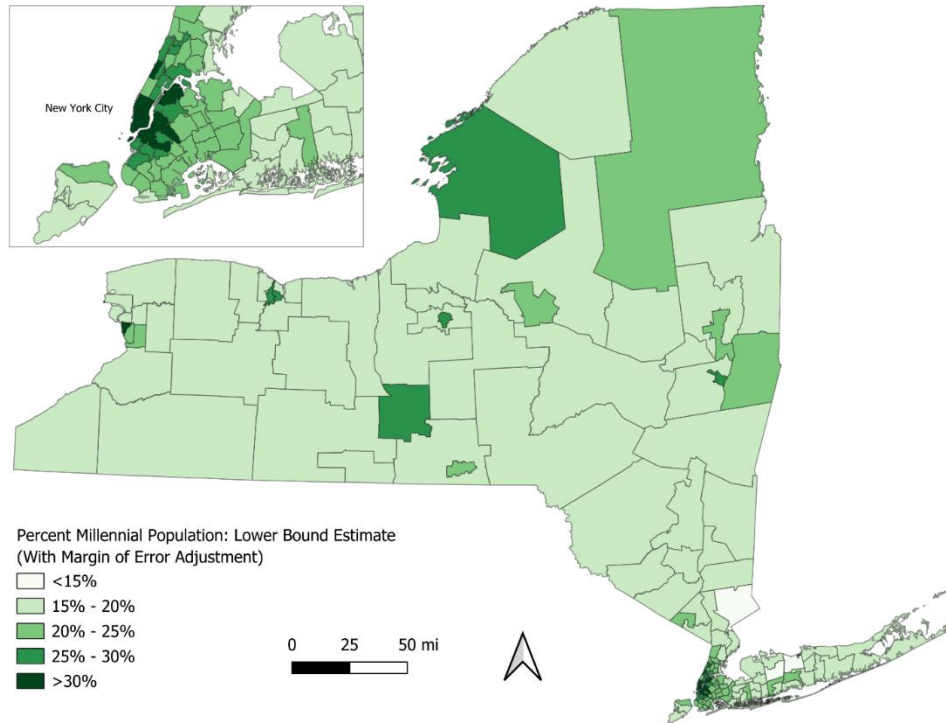


Figure 2-4. Lower bound estimate of millennial population across NYS by PUMS region (2017 ACS PUMS data).

Using the ACS data, NYS population distributions by generation and gender for 2009 and 2017 are illustrated in Figure 2-5. Millennials accounted for 25% of males and 23% of females in NYS in 2009, and those numbers remained stable in 2017. The rest of the US shows a similar pattern in both years. The numbers are identical to the NHTS data, showing the continued presence of millennials in NYS. Now in their prime working and family-rearing years, millennials are becoming the driving force of the economy. Their consistent presence in the population indicates they are aging but remain a significant, active part of society. This demographic is likely contributing to key economic sectors, influencing market trends, and driving demand for housing, technology, and other services. On the other hand, the slight decline in Gen Xers and baby boomers reflects natural aging and the beginning of a demographic shift as these generations move closer to retirement. As baby boomers, in particular, begin to exit the workforce and transition into retirement, their influence on the economy might diminish. The decline in the “Other” category further illustrates the aging population, as older generations pass on and their share of the population reduces.

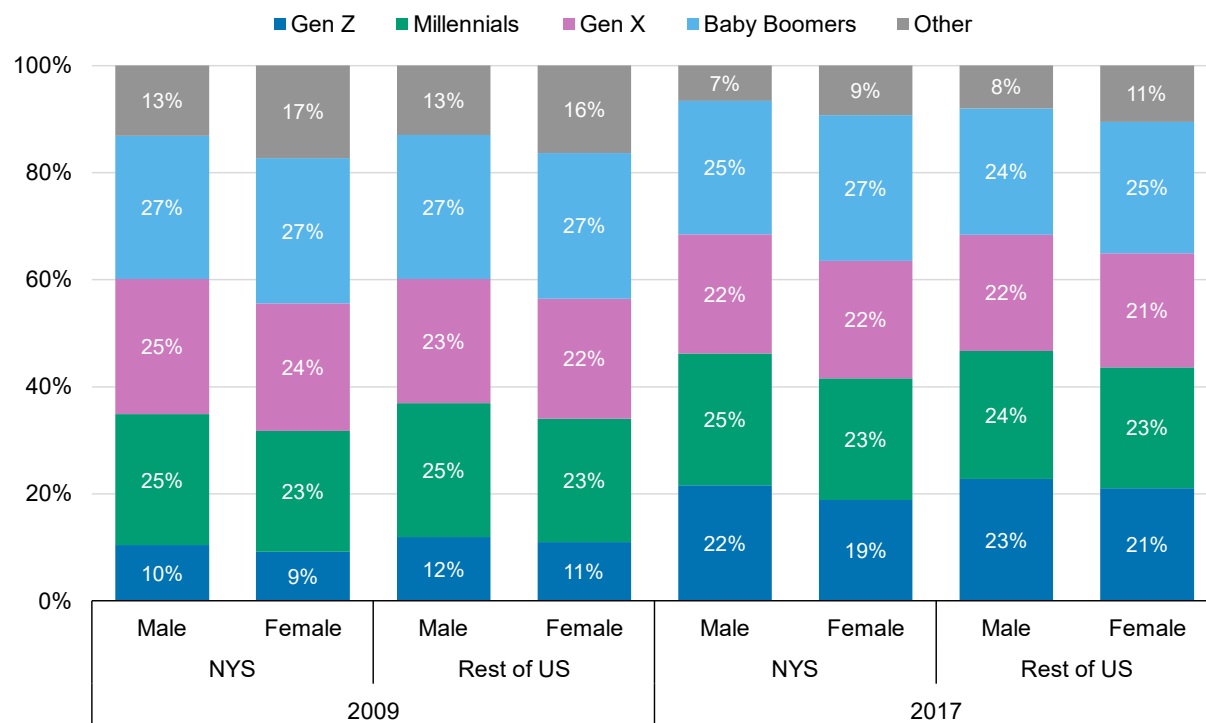


Figure 2-5. Distribution of population by generation and gender in NYS and the rest of the US (ACS data).

2.2 HOUSEHOLD CHARACTERISTICS

2.2.1 Millennial Household Count and Proportion

Figure 2-6 illustrates the household count and proportion by generation of the oldest household member. In 2009, millennials constituted a small fraction of households across all regions, with only 3%–4% in NYS, including NYC, other urban areas, and nonurban areas, as well as in the rest of the US. By 2017, however, a substantial increase in the proportion of millennial-led households had taken place. NYC saw the most significant jump, with millennials now representing 19% of households, compared with just 4% in 2009. Similarly, in other urban areas of NYS, the proportion increased from 3% to 14%, and in nonurban areas, from 4% to 14%. Across NYS as a whole, millennials made up 16% of households in 2017, a notable rise from 3% in 2009. The rest of the US followed this trend. The particularly high percentage in NYC might be attributed to several factors, including the city’s appeal to younger, career-driven individuals who are drawn to its diverse job market, cultural amenities, and urban lifestyle. The trend also reflects broader demographic shifts as millennials move out of their parents’ homes and establish their own households. For Gen Xers, the proportion of households increased moderately from 2009 to 2017 across all regions. The Baby Boomer Generation, while still holding a significant share of households, saw a slight decline over the same period. Generation Z is notably absent from the data as the oldest household member, which is expected given the generational timeline.

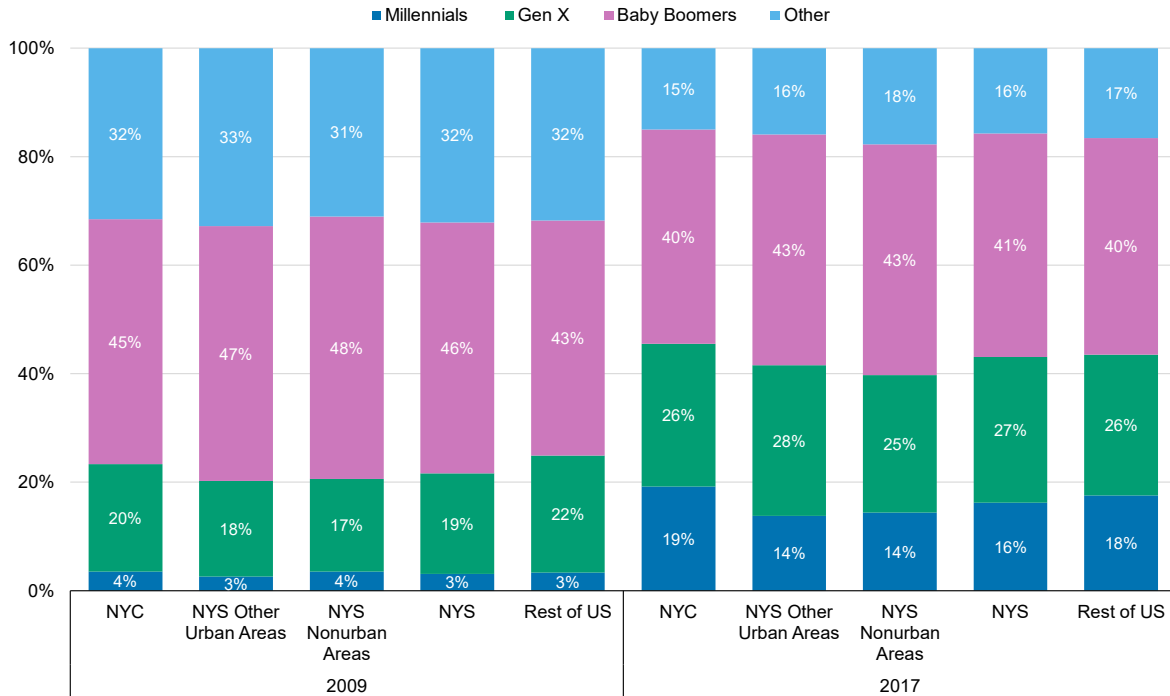


Figure 2-6. Household distribution by generation of the oldest household member.

As shown in Figure 2-6, the proportion of millennial-led households increased significantly from 2009 to 2017 is further demonstrated in Table 2-2. This table reinforces the earlier observation that millennials are increasingly taking on the role of household heads as they age into adulthood and gain financial independence. Meanwhile, the proportion of households where millennials live with older household members decreased slightly from 2009 to 2017 across all regions. In NYC, this percentage dropped from 22% in 2009 to 17% in 2017. Similarly, in other urban areas of NYS, the proportion decreased from 22% to 19%, and in nonurban areas, from 20% to 17%. In the rest of the US, the percentage declined from 21% to 15%. The data also indicate a decline in the proportion of households with no millennials from 2009 to 2017, further supporting the rising number of millennial-headed households and suggesting that millennials are becoming more prevalent as household leaders across various regions.

Table 2-2. Distribution of millennial household count by household status across geographic regions (NHTS data).

Year	Region	Households with no millennials	Millennials living with older household members	Millennials as heads of household	Total
2009	NYC	74%	22%	4%	3,055,273
	NYS other urban areas	76%	22%	3%	3,380,154
	NYS nonurban areas	76%	20%	4%	702,055
	NYS	75%	22%	3%	7,137,482
	Rest of US	76%	21%	3%	105,963,848
2017	NYC	64%	17%	19%	3,129,147
	NYS other urban areas	67%	19%	14%	3,379,813
	NYS nonurban areas	69%	17%	14%	724,734
	NYS	66%	18%	16%	7,233,694
	Rest of US	68%	15%	18%	110,974,557

Note: Percentages may not total 100 due to rounding.

2.2.2 Household Size

The distribution of household size among the millennial population based on household status and geographic region is shown in Figure 2-7. The data show a significant variation in household size for millennials who are heads of households. Generally speaking, millennials who are heads of households are more likely than millennials living with older household members to be part of one- or two-person households. This is clearly shown in Figure 2-7. In NYC, a high proportion of one-person households (37%) and two-person households (45%) suggests that many millennial-headed households consist of individuals living alone or with a partner. This is consistent with the high cost of living in NYC, which might limit the ability for some people to afford larger households. In contrast, NYS nonurban areas show a notable proportion of larger millennial-headed households, with 46% of households having four or more persons. This could reflect the availability of more affordable housing and a greater tendency for millennials in nonurban areas to have families with children. The rest of the US and NYS as a whole present a more balanced distribution of household sizes, with a fairly even spread across the different household sizes for millennial-headed households. On the other hand, across all regions, households where millennials live with older household members tend to have larger household sizes. For example, 50% of such households in the rest of the US and 51% in NYS have four or more persons. The proportions are even higher in NYS nonurban areas (58%) and NYS other urban areas (59%). NYC is somewhat of an outlier, where a significant proportion (24%) of these households have two persons, which might indicate situations where millennials live with an older family member. Overall, the data reflect the clear distinction between urban and nonurban living conditions. In NYC and other urban areas, millennial households tend to be smaller, likely because of higher housing costs and space constraints. In nonurban areas, larger households are more common, driven by lower living costs and different lifestyle preferences. Also, the prevalence of larger households among millennials living with older household members suggests cultural factors, such as multigenerational living, as well as economic factors.

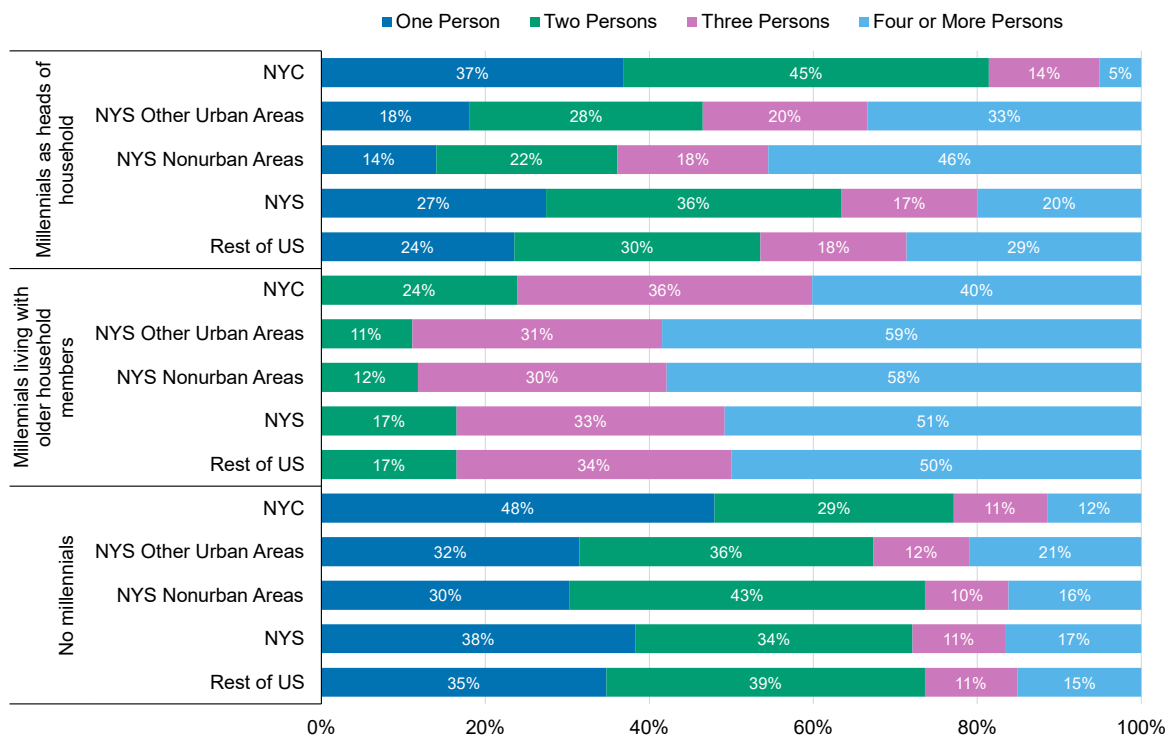


Figure 2-7. Household size distribution of the millennial population based on their household status and geographic region (2017 NHTS data).

2.2.3 Household Income

Household income distribution by household status for 2017 is depicted in Figure 2-8. Based on observations, millennials as heads of households in NYC show a significant skew toward higher income, with 35% earning \$100,000 or more. However, a substantial proportion (16%) earns less than \$25,000, reflecting the diverse economic circumstances of millennials in the city. In NYS other urban areas, the income distribution is more varied, with 27% earning less than \$25,000 and 22% earning \$100,000 or more. This indicates a broader income spectrum for millennial-headed households in these regions. In contrast, in NYS nonurban areas, a striking 37% of millennial-headed households earn less than \$25,000, suggesting the presence of economic challenges in nonurban areas. Only 11% earn \$100,000 or more, highlighting a lower prevalence of high-income millennial households in these regions. Meanwhile, the income distribution is more balanced in the rest of the US, with 23% earning less than \$25,000 and 21% earning \$100,000 or more. This reflects the overall economic diversity of millennial-headed households across the country.

Among millennials living with older household members in NYC, 34% are in households earning \$100,000 or more, indicating a significant presence of higher-income households. This could reflect multigenerational households where multiple earners contribute to a higher overall income. Interestingly, 14% of these households earn less than \$25,000, highlighting a significant income gap. Whereas in NYS other urban areas, a significant proportion (42%) of millennials living with older household members are in households earning \$100,000 or more, which may indicate a higher concentration of wealth in these multigenerational households compared with those of other regions. In addition, the income distribution is more balanced in NYS nonurban areas, with 25% of these households earning less than \$25,000 and 22% earning \$100,000 or more.

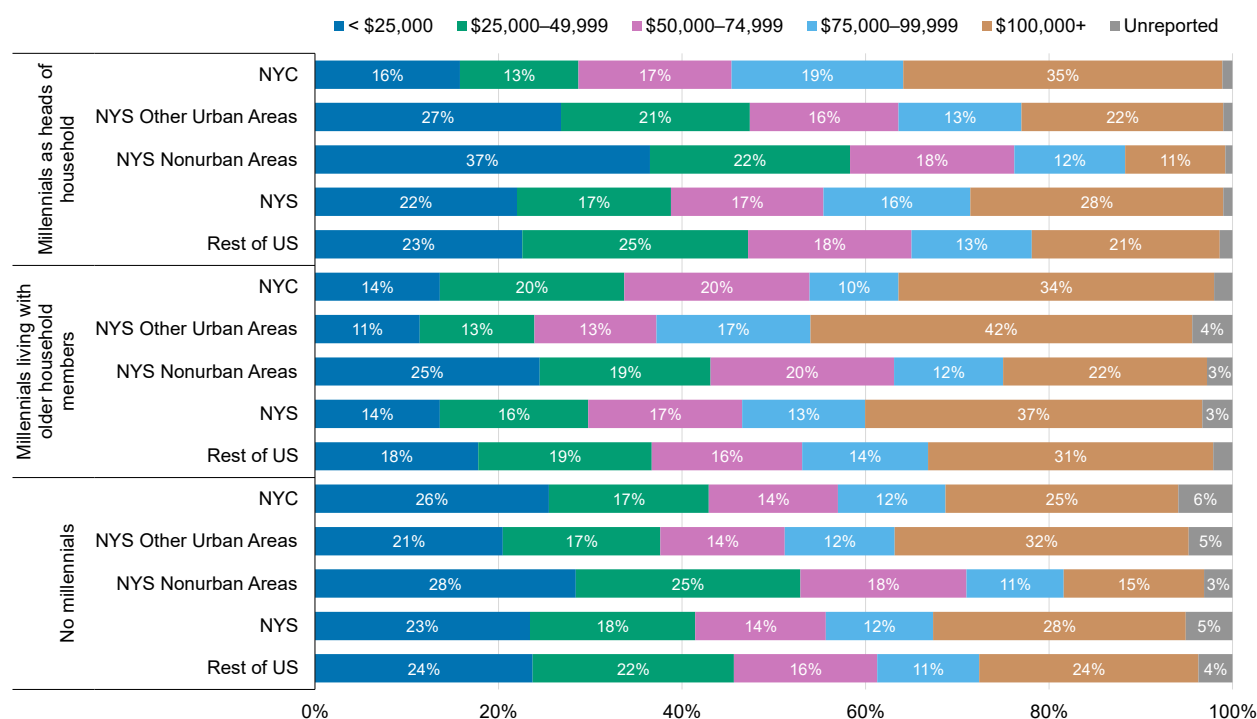


Figure 2-8. Household income distribution by household status (2017 NHTS data).

2.2.4 Household Vehicle Ownership

Table 2-3 provides insights into the distribution of millennial household status by vehicle ownership across different geographic regions using 2017 NHTS data. In NYC, more than half of the millennial-headed households have zero vehicles (64%), which is considerably higher than in other regions as well as in households without millennials (53%). This is likely due to the extensive public transportation system in NYC, where many residents do not require personal vehicles. Also, many millennials in NYC may prefer or are economically compelled to rely on public transportation, potentially owing to high costs associated with vehicle ownership in the city. In contrast, in other urban areas in NYS, the vehicle ownership pattern is more balanced: 43% of millennial-headed households own one vehicle, and 37% own two vehicles. In nonurban areas, 37% own two vehicles, with a smaller percentage (14%) owning three or more vehicles, reflecting a moderate need for personal transportation. The vehicle ownership distribution in the rest of the country shows that 40% of millennial-headed households own two vehicles, which is the highest percentage across all vehicle ownership categories. This suggests that many millennials, particularly outside of major urban centers, rely on having at least one or two personal vehicles for daily commuting and other transportation needs.

Among millennials living with older household members, 33% of households have no vehicle, which is lower than that of the no millennials and millennials as heads of household categories in NYC but still reflects a significant reliance on public transportation. A considerable number (36%) own one vehicle, and 21% own two vehicles, indicating some level of dependence on personal transportation. Importantly, a noticeable pattern is observed in these households, where a substantial proportion own three or more vehicles (56% in other urban areas and 50% in nonurban areas). This likely indicates larger, multigenerational households with multiple working adults requiring their own vehicles.

Overall, researchers have observed a clear distinction between urban and nonurban areas in terms of vehicle ownership. Urban areas, particularly NYC, show a higher reliance on public transportation and other alternative modes of transportation (i.e., walking and biking), with a significant proportion of households owning no vehicles. In contrast, nonurban areas exhibit higher rates of vehicle ownership, reflecting the necessity of personal vehicles in regions with less developed public transportation infrastructure. Millennials living with older household members tend to have larger vehicle fleets, especially in nonurban areas, likely owing to the presence of multiple working adults in these households. This contrasts with millennial-headed households, particularly in NYC, where vehicle ownership is lower, possibly owing to economic constraints and the availability of alternative transportation options.

Table 2-3. Distribution of millennial household status by number of vehicles and geographic region (2017 NHTS data).

Household status	Region	Zero vehicles	One vehicle	Two vehicles	Three or More vehicles	Total number of households
Millennials as heads of household	NYC	64%	33%	3%	0%	599,394
	NYS other urban areas	11%	43%	37%	10%	464,955
	NYS nonurban areas	20%	29%	37%	14%	104,418
	NYS	39%	37%	19%	5%	1,168,767
	Rest of US	8%	39%	40%	13%	19,461,970
Millennials living with older household members	NYC	33%	36%	21%	10%	537,786
	NYS other urban areas	5%	10%	29%	56%	641,047
	NYS nonurban areas	7%	18%	24%	50%	123,474
	NYS	17%	21%	25%	37%	1,302,306
	Rest of US	3%	16%	32%	49%	16,117,678

Table 2-3. Distribution of millennial household status by number of vehicles and geographic region (2017 NHTS data) (continued).

Household status	Region	Zero vehicles	One vehicle	Two vehicles	Three or More vehicles	Total number of households
No millennials	NYC	53%	35%	11%	2%	1,991,968
	NYS other urban areas	11%	38%	35%	17%	2,273,811
	NYS nonurban areas	11%	36%	31%	23%	496,842
	NYS	29%	36%	24%	11%	4,762,621
	Rest of US	9%	36%	32%	23%	75,394,909

Note: Percentages may not total 100 due to rounding.

Additionally, for the Mid-Atlantic Division, Figure 2-9 provides a valuable comparison of vehicle ownership trends for millennial households using 2009, 2017, and 2022 NHTS data. The figure shows that vehicle ownership grew steadily between 2009 and 2022. The percentage of households with no vehicles decreased, whereas ownership of one, two, and even three or more vehicles increased. This shift reflects changing life stages for millennials as they moved into their 30s and 40s, often requiring more transportation options for family and work. Moreover, millennials living with older household members tend to have higher rates of multivehicle ownership, particularly in recent years. By 2022, 44% of these households owned two vehicles, and 36% owned three or more vehicles, underscoring the transportation needs of larger, multigenerational households.

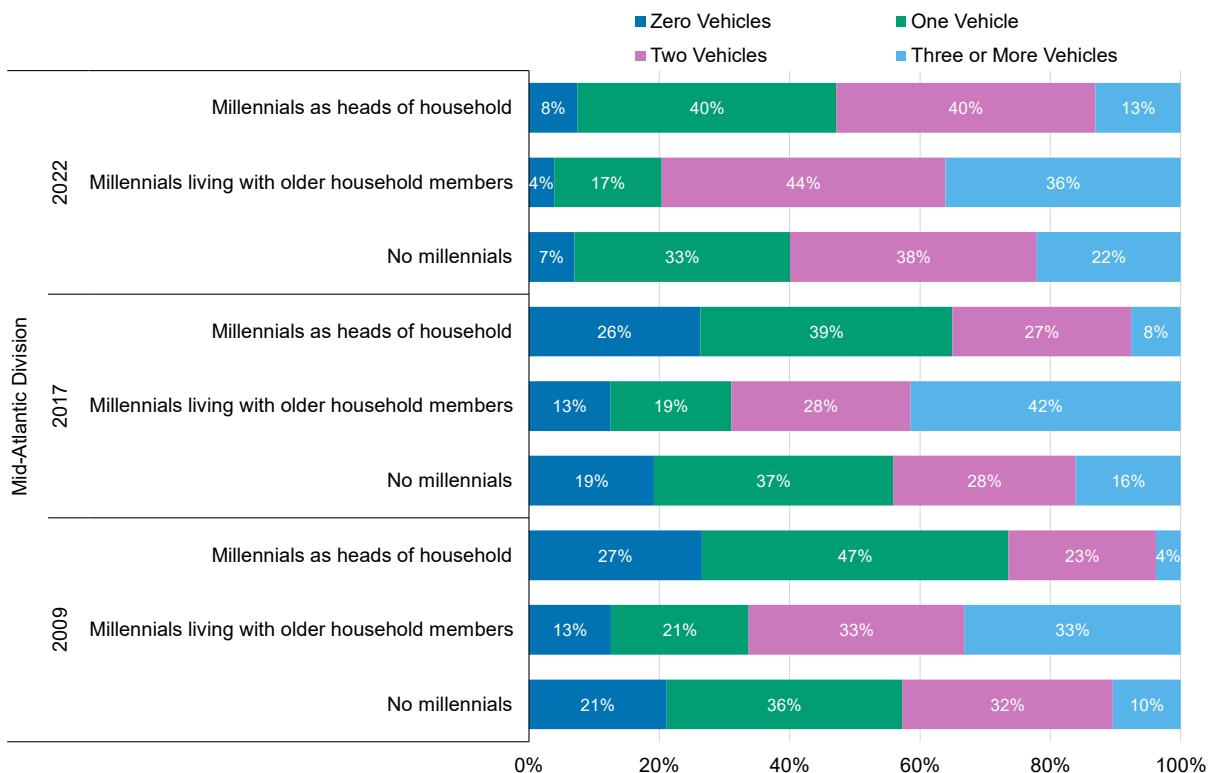


Figure 2-9. Distribution of millennial household status by number of vehicles over the years in the Mid-Atlantic Division (NHTS data).

An overview of the distribution of millennial household status by age of vehicles owned across different geographic regions using 2017 NHTS data is illustrated in Figure 2-10. In households where millennials

are the heads, NYC shows a unique pattern where 35% of vehicles are 11 years or older, but there is also a significant proportion of newer vehicles, with 25% being 2–3 years old and 12% being up to 1 year old. The proportion of vehicles 3 years old or newer in NYC is higher than that in the other NYS regions. This may suggest that millennials in NYC are inclined to invest in newer vehicles, possibly because of the urban environment’s demands or the economic status of millennial-headed households in the city.

In the households of millennials living with older household members, 33% of vehicles in NYC are 11 years or older, with a notable 22% being 2–3 years old. This higher proportion of newer vehicles may indicate a preference for newer models in urban settings. In the nonurban areas, 39% of vehicles are 11 years or older, whereas 26% are 6–10 years old, again indicating a mix of older vehicles similar to other urban areas. In the rest of the US and NYS overall, the vehicle age distribution is quite similar to that of households without millennials, with 45% of vehicles in the rest of the US and 33% of vehicles in NYS being 11 years or older. This suggests that the presence of millennials in multigenerational households does not significantly alter the age of vehicles, likely because of shared economic circumstances.

Overall, the presence of millennials, whether living with older household members or as heads of households, does not appear to change the age distribution of vehicles compared with households without millennials. However, millennials in urban areas, especially in NYC, appear to favor newer vehicles more than those in nonurban areas.

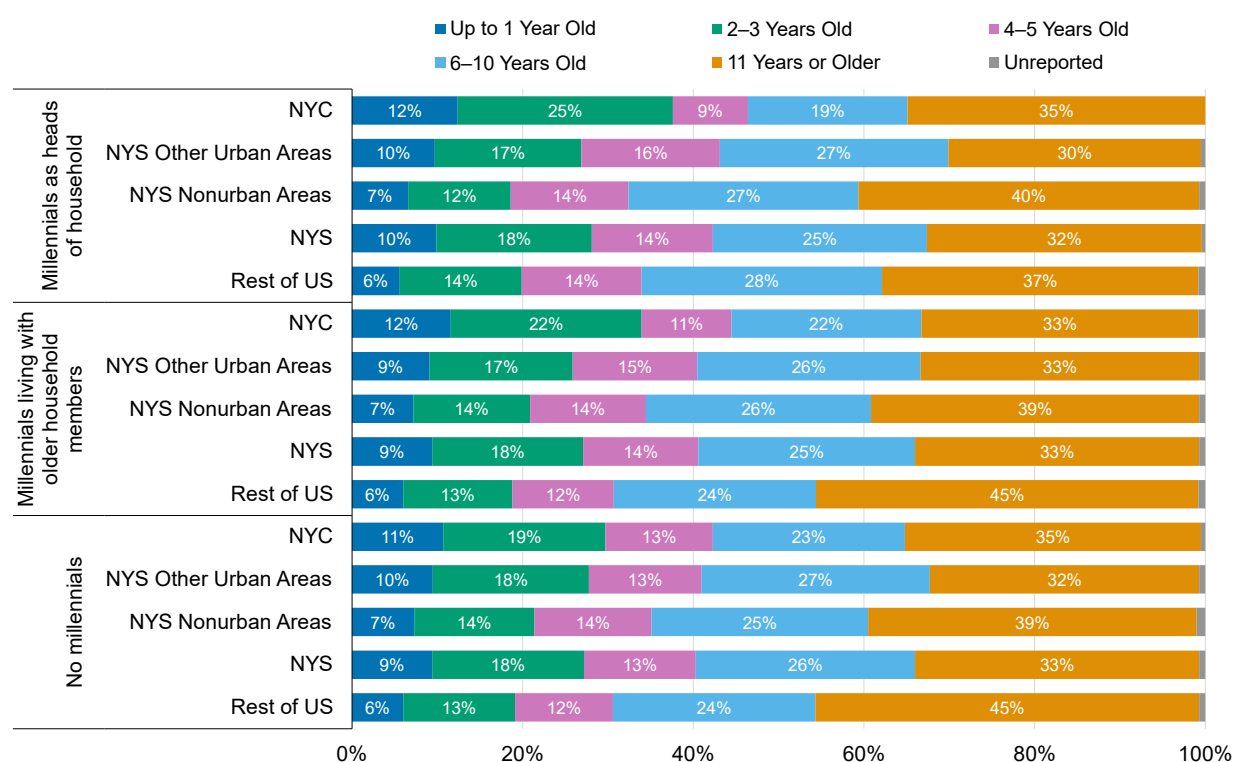


Figure 2-10. Distribution of millennial household status by vehicle age and geographic region (2017 NHTS data).

2.2.5 Driving Status

Figure 2-11 and Table 2-4 provide a comprehensive breakdown of the percentage of drivers by generation in 2009 and 2017, further subdivided by gender. In 2009, millennials represented a relatively small portion of the driving population across all regions. For example, in NYC, millennials made up 18% of

drivers, whereas in NYS other urban areas, they accounted for 20%. The rest of the US showed similar trends, with millennials comprising 19%–21% of drivers. By 2017, millennials had become a much larger share of the driving population. In NYC, the percentage of millennial drivers increased to 31%, matching the proportion of Gen X drivers. This shift is seen across other regions as well, with 27%–31% of drivers being millennials. This increase reflects the aging of the millennial generation into their late 20s and 30s, a period when car ownership and driving become more prevalent. Looking at other generations, Gen X was the dominant group of drivers in 2009, comprising 28%–38% of drivers across all regions. Baby boomers were also a significant portion of the driving population in 2009, accounting for 32%–36% of drivers in most regions. These generations were at the peak of their driving years, balancing careers and family life, which typically require regular driving. By 2017, the share of Gen X drivers had decreased slightly, aligning more closely with millennials. Also, the percentage of baby boomers driving decreased across all regions, with their share dropping to 30%–34%. This decline is due to the aging of this generation, with many retiring or reducing their driving because of health or lifestyle changes.

Table 2-4 indicates that the rise in millennial drivers is consistent across both genders, although the rate of increase and the proportions slightly differ. By 2017, both male and female millennials had become a significant part of the driving population, reflecting their growing economic independence and mobility. For example, in 2017, millennial male drivers had increased their share to 27%–31% across regions, whereas females made up to 24%–30% in all regions. Overall, the data reflect a clear generational shift in driving patterns, with millennials increasingly becoming the dominant driving cohort by 2017, overtaking Gen Xers in some regions and significantly reducing the dominance of baby boomers.

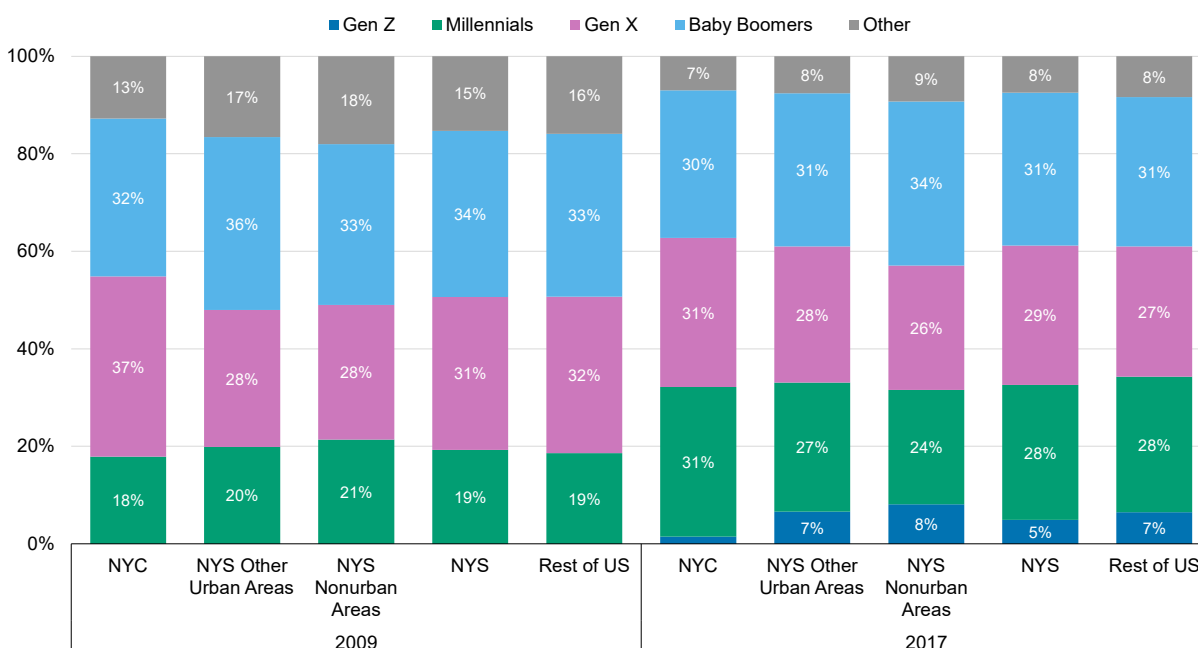


Figure 2-11. Share of drivers by generation (NHTS data).

Table 2-4. Percent of driver populations over time by gender and region (NHTS data).

Year	Gender	Region	Gen Z	Millennials	Gen X	Baby boomers	Other
2009	Male	NYC	0%	18%	38%	32%	12%
		NYS other urban areas	0%	21%	28%	36%	16%
		NYS nonurban areas	0%	22%	29%	33%	17%
		NYS	0%	20%	32%	34%	14%
		Rest of US	0%	19%	33%	33%	15%
	Female	NYC	0%	18%	36%	33%	13%
		NYS other urban areas	0%	19%	28%	35%	18%
		NYS nonurban areas	0%	21%	27%	33%	19%
		NYS	0%	19%	31%	34%	16%
		Rest of US	0%	18%	31%	34%	17%
2017	Male	NYC	1%	31%	30%	31%	7%
		NYS other urban areas	6%	27%	27%	31%	8%
		NYS nonurban areas	9%	23%	26%	33%	9%
		NYS	5%	28%	28%	31%	8%
		Rest of US	7%	28%	27%	30%	8%
	Female	NYC	2%	30%	31%	30%	7%
		NYS other urban areas	7%	26%	28%	32%	8%
		NYS nonurban areas	7%	24%	25%	34%	10%
		NYS	5%	27%	29%	31%	8%
		Rest of US	6%	28%	27%	31%	9%

Note: Percentages may not total 100 due to rounding.

Additionally, Figure 2-12 presents a breakdown of the driving population by generation and gender in the Mid-Atlantic Division. Observation indicates that millennials saw a substantial rise in their share of the driving population between 2009 and 2017. By 2022, this percentage remained stable, suggesting millennials are now an established driving group.

Furthermore, the percentage of drivers by millennial household status across different regions for the years 2009 and 2017 is illustrated in Figure 2-13. Millennials as heads of households represented a small portion of the driving population across all regions in 2009. For example, in NYC, only 4% of drivers were millennials who headed their own households. This low percentage is consistent across other regions as well, with 3%–4% of drivers being millennial heads of households. In 2017, a significant increase took place in the proportion of drivers who were millennial heads of households. In NYC, this group accounted for 19% of drivers, a substantial increase from 2009. Other regions also saw similar increases, with 13%–20% of drivers being millennials who were heads of households.

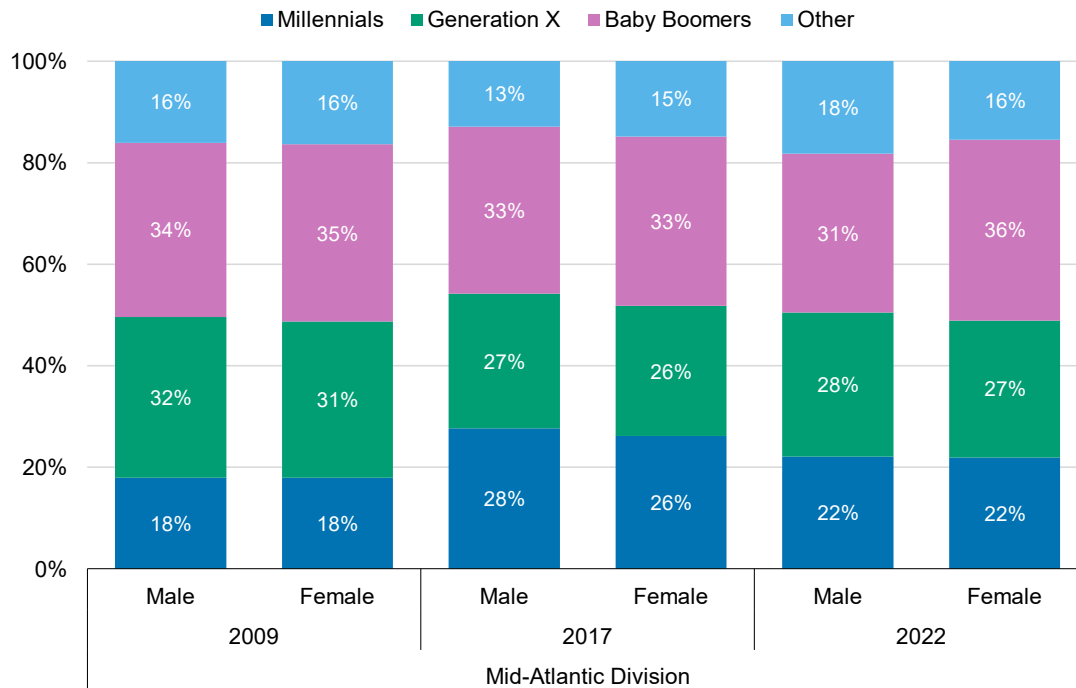


Figure 2-12. Share of drivers by generation and gender in the Mid-Atlantic Division (NHTS data).

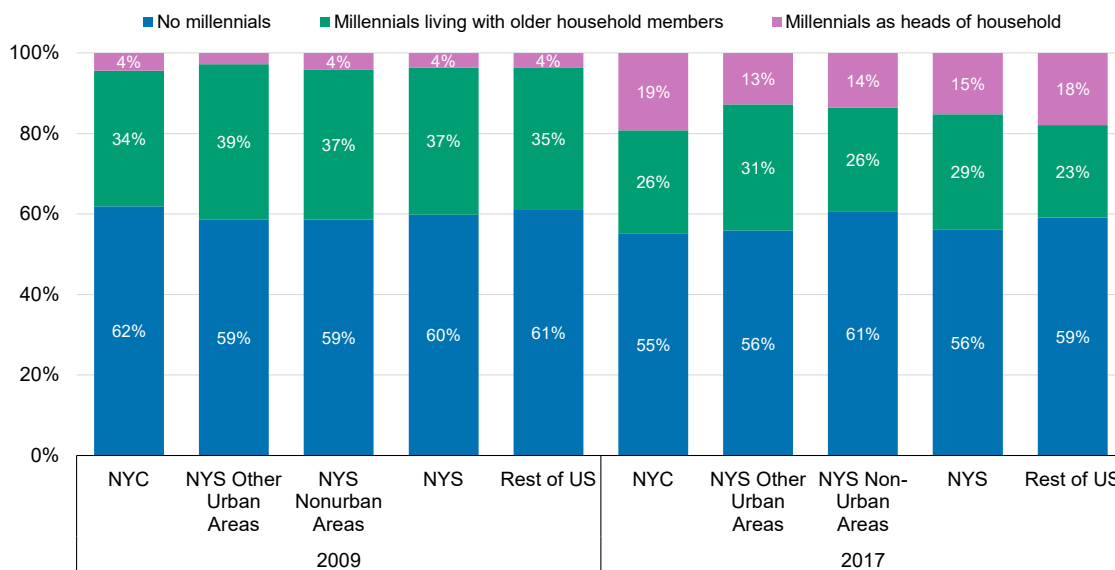


Figure 2-13. Percentages of drivers by millennial household status (NHTS data).

2.2.6 Worker Status

The percentage of the population that worked during 2009 and 2017 is presented in Figure 2-14, broken down by generation and geographic region. In 2009, millennials represented a smaller portion of the working population across all regions. In NYC, millennials made up 20% of the workforce, whereas in NYS other urban areas, they accounted for 23%. The rest of the US showed similar percentages, with

millennials comprising 20% of the workforce. This relatively low proportion can be attributed to the fact that many millennials were either still completing their education or just entering the workforce during this period. The 2017 data show a significant increase in the proportion of millennials in the workforce across all regions. In NYC, millennials' representation nearly doubled to 41%. A similar trend in millennials in the workforce is observed in other regions, with 37% in NYS and 36% in the rest of the US. This dramatic rise reflects the aging of the millennial generation into their prime working years, taking on a larger share of the workforce as Gen Xers and baby boomers begin to exit the workforce. Regarding other generations, there is a slight decline in Gen X's workforce share (for example, in NYC, from 42% in 2009 to 35% in 2017). However, they still hold a significant portion, indicating that they remain a crucial part of the workforce, especially in leadership and experienced roles. However, baby boomers' percentages showed a substantial decline, underscoring the ongoing retirement of this generation. Meanwhile, the small but growing presence of Generation Z in the workforce marks the beginning of their economic influence, which is expected to grow in the coming years.

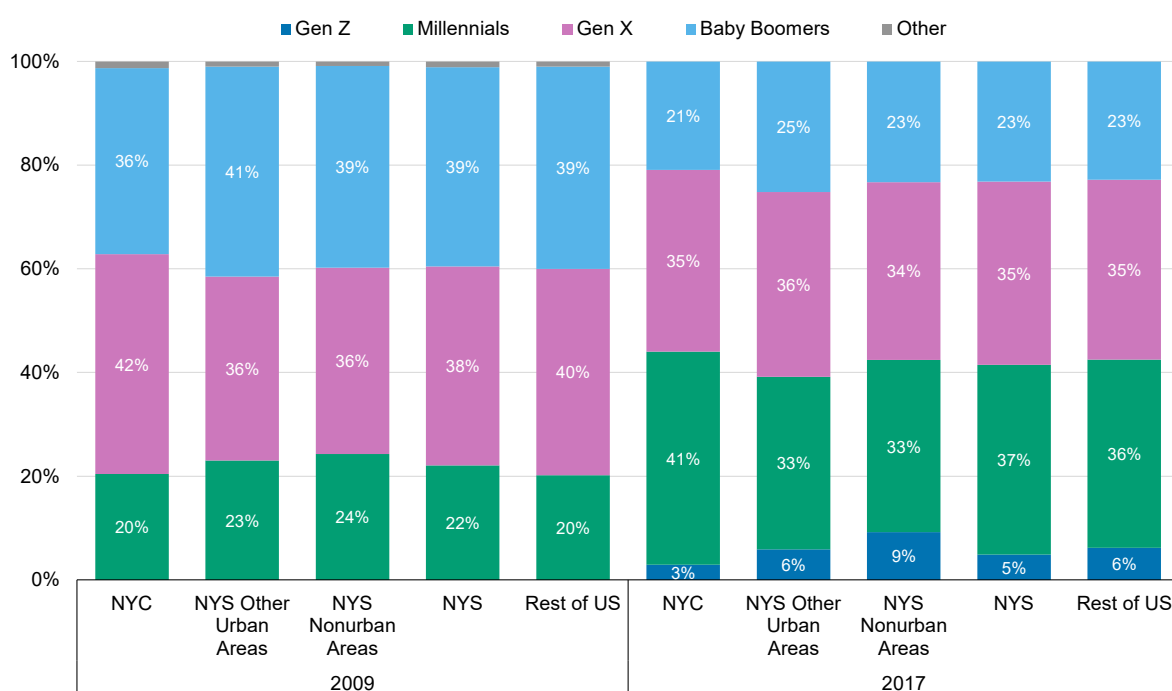


Figure 2-14. Workers by generation (NHTS data).

Table 2-5 further breaks down workforce participation data by gender, offering a more detailed view of how different generations, particularly millennials, contributed to the workforce in 2009 and 2017 across various regions. It additionally illustrates a clear trend where millennials have grown to dominate the workforce by 2017, replacing Gen Xers and baby boomers as the leading working generation. This transition is evident in both male and female labor markets, with millennials becoming the largest generational cohort across nearly all regions. The significant increase from 2009 (18%–26% of males, 20%–23% of females) to 2017 (33%–42% of males and 32%–40% of females) underscores the effect of this generation as they moved into their prime working years. The rise of millennials in the workforce is consistent across gender lines, though the timing and extent of their entry slightly vary. In 2009, the gender distribution showed that females had a slightly higher workforce participation rate among millennials, but by 2017, the gender gap narrowed as male millennials caught up.

Table 2-5. Percent of worker populations over time by gender and region (NHTS data).

Year	Gender	Region	Gen Z	Millennials	Gen X	Baby boomers	Other
2009	Male	NYC	0%	18%	45%	35%	2%
		NYS other urban areas	0%	23%	37%	40%	1%
		NYS nonurban areas	0%	26%	37%	36%	1%
		NYS	0%	21%	40%	38%	1%
		Rest of US	0%	20%	42%	37%	1%
	Female	NYC	0%	23%	40%	37%	1%
		NYS other urban areas	0%	23%	34%	41%	1%
		NYS nonurban areas	0%	22%	34%	43%	1%
		NYS	0%	23%	37%	40%	1%
		Rest of US	0%	20%	38%	41%	1%
2017	Male	NYC	2%	42%	36%	20%	0%
		NYS other urban areas	6%	33%	36%	26%	0%
		NYS nonurban areas	9%	34%	35%	23%	0%
		NYS	4%	37%	36%	23%	0%
		Rest of US	6%	36%	35%	23%	0%
	Female	NYC	4%	40%	34%	22%	0%
		NYS other urban areas	6%	34%	35%	24%	0%
		NYS nonurban areas	10%	32%	34%	24%	0%
		NYS	6%	36%	35%	23%	0%
		Rest of US	6%	36%	35%	23%	0%

Note: Percentages may not total 100 due to rounding.

Figure 2-15 provides insights into the percentage of workers by millennial household status across different regions for the years 2009 and 2017. This additional layer of analysis offers a closer look at how millennials, particularly those who are heads of households, contributed to the workforce compared with other household compositions. In 2009, millennials as heads of households represented a small fraction of the working population across all regions. For instance, in NYC, only 5% of workers were millennials who headed their own households. This pattern was similar in other regions, with 3%–5% in NYS other urban and nonurban areas and 4% in the rest of the US. The low proportion can be attributed to the fact that many millennials were still relatively young, likely still in education or early in their careers, and thus less likely to be household heads at that time. A significant shift occurred by 2017, with millennials as heads of households forming a much larger part of the workforce. In NYC, this group accounted for 27% of the workforce, a substantial increase from 2009. Other regions also saw notable increases, with 16%–20% in NYS other urban and nonurban areas, and 21% across NYS overall. The rest of the US followed a similar trend, with 24% of the workforce being millennial-headed households. This growth reflects the aging of millennials into their late 20s and 30s, when they are more likely to have established their own households and careers.

In 2009, households without millennials comprised the majority of the workforce in all regions, with 57% in NYC and 52%–56% in other areas. In 2017, the proportion of workers in nonmillennial households declined across the board, most notably in NYC, where it dropped to 44%. This decline mirrors the overall aging population and the exit of older generations from the workforce as millennials stepped into these roles. Millennials living with older household members saw a more modest shift compared with millennial-headed households. In 2009, millennials living with older household members made up 39% of the workforce in NYC and 40%–45% in other regions. These figures had decreased by 2017, with 29% in

NYC and 26%–36% in other areas. The reduction in this category may indicate that as millennials aged and became more financially independent, they moved out of multigenerational homes to form their own households, thus shifting from this category to becoming heads of households themselves.

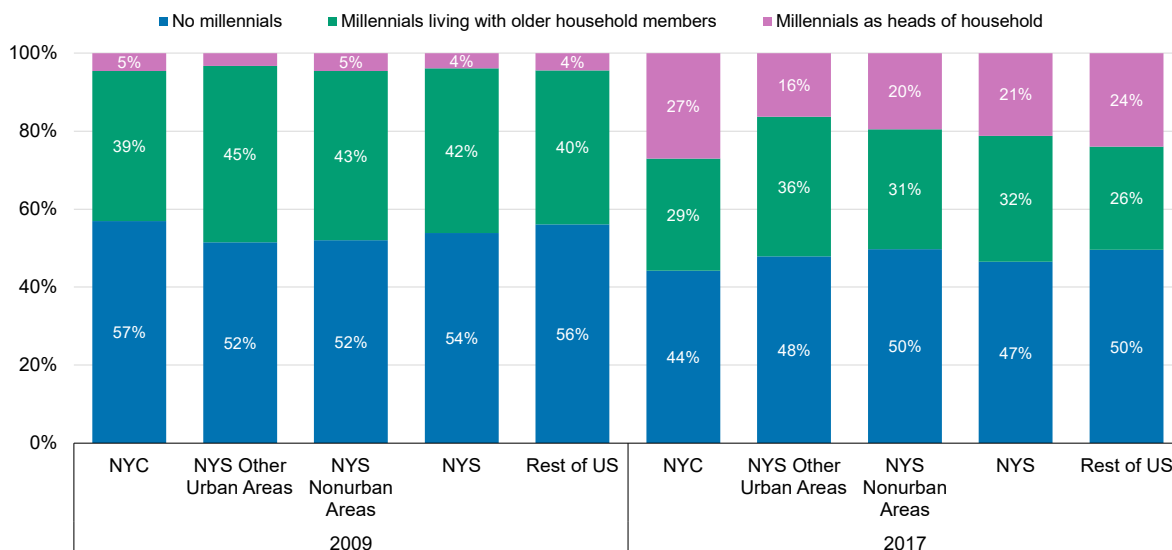


Figure 2-15. Percentages of workers by millennial household status (NHTS data).

Additionally, Figure 2-16 illustrates the share of workers by millennial household status in the Mid-Atlantic Division for the years 2009, 2017, and 2022. It shows a steady decline in the number of millennials living with older household members from 2009 to 2022. Meanwhile, the percentage of workers in millennial-headed households increased substantially between 2009 and 2017 but remained stable from 2017 to 2022. This suggests that by 2017, many millennials had already transitioned into independent household leadership, and the proportion of these households has since plateaued.

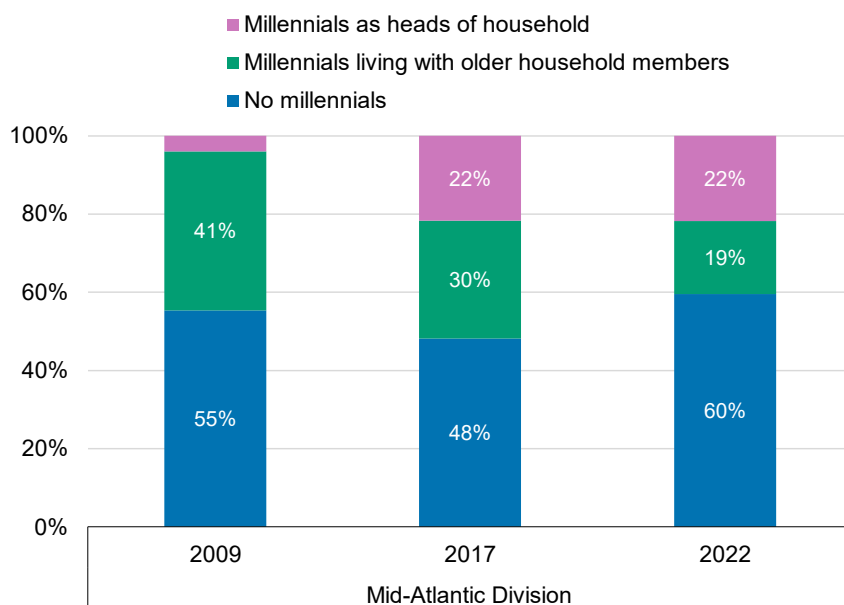


Figure 2-16. Distribution of workers by millennial household status in the Mid-Atlantic Division (NHTS data).

3. TRIP PATTERNS OF MILLENNIALS

3.1 TRAVEL STATISTICS BY PERSON TRIPS

3.1.1 Average Daily Person-Trip Rate

The average daily person trips per person by generation across various regions, comparing NHTS data from 2009 and 2017, is shown in Figure 3-1. In 2009, millennials had an average of 3.0–3.5 daily person trips across different regions. For example, millennials in NYC had 3.0 daily trips, whereas in NYS other urban areas, the average was 3.5 trips. Observations show that millennials in NYC made fewer trips compared with those in other regions, which may reflect the higher reliance on public transportation and the dense urban structure of NYC, where walking or short-distance trips are more common. However, factoring in the MOE is critical when assessing whether the observed differences among generations, regions, and years are statistically significant. The MOE provides a range of potential variation in the estimate and helps to determine the reliability of the differences between groups. For example, in 2009 NYC, millennials averaged 3.0 trips per day, compared with 4.2 trips for Gen Xers. After adjusting for their MOEs, millennials' trip range become 2.7 to 3.3, whereas Gen Xers' range is 3.9 to 4.5. Because these ranges don't overlap, the difference in trips between millennials and Gen Xers in NYC is statistically significant, showing that Gen Xers had a notably higher number of daily trips compared with those of millennials. Similarly, in other regions, millennials made significantly fewer trips than Gen Xers and baby boomers made, indicating that millennials were still relatively young and less likely than older generations to be traveling for work or family responsibilities. In contrast, whereas millennials averaged more trips than those of Gen Zers, these differences were not statistically significant. However, when comparing millennials' trips across regions, the rest of the US consistently showed a statistically higher number of daily trips compared with those in NYC and NYS areas.

Similarly, in 2017, millennials showed significant higher number of trips compared with those of Gen Zers. Although millennials made fewer trips than Gen Xers and baby boomers made, the difference between millennials and Gen Xers was not statistically significant in NYC. This suggests that by 2017, millennials' travel patterns in NYC began to align more closely with those of Gen Xers. However, in other regions like NYS other urban areas, NYS nonurban areas, and the rest of the US, the difference between millennials and Gen Xers remained statistically significant. Millennials in these regions continued to make fewer trips than those of Gen Xers, indicating that older generations still traveled more frequently, likely because of their stage of life involving more work or family-related responsibilities.

A comparison of millennials' travel behavior between 2009 and 2017 shows slight increases in NYC, where daily trips rose from 3.0 in 2009 to 3.4 in 2017. However, this increase was not statistically significant. In other areas, like NYS nonurban areas, millennials' travel behavior remained stable, with average daily trips hovering around 3.0 in both years, suggesting little change in travel patterns for millennials in less urbanized settings.

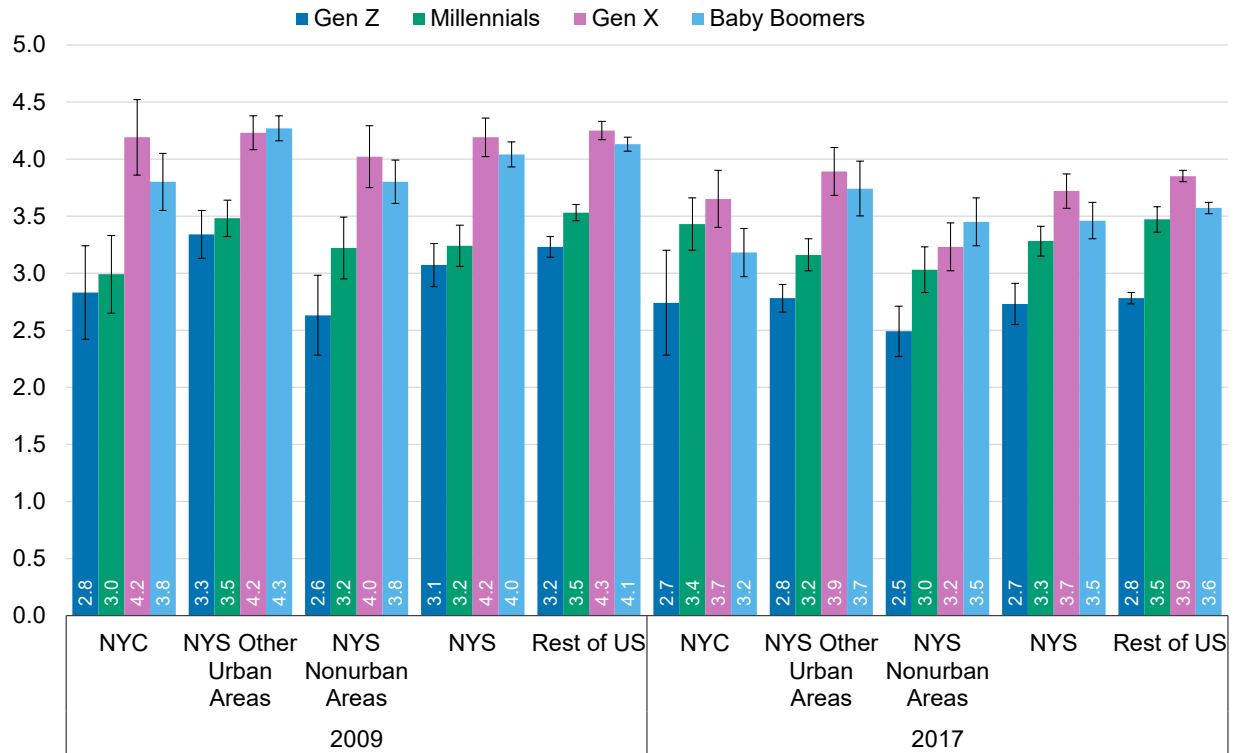


Figure 3-1. Average daily person trips per person by generation and region over time (NHTS data).

Additionally, we observed the average daily person trips per person by generations for the Mid-Atlantic region over time (Figure 3-2). In 2009, millennials in the Mid-Atlantic region averaged 3.2 trips per day, which was slightly fewer than those of Gen Xers and baby boomers (both averaging 4.2 trips per day). This is statistically significant and consistent with the national trend where millennials made fewer trips than those of older generations, likely because of their life stage, with fewer responsibilities such as commuting for work or driving children to activities. In 2017, millennials in the Mid-Atlantic region experienced a slight increase in their average number of daily trips, rising to 3.3 trips per day, whereas Gen Xers and baby boomers saw declines to 3.7 and 3.6 trips, respectively. Although millennials' travel behavior in this region was becoming closer to that of older generations, the gap remained, reflecting the generational differences in travel demands, particularly for work- or family-related travel. In the rest of the US, millennials averaged 3.5 trips per day, slightly higher than in the Mid-Atlantic, indicating that millennials in other regions might be more reliant on personal travel. In 2022, a dramatic decrease took place in the average daily trips for all generations across the Mid-Atlantic region and the rest of the US. Millennials in the Mid-Atlantic dropped to 2.3 trips per day, down from 3.3 in 2017. This drop aligns with a broader trend of reduced travel across all generations, likely influenced by societal changes such as the COVID-19 pandemic, increased remote work, and shifting lifestyles (Patwary & Khattak, 2022). For comparison, Gen Xers averaged 2.5 trips per day in 2022, and baby boomers dropped to 2.6 trips, narrowing the gap between generations.

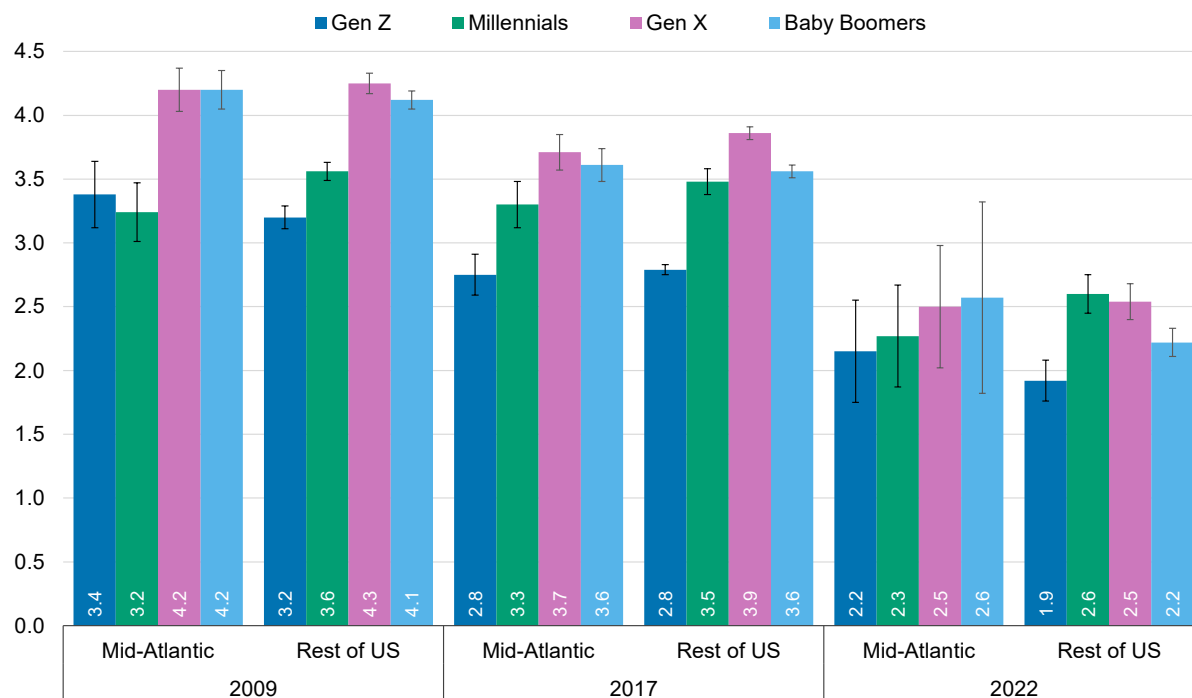


Figure 3-2. Average daily person trips per person by generation in the Mid-Atlantic region and the rest of the US over time (NHTS data).

Statistics presented in Figure 3-3 revealed the average daily person trips per person by gender for millennials across various regions, comparing 2009 and 2017 NHTS data. Statistically significant differences between male and female millennials were observed only in the rest of the US, both in 2009 and 2017, in which females consistently made more trips than those of males. In all other regions, the differences between genders were not statistically significant. For example, in 2017, male millennials in NYC averaged 3.3 trips, whereas females averaged 3.6 trips, with MOEs of 0.4 for males and 0.2 for females. The difference between male and female millennials' trips is not statistically significant, although the gap has narrowed compared with the data from 2009. The average trips for millennials in NYC increased slightly between 2009 and 2017, although the increase was not statistically significant. NYS other urban areas and NYS nonurban areas showed little change in the number of daily trips between 2009 and 2017 for both genders, and these changes were also not statistically significant.

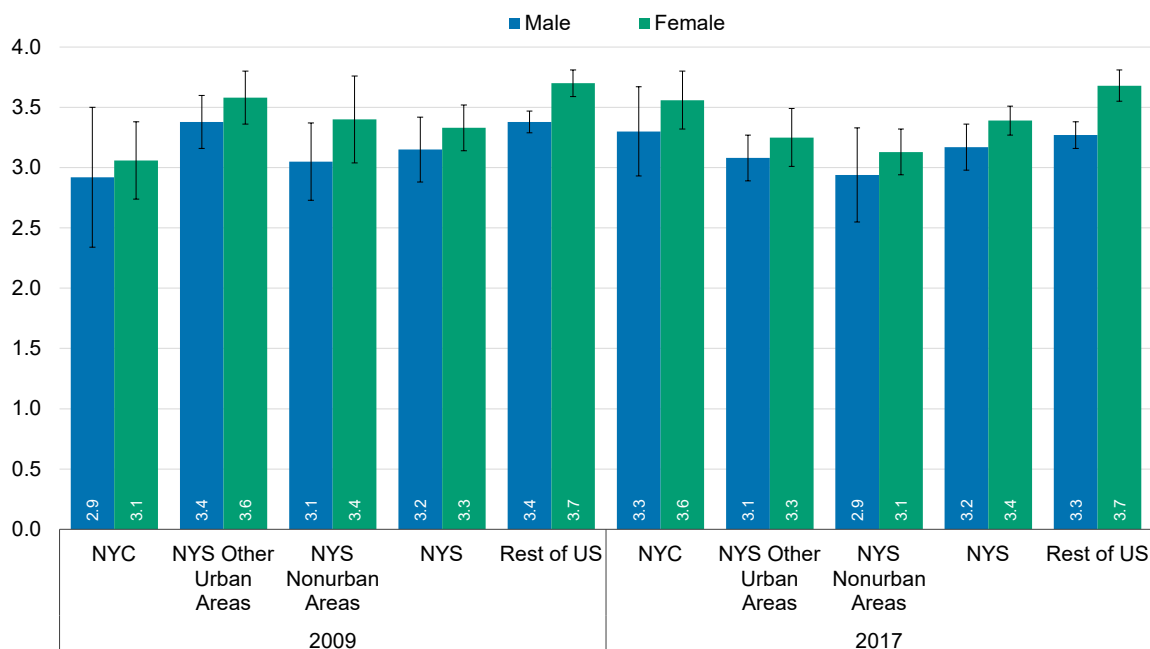


Figure 3-3. Average daily person trips per person for millennial by region and gender over time (NHTS data).

The average daily person trips per person by millennial household status and region is presented in Table 3-1. NYC shows a notable increase in the number of trips for millennials as heads of households in 2017 compared with 2009, with a statistically significant difference from other household types in 2017. In NYS other urban areas, millennials as heads of households made fewer trips compared with households without millennials in 2017, marking a statistically significant change. NYS nonurban areas saw no statistically significant difference in trips for millennials as heads of households between 2009 and 2017, indicating stable travel behavior. In the rest of the US, millennials as heads of households showed similar trip rates to those of no-millennials households, with no significant changes from 2009 to 2017, reflecting consistency in travel patterns across these years.

Additionally, Figure 3-4 provides insights into the average daily person trips made by male and female millennials as heads of household across different regions in 2009 and 2017. A statistically significant difference was observed in 2009 in NYS other urban areas, where female millennials made significantly more trips than male millennials made. By 2017, however, the differences between male and female millennials were not statistically significant in most regions. In NYC, both males and females experienced a decrease in the number of trips from 2009 to 2017, but the differences between genders were not statistically significant in either year. Although slight changes took place in trips between 2009 and 2017 in NYS nonurban areas, these changes were not statistically significant for either gender. In the rest of the US, female millennials consistently made more trips than males made, with a statistically significant difference in both 2009 and 2017.

Table 3-1. Average daily person trips per person by millennial household status and region (NHTS data).

Year	Region	Estimate			Margin of error		
		No millennials	Millennials living with older household members	Millennials as heads of household	No millennials	Millennials living with older household members	Millennials as heads of household
2009	NYC	3.68	3.13	3.60	0.22	0.23	1.03
	NYS other urban areas	3.86	3.78	3.84	0.08	0.14	0.42
	NYS nonurban areas	3.54	3.41	4.01	0.15	0.23	0.66
	NYS	3.75	3.47	3.74	0.11	0.13	0.51
	Rest of US	3.83	3.77	3.88	0.04	0.07	0.19
2017	NYC	3.32	2.64	3.85	0.20	0.36	0.28
	NYS other urban areas	3.57	3.02	3.29	0.16	0.23	0.22
	NYS nonurban areas	3.14	2.77	3.17	0.18	0.30	0.32
	NYS	3.42	2.84	3.55	0.09	0.25	0.17
	Rest of US	3.44	3.08	3.54	0.03	0.09	0.13

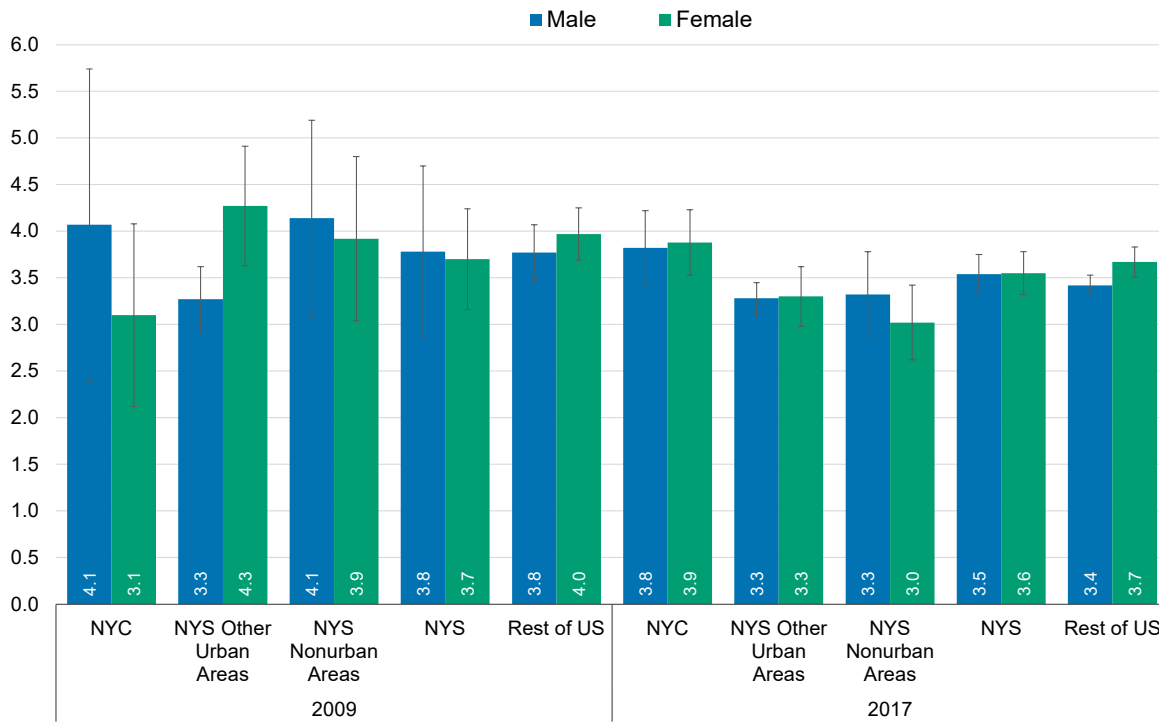


Figure 3-4. Average daily person trips per person by millennials as heads of household by gender across regions over time (NHTS data).

3.1.2 Average Daily Person-Trip Length

Figure 3-5 illustrates the average person-trip length in miles for different generations across regions in 2009 and 2017. Millennials' trip lengths increased substantially across all regions between these years. In NYC, for instance, the average trip length for millennials nearly doubled from 5.8 miles in 2009 to 10.3

miles in 2017, indicating a potential shift in travel behavior. However, the large MOE in 2017 (± 4.2 miles) makes it statistically not significant.

In 2009, the differences in trip length between millennials in NYC and other regions (NYS other urban areas, NYS nonurban areas, and the rest of the US) were statistically significant. Millennials in NYC took considerably shorter trips compared with those in other areas. For instance, millennials' trip lengths in NYC (5.8 miles) were 48% shorter than those in NYS other urban areas (8.6 miles). Similarly, the difference between NYC and NYS nonurban areas (where millennials averaged 12.0 miles per trip) was statistically significant, reflecting the generally longer distances associated with nonurban travel. By 2017, despite the observed increase in millennials' trip lengths in NYC and other regions, the differences across regions were not statistically significant. This suggests that although millennials were traveling farther across all regions, the variations between regions could not be confirmed with confidence. For example, in NYS other urban areas, millennials averaged 13.3 miles per trip in 2017, but the MOE (± 4.34 miles) creates overlap with the trip lengths observed in other regions.

In NYC and NYS nonurban areas, millennials consistently made longer trips in 2009 compared with those of other generations. This trend continued in NYC in 2017. However, in NYS nonurban areas, Gen Xers and baby boomers exhibited slightly longer trips. There was no statistically significant difference between millennials' and other generations' trips. In the rest of the US, millennials' trip lengths increased from 9.5 miles in 2009 to 13.0 miles in 2017. By 2017, millennials were traveling farther than baby boomers, whose average trip length was 11.9 miles.

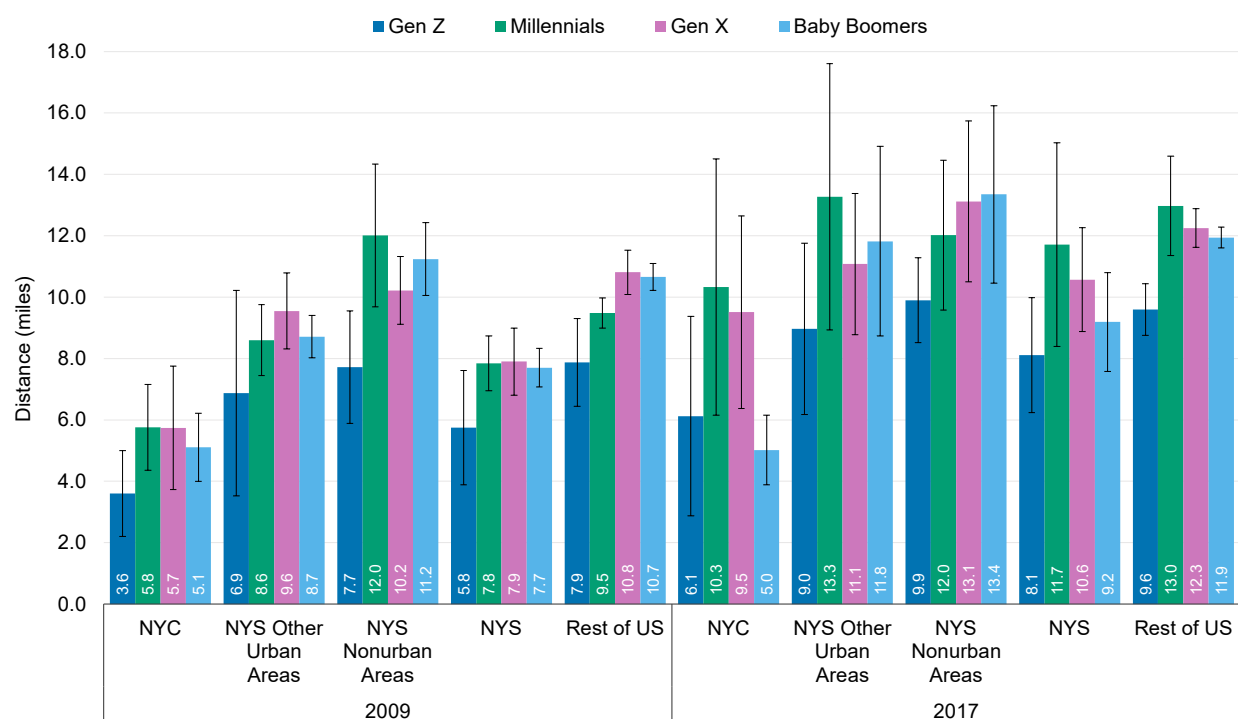


Figure 3-5. Average person-trip length by generation and region over time (NHTS data).

Additionally, a breakdown of average person-trip lengths by millennials and gender across different regions for 2009 and 2017 is presented in Table 3-2. In 2009, trip lengths among millennial males were consistently higher than those of their female counterparts across most regions. For example, in NYC, millennial males traveled an average of 7.2 miles, whereas females averaged 4.3 miles. However, the large MOE for males (± 2.6 miles) means that this difference might not be statistically significant. This

uncertainty is reflected in other regions as well. Males' trip lengths were also higher in 2017 than those of females; however, this difference is found not to be statistically significant.

Table 3-2. Average person-trip length by millennials' gender and region over time (NHTS data).

Year	Region	Estimate		Margin of error	
		Male	Female	Male	Female
2009	NYC	7.2	4.3	2.6	1.5
	NYS other urban areas	9.3	7.9	1.9	1.2
	NYS nonurban areas	11.8	12.2	2.8	2.5
	NYS	8.7	7.0	1.5	0.9
	Rest of US	9.7	9.2	0.6	0.7
2017	NYC	10.4	10.3	6.6	5.6
	NYS other urban areas	15.6	11.0	8.2	1.3
	NYS nonurban areas	13.6	10.4	3.8	2.6
	NYS	12.9	10.6	4.6	3.2
	Rest of US	14.2	11.9	1.6	1.9

Figure 3-6 examines gender differences in trip lengths between the Mid-Atlantic region and the rest of the US. Overall, millennial males generally had longer trips than those of females. When comparing regions within each gender, several key observations emerge. In 2017, millennial males in the Mid-Atlantic had longer trip lengths compared with their counterparts in the rest of the US, whereas the opposite pattern was observed for females. However, these differences were not statistically significant. By 2022, millennial females in the Mid-Atlantic drove fewer miles than females drove in the rest of the US, a statistically significant finding. Conversely, the regional differences in trip lengths among millennial males in 2022 remained statistically insignificant.

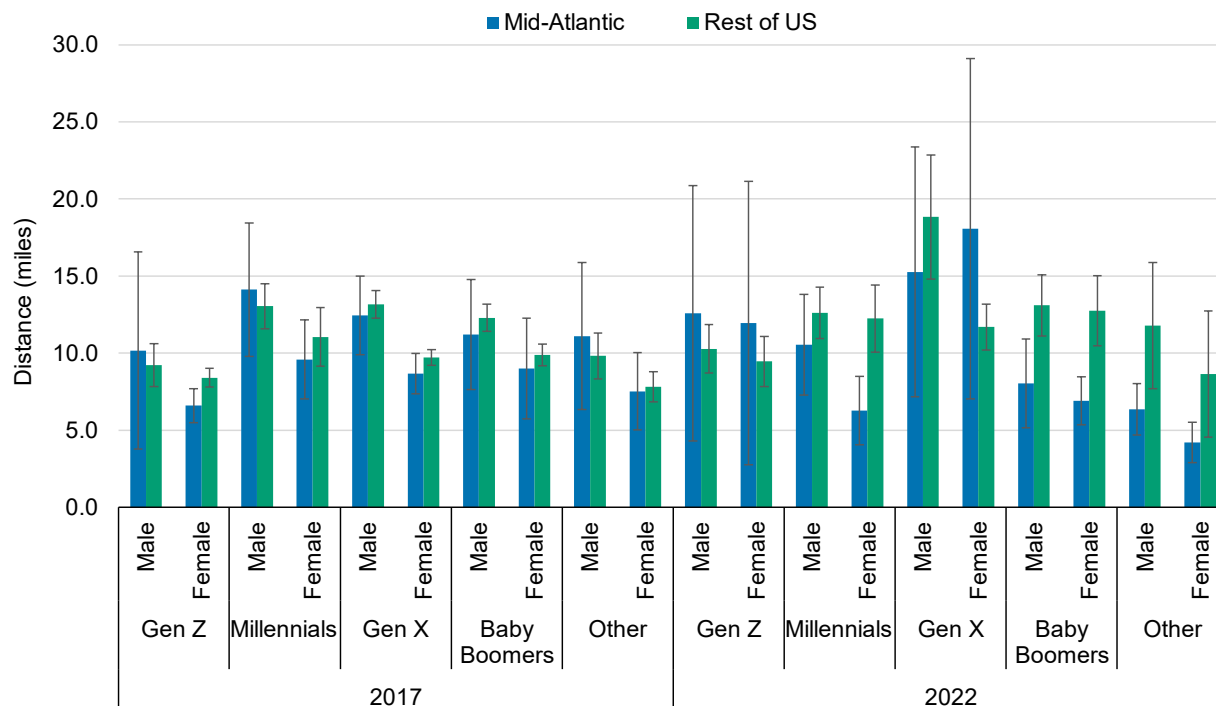


Figure 3-6. Average person-trip length by generation and gender in the Mid-Atlantic region vs. the rest of the US over time (NHTS data).¹

Several insights emerge when comparing the trip lengths by millennial household status. In NYC in 2009, millennials who were heads of household had the shortest average trip length at 3.7 miles, which was shorter than those living with older household members (5.9 miles) and households without millennials (4.9 miles), as illustrated in Figure 3-7. However, because of the MOE for millennials as heads of household (± 1.5 miles), these differences might not be statistically significant.

Across other regions, trip lengths were generally higher than those in NYC for all household statuses. For instance, the average trip length for millennials as heads of household in NYS nonurban areas was 14.5 miles, far exceeding the 3.7 miles in NYC—a difference of 292%. This significant difference reflects the stark contrast between urban and nonurban travel patterns, where nonurban regions typically require longer travel distances for daily activities. Similarly, trip lengths for millennials as heads of household in the rest of the US were 9.2 miles, more than double the distance seen in NYC.

By 2017, trip lengths for millennials as heads of household increased in all regions except the NYS nonurban areas, where there was a slight decrease. In NYC, the average trip length for millennials as heads of household rose from 3.7 miles in 2009 to 10.0 miles in 2017. However, because of the large MOE (± 5.8 miles), this increase may not be statistically significant. Similarly, other regions saw increases in trip lengths, such as the NYS other urban areas, where trip lengths grew from 7.6 miles to 9.3 miles, though again, the MOE suggests that these changes should be interpreted with caution.

¹ The 2009 NHTS data are excluded from this chart owing to methodological differences. Specifically, in 2009, respondents self-reported distances for their trips, whereas in the 2017 and 2022 NHTS, the distances were calculated using the Google API, which computes the shortest route between geocoded origin and destination points.

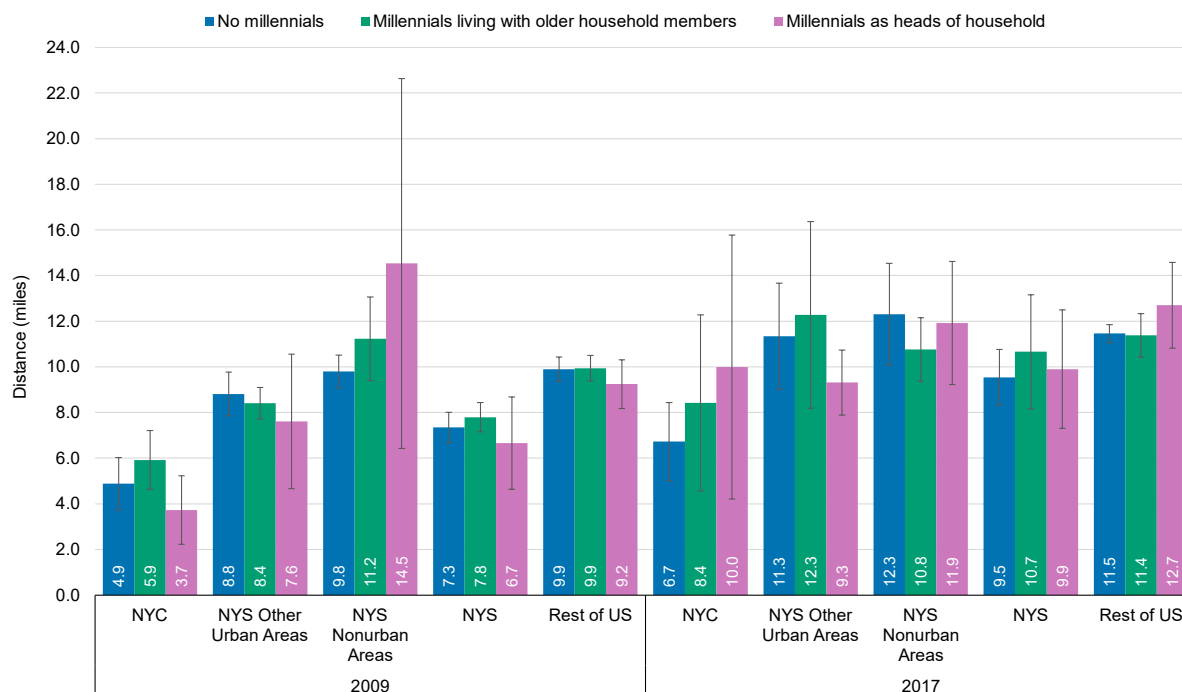


Figure 3-7. Average person-trip length by millennial household status and region over time (NHTS data).

3.2 MODE SHARES

3.2.1 Primary Mode Share

Except for residents of NYC, millennials largely rely on privately owned vehicles (POVs) for the majority of their daily travels (as measured by person trips). In NYS other urban areas, NYS nonurban areas, and the rest of the US, millennials used POVs (either as a driver or passenger) in 80% to over 90% of their trips. This strong dependence on personal vehicles reflects the limited availability of alternative transportation modes in less urbanized regions, where public transit networks are often less developed, and distances between destinations are greater. In contrast, NYC millennials exhibit significantly different travel behavior. As illustrated in Figure 3-8, they relied on POVs for less than one-third of their daily trips in 2017, with only 16% of trips taken as a driver and 9% as a passenger. Walking and public transit account for a much larger share of daily trips—41% and 28%, respectively. This reliance on nonmotorized modes and public transit highlights millennials' preference for or dependence on urban mobility options in NYC, where the density of the city, efficient public transit infrastructure, and the high cost of car ownership reduce the need for personal vehicles. NYC's urban landscape, with its mix of public transit options and walkability, makes owning a car less practical or necessary compared with suburban or rural areas.

In 2017, a slight but noticeable increase is observed in baby boomers' reliance on POVs compared with that of millennials. Across all regions, baby boomers were more likely to use POVs as drivers. For instance, in NYC, 21% of baby boomers drove, compared with 16% of millennials. Additionally, baby boomers generally have a lower percentage of passenger trips, which might indicate their preference for driving themselves rather than relying on others for transport. For example, baby boomers in NYS nonurban areas had 72% POV-driver share, whereas millennials had a slightly lower 64% driver share and a higher 20% passenger share.

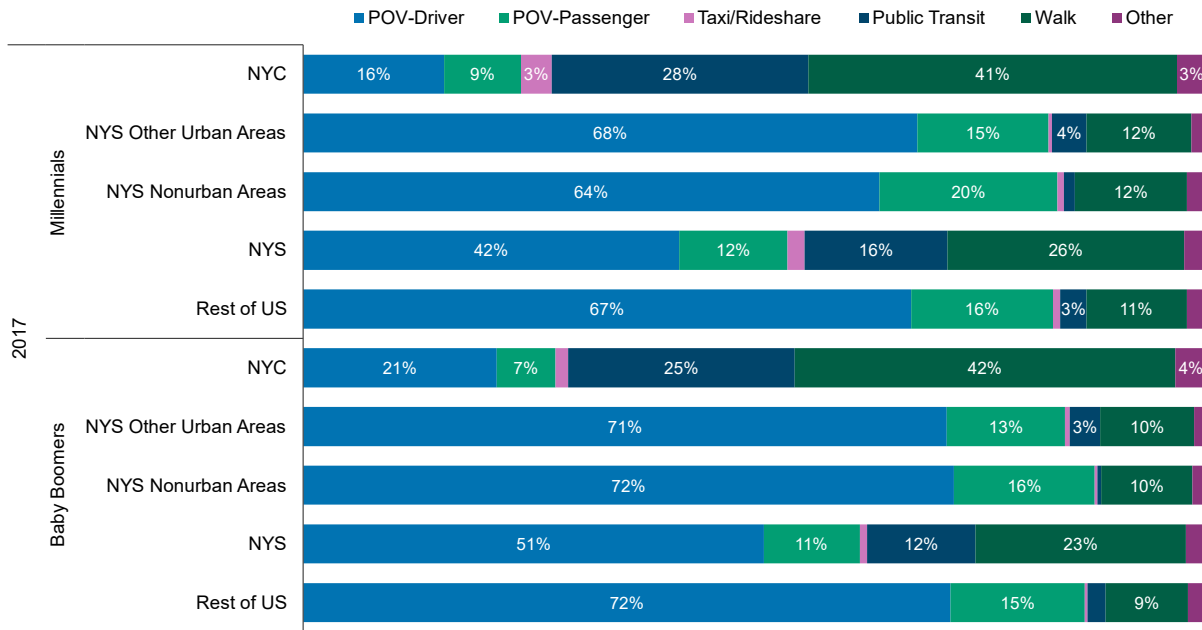


Figure 3-8. Mode share distribution for millennials and baby boomers by region (NHTS data).

Millennials who were heads of households had the lowest share of POV-driver trips compared with other millennial households in both the Mid-Atlantic and the rest of the US, as illustrated in Figure 3-9. Across all household statuses, the Mid-Atlantic consistently reported a lower share of POV-driver trips and a higher share of public transit usage than the rest of the US. This trend highlights the region's greater reliance on public transportation, particularly in urban centers. Notably, millennials as heads of households also exhibited a higher percentage of walking and public transit trips compared with other household types over the years. This pattern likely reflects the influence of well-developed public transportation systems in major Mid-Atlantic cities, such as NYC, where robust transit infrastructure supports alternatives to driving. The increased use of walking and public transit among this group suggests that urban living, characterized by accessibility to transit and proximity to destinations, plays a key role in shaping their travel behavior.

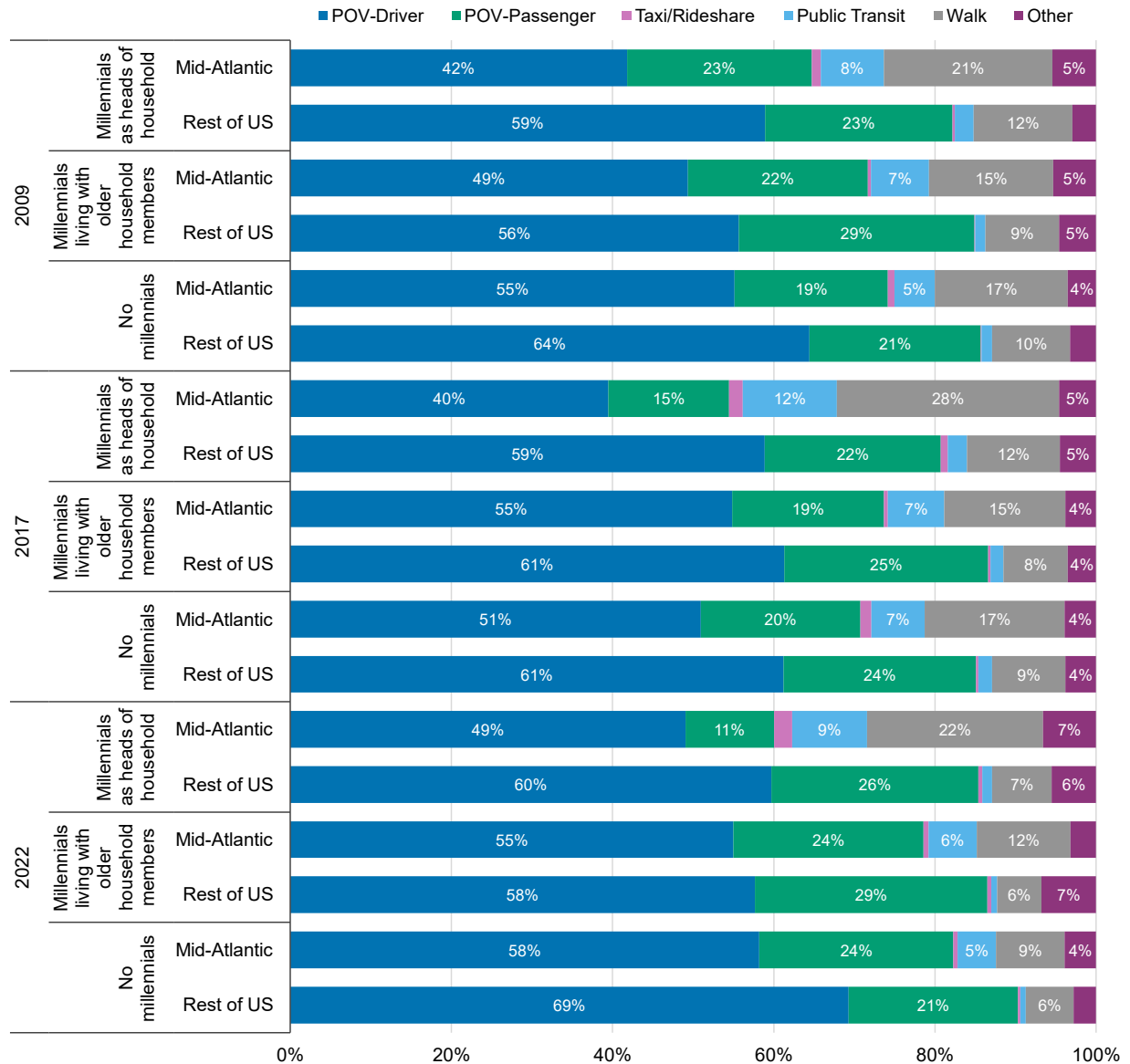


Figure 3-9. Mode share distribution by millennial household status in the Mid-Atlantic Division over time (NHTS data).

Similarly, as seen in Figure 3-10, millennial-headed households across most regions are still largely dependent on POVs for their daily trips, with the key exception being NYC, where walking and public transit are the dominant modes of transportation. In 2009, millennials heading households in NYC demonstrated a low reliance on POVs, with only 16% as drivers and 7% as passengers. Instead, walking accounted for 50% of trips, making it the most common mode, whereas public transit was also heavily used, representing 19% of all trips. By 2017, a noticeable shift toward public transit occurred, with 28% of trips being taken by public transportation, suggesting potential improvements in public transit infrastructure or changing commuting preferences among millennials in NYC have occurred. Despite this shift, walking remained a key mode of transportation, although it saw a slight decrease from 50% to 41%, possibly owing to the growing use of public transit. This trend reflects millennials' evolving transportation preferences in NYC, where car ownership is often less practical or necessary owing to high costs, traffic, and the availability of robust public transit options.

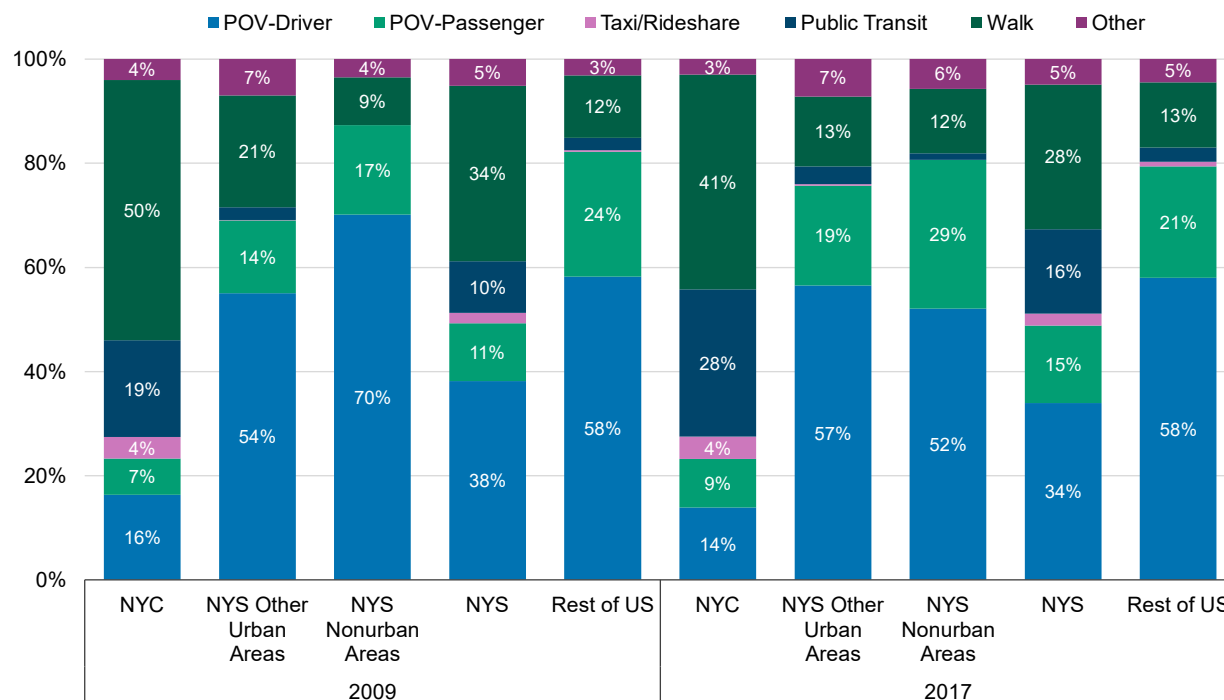


Figure 3-10. Mode share distribution by millennials as heads of household by region over time (NHTS data).

3.2.2 Alternative Mode Share

In 2017, millennials displayed notable variations in their alternative mode share preferences across different regions, with a significant reliance on cars and SUVs, except in NYC, where public transit and nonmotorized modes dominated (Table 3-3). In NYC, only 15% of millennials relied on cars, and 8% used SUVs, indicating low dependence on personal vehicles. Instead, public transit was a major mode of transportation, accounting for 28% of trips, whereas 47% of trips fell into the “other” category, which includes walking and biking. In NYS other urban areas, the situation was quite different; 51% of millennials primarily used cars, and 21% drove SUVs. Public transit usage dropped significantly to only 4%, whereas alternative modes like walking or biking were also less common, reflected in the 14% under “other.” This highlights the higher dependence on personal vehicles in urban areas outside NYC, where public transit infrastructure is not as extensive. In NYS nonurban areas, millennials were even more dependent on personal vehicles; 40% of them used cars, and 21% relied on SUVs. Pickup trucks also accounted for a notable 12% of mode share, likely reflecting the rural or suburban nature of these areas where such vehicles are more practical. Public transit use was negligible at 1%, further emphasizing the lack of transit options in these regions. Gen Xers and baby boomers had higher reliance on cars and SUVs than millennials had across most regions. In NYC, both Gen Xers and baby boomers showed similarly low reliance on personal vehicles, with public transit and other modes dominating, though millennials had a slightly higher proportion of “other” modes, which could reflect differences in lifestyle choices, such as walking or biking more than older generations. Pickup trucks were more popular among baby boomers and Gen Xers in nonurban areas, with 16% of baby boomers and 13% of Gen Xers using them in NYS nonurban areas, compared with only 12% for millennials in those areas.

Table 3-3. Alternative mode share distributions by generation and region (2017 NHTS data).

Generation	Region	Car	SUV	Van	Pickup truck	Motorcycle/moped	RV	Rental car	Public transit	Other
Gen Z	NYC	11%	11%	8%	0%	0%	0%	1%	23%	47%
	NYS other urban areas	30%	24%	10%	2%	0%	0%	0%	2%	33%
	NYS nonurban areas	33%	18%	14%	6%	0%	0%	0%	1%	29%
	NYS	23%	19%	10%	2%	0%	0%	0%	9%	37%
	Rest of US	35%	23%	12%	6%	0%	0%	0%	1%	23%
Millennials	NYC	15%	8%	2%	0%	0%	0%	0%	28%	47%
	NYS other urban areas	51%	21%	5%	6%	0%	0%	0%	4%	14%
	NYS nonurban areas	40%	21%	10%	12%	0%	0%	0%	1%	15%
	NYS	32%	15%	4%	3%	0%	0%	0%	16%	31%
	Rest of US	49%	21%	5%	8%	0%	0%	0%	3%	14%
Gen X	NYC	16%	11%	3%	0%	0%	0%	1%	25%	44%
	NYS other urban areas	37%	31%	10%	6%	0%	0%	0%	4%	12%
	NYS nonurban areas	34%	28%	12%	13%	0%	0%	0%	2%	11%
	NYS	28%	22%	7%	4%	0%	0%	1%	13%	26%
	Rest of US	41%	26%	9%	11%	0%	0%	0%	2%	11%
Baby boomers	NYC	17%	8%	3%	1%	0%	0%	0%	25%	47%
	NYS other urban areas	44%	28%	6%	6%	0%	0%	0%	3%	12%
	NYS nonurban areas	41%	25%	6%	16%	0%	0%	0%	0%	12%
	NYS	32%	20%	5%	5%	0%	0%	0%	12%	26%
	Rest of US	43%	24%	6%	13%	0%	0%	0%	2%	11%

Note: Percentages may not total 100 due to rounding.

As shown in Figure 3-11, millennials as heads of households across all regions showed lower reliance on cars and SUVs compared with millennials living with older household members and households without millennials, particularly in NYC and NYS other urban areas. For example, in NYC, 13% of the millennials as heads of households used cars, and 7% relied on SUVs. This contrasts sharply with households without millennials, where 16% of the trips were made using cars and 9% using SUVs. Millennials living with older household members also showed slightly higher usage of cars (17%) and SUVs (12%) than millennial-headed households. Public transit and nonmotorized modes played a larger role for millennial-headed households, especially in NYC, where walking and biking dominated transportation choices, reflecting a more urban lifestyle. In NYS nonurban areas, millennial-headed households exhibited a greater reliance on pickup trucks compared with other household statuses, highlighting the practicality of these vehicles in less densely populated areas with longer travel distances and less-developed road infrastructure. Multigenerational households with millennials living with older household members consistently showed the highest car usage across all regions, suggesting that households with more members may have a greater need for multiple vehicles, especially in areas with limited public transportation options.

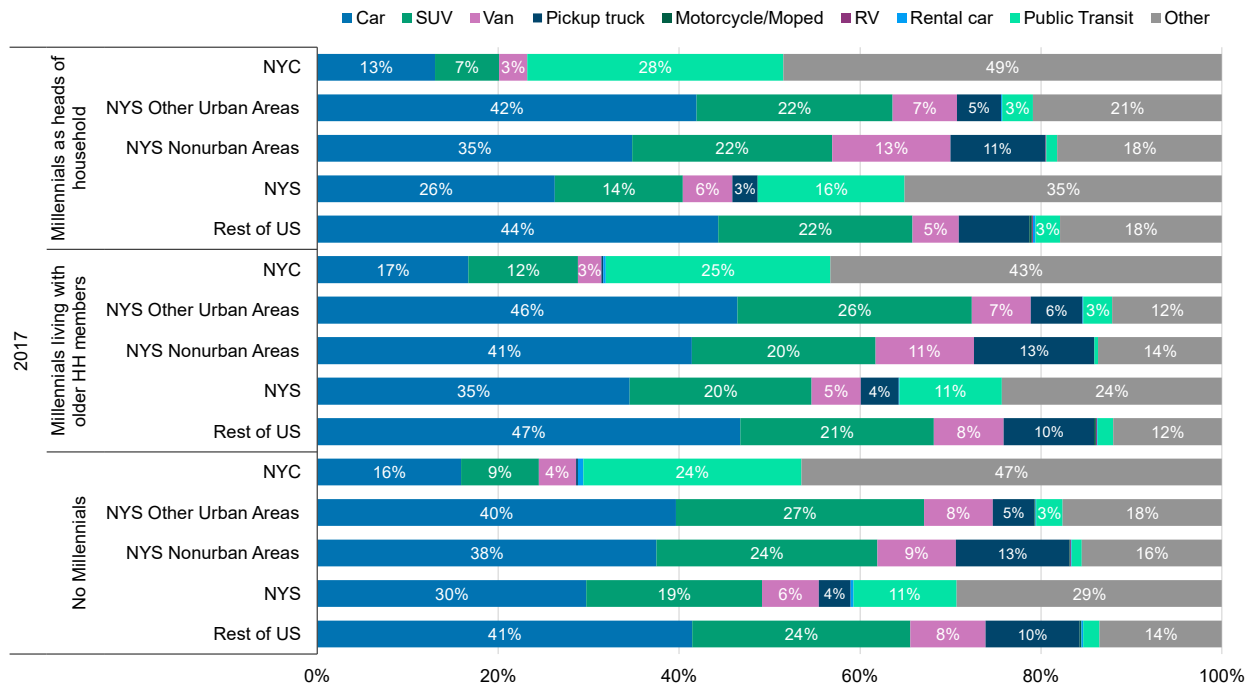


Figure 3-11. Alternative mode share distribution by household status and region (2017 NHTS data).

3.3 TRIP PURPOSE

Figure 3-12 illustrates the distribution of daily average person trips by trip purpose across different generations and regions in 2017, focusing on categories like earning a living; family and personal business; civic, educational, and religious activities; social and recreational activities; and others. Millennials across all regions dedicated a significant portion of their trips to earning a living, especially in NYC where 28% of their trips were work-related. This was notably higher compared with baby boomers in NYC, who only allocated 18% of their trips to this purpose. Similarly, millennials in NYS other urban areas and NYS also had 28% of their trips dedicated to earning a living, compared with 18%–20% for baby boomers in these areas. When compared with Gen Xers, millennials had slightly higher or similar shares of work-related trips, with Gen Xers allocating around 24%–27% of their trips to earning a living across regions. Gen Zers, being the youngest generation and likely still in school or early in their careers, understandably had the lowest percentages of work-related trips, ranging from 5% to 10% across all regions. This distribution suggests that millennials, being in their prime working years, are more engaged in the workforce, leading to a higher share of trips dedicated to employment compared with those of older generations like baby boomers.

In contrast, millennials made fewer trips for family and personal business compared with baby boomers. For example, in NYC, millennials allocated 31% of their trips to family and personal matters, whereas baby boomers dedicated a much larger portion at 46%. This trend is consistent across regions, with baby boomers showing higher shares for family-related trips in NYS nonurban areas (48%) and NYS other urban areas (47%), compared with millennials' 33%–40% in these areas. This difference could be attributed to baby boomers having more family obligations, such as caretaking, running errands for dependents, or household-related tasks, whereas millennials may be more focused on professional responsibilities. Gen Xers also made more family and personal business trips than those of millennials, suggesting that as people age and settle into family life, their trips become more focused on household and caregiving needs.

Millennials exhibited moderate participation in civic, educational, and religious trips. For instance, in NYC, millennials made 4% of their trips for these purposes, which is slightly lower than that of baby boomers, who made 5% of their trips for similar activities. However, in NYS other urban areas, millennials allocated 6% of their trips to these purposes, which was higher than baby boomers' 4% in the same region. This suggests that millennials, especially in urban areas, are more engaged than baby boomers are in educational or community-oriented activities. Gen Zers, on the other hand, had the highest percentage of trips for civic, educational, and religious purposes across all generations, particularly in NYC and NYS other urban areas, where they dedicated 39% and 36% of their trips, respectively. This trend likely reflects the younger generation's focus on education because many Gen Z individuals are still in school or early in their educational journey.

Social and recreational activities formed a significant part of millennials' travel. In NYC, millennials allocated 33% of their trips to social and recreational activities, which was higher than that of baby boomers, who allocated 27% of their trips for this purpose. This trend is seen across various regions, where millennials show a slightly higher tendency toward social and recreational activities compared with baby boomers. The higher percentage of trips for social and recreational activities among millennials may indicate their inclination toward more active social lives or recreational pursuits, which could be attributed to lifestyle preferences. This is consistent with the perception of millennials as a generation that values experiences and social engagements. Even in NYS other urban areas and NYS nonurban areas, millennials displayed higher percentages for these types of trips compared with older generations.

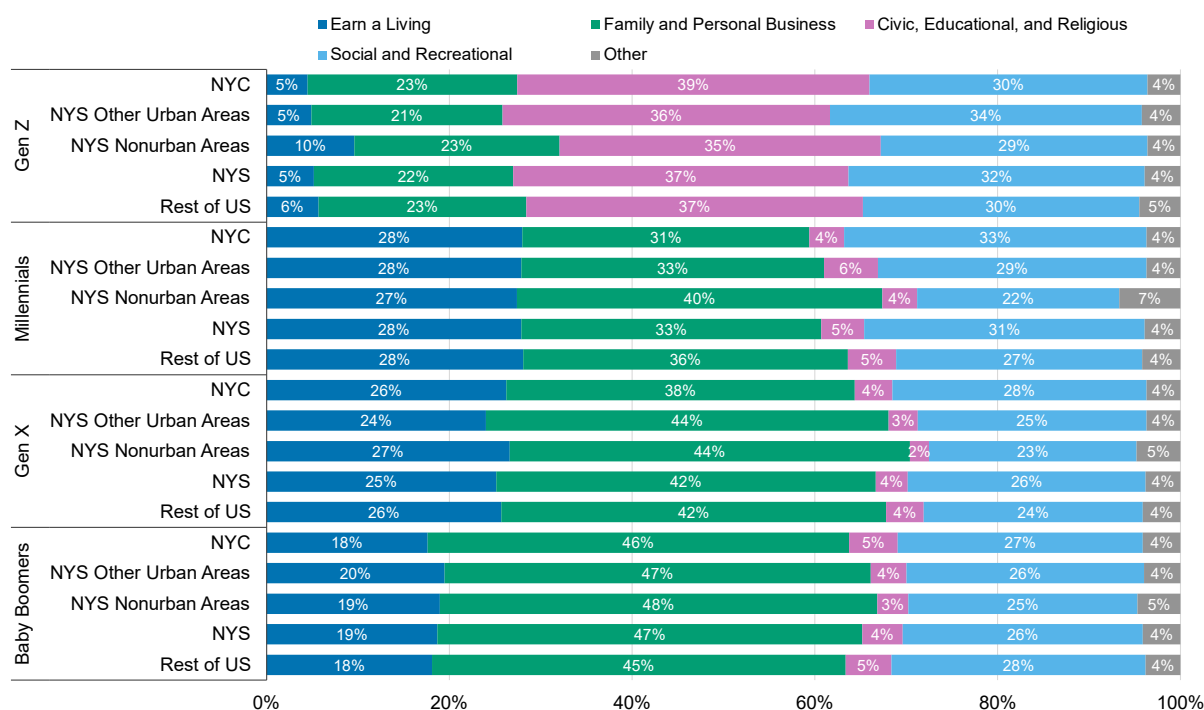


Figure 3-12. Distribution of daily average person trips by generation, trip purpose, and region (2017 NHTS data).

Millennial travel purpose in the Mid-Atlantic region in 2017 shows a similar pattern, as illustrated in Figure 3-13, with the largest portion of trips being work-related, primarily for earning a living. This pattern was also evident in the rest of the US, indicating that millennials across regions prioritize work-related travel. By 2022, this trend became even more pronounced in the Mid-Atlantic, where 31% of all millennial trips were dedicated to earning a living, up from 19% in 2009 and 28% in 2017. This steady

rise reflects the growing importance of work-related mobility among millennials in this region. In fact, by 2022, millennials in the Mid-Atlantic had the highest share of work-related trips of all generations, underscoring their unique role in the labor force and travel behavior.

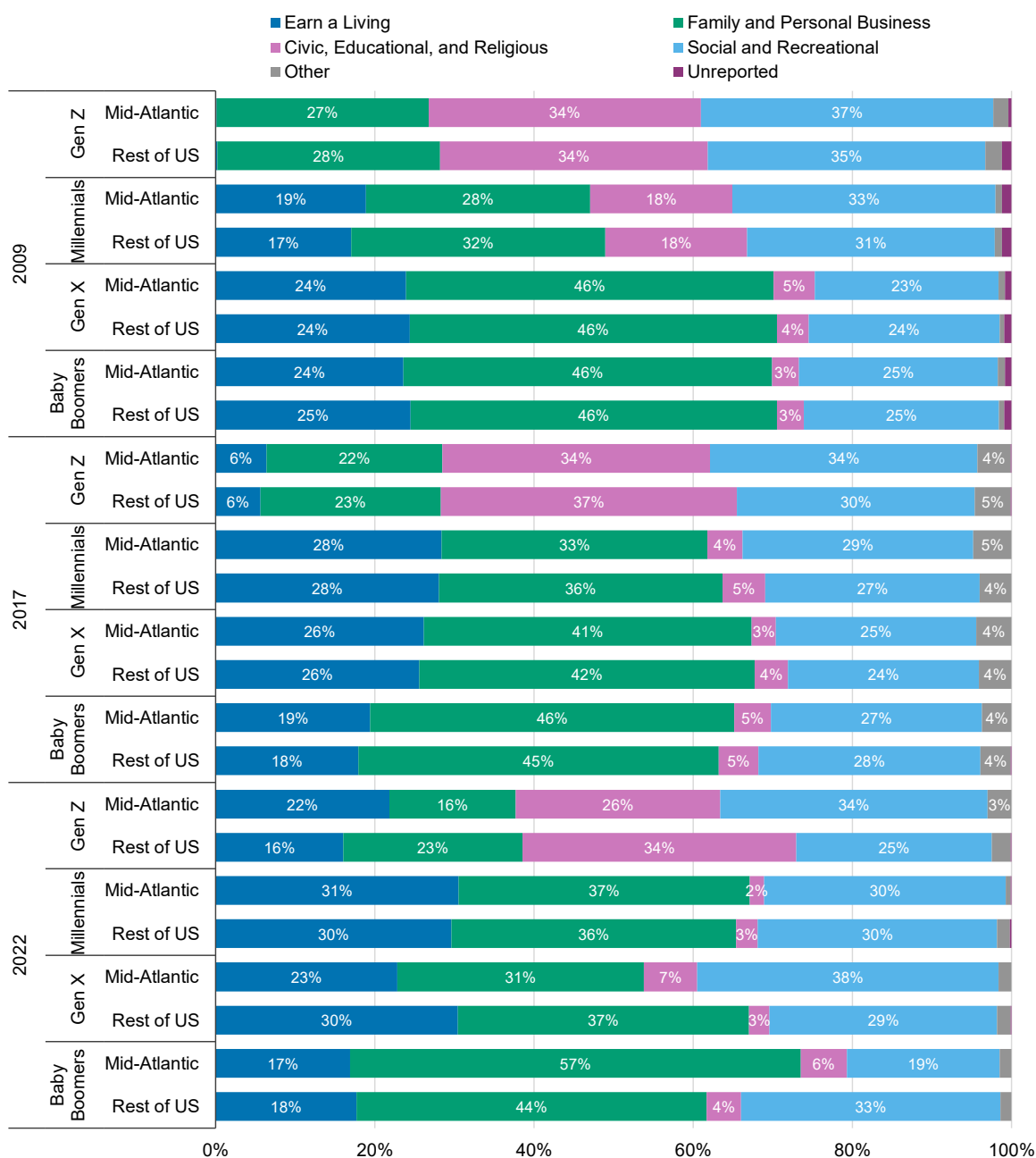


Figure 3-13. Distribution of daily average person trips by generation and trip purpose in the Mid-Atlantic Division over time (NHTS data).

In a close examination of millennials by their household status, distinct nuances emerge in their trip purposes, as highlighted in Table 3-4. In 2017, millennials who were heads of household exhibited distinct trip patterns compared with those of other household types, particularly in NYC. These households allocated 27% of their trips to earning a living, a relatively high proportion, comparable to millennials living with older household members (24%) but significantly higher than households without

millennials (17%). This highlights that millennials heading their own households are more engaged in work-related travel, particularly in urban areas like NYC, compared with travel of nonmillennial households.

Households without millennials allocated an even larger proportion of their trips to family and personal business, with 41%–42% of trips across most regions dedicated to this category. This is a higher share than that of the millennial-headed households. For example, in NYC, millennials as heads of household allocated 30% of their trips to family and personal business, lower than the 36% and 41% seen in other categories. In terms of civic, educational, and religious trips, millennial-headed households in NYC had the lowest share (5%) compared with other household types. However, in NYS other urban areas, millennials heading their own households allocated 13% of their trips to these purposes, the highest share across all regions and household statuses. This suggests a regional variation in how millennials engage with community, educational, or religious activities, particularly in urban areas outside of NYC where such activities may be more prevalent.

Social and recreational activities accounted for a significant portion of trips for millennials as heads of household. In NYC, 35% of their trips were for social and recreational purposes, a higher share than that of millennials living with older household members (29%) and households without millennials (28%). This reflects the social preferences of millennials, who may prioritize leisure and social engagements, especially in urban environments where such activities are easily accessible. Across all regions, nonmillennial households consistently made fewer social and recreational trips compared with those of millennials, further emphasizing that younger generations, particularly millennials, are more inclined toward social engagement and recreational activities.

Table 3-4. Distribution of daily average person trips by trip purpose and millennial household status (2017 NHTS data).

Household status	Region	Earn a living	Family and personal business	Civic, educational, and religious	Social and recreational	Other
Millennials as heads of household	NYC	27%	30%	5%	35%	3%
	NYS other urban areas	22%	34%	13%	28%	4%
	NYS nonurban areas	21%	40%	10%	23%	7%
	NYS	24%	33%	9%	31%	4%
	Rest of US	24%	34%	10%	27%	5%
Millennials living with older household members	NYC	24%	36%	7%	29%	5%
	NYS other urban areas	25%	36%	9%	27%	4%
	NYS nonurban areas	26%	39%	9%	21%	5%
	NYS	24%	36%	8%	27%	4%
	Rest of US	23%	36%	11%	25%	4%
No millennials	NYC	17%	41%	11%	28%	4%
	NYS other urban areas	15%	41%	11%	29%	4%
	NYS nonurban areas	17%	42%	10%	28%	4%
	NYS	16%	41%	11%	28%	4%
	Rest of US	16%	41%	11%	28%	4%

Note: Percentages may not total 100 due to rounding.

3.4 DAILY PERSON TRIPS BY TIME OF TRAVEL

The temporal distribution of person trips by time of travel, as shown in Table 3-5, reveals several interesting trends, particularly for millennials in comparison with baby boomers and other generations across regions.

- **Morning Rush (6–9 a.m.):** Millennials in NYC make 13% of their trips during the morning rush hour, lower than the 20% for Gen Zers and 18% for Gen Xers in the same region. Millennials in NYC are on par with baby boomers, who also make 13% of their trips during this time. However, in other regions, such as NYS other urban areas and NYS nonurban areas, millennials exhibit slightly higher morning activity (15%–16%) compared with baby boomers, indicating that millennials outside NYC have a more pronounced morning commuting pattern.
- **Midday and Early Afternoon (9 a.m.–3 p.m.):** Millennials tend to distribute their trips more evenly between 9 a.m.–12 p.m. and 12–3 p.m., particularly in urban areas like NYC and NYS other urban areas. For instance, millennials in NYC take 15% of their trips between 9 a.m. and 12 p.m. and 20% between 12 and 3 p.m., reflecting a balanced daytime travel pattern. In contrast, baby boomers in NYC concentrate 23% of their trips between 9 a.m. and 12 p.m. and 24% between 12 and 3 p.m., suggesting that baby boomers may focus more on non-work-related errands or activities during these hours. The pattern is similar in NYS nonurban areas, where millennials allocate 16% and 19% of their trips to the same time slots, whereas baby boomers show higher shares of 23% and 25%, respectively.
- **Afternoon and Evening (3–9 p.m.):** Millennials in NYC spread their travel relatively evenly during the afternoon and evening periods, with 20% of trips occurring between 3 and 6 p.m. and 19% between 6 and 9 p.m. This contrasts with baby boomers, who allocate 23% of their trips between 3 and 6 p.m. and only 12% between 6 and 9 p.m. Millennials' evening travel patterns may reflect a stronger tendency toward social or recreational activities after work, particularly in urban settings. In NYS nonurban areas, millennials show slightly higher activity between 3 and 6 p.m. (26%) compared with baby boomers (21%), with 13% of millennials' trips occurring between 6 and 9 p.m., compared with 11% for baby boomers.
- **Late Evening (9 p.m.–12 a.m.):** Millennials are more likely to travel during the late evening hours (9 p.m.–12 a.m.) than baby boomers, particularly in NYC, where 9% of millennials' trips occur during this period, compared with just 4% for baby boomers. This suggests that millennials in urban environments like NYC engage more in nighttime social or recreational activities. Across regions, millennials tend to make more late-evening trips than baby boomers, whose travel tapers off after 9 p.m.
- **Midnight and Early Morning Hours (12–6 a.m.):** Millennials show travel patterns similar to those of baby boomers during the midnight and early morning hours (12–6 a.m.). In NYC, 2% of millennials travel between 12 and 3 a.m. and 1% travel between 3 and 6 a.m., comparable to 1% for baby boomers in both the same time slots. Across regions, millennials exhibit similar early morning travel activity to other generations, except for Gen Zers, who make no trips between 3 and 6 a.m. across most regions.

Table 3-5. Temporal distribution of person trips by time of travel, generation, and region (2017 NHTS data).

Generation	Region	12–3 a.m.	3–6 a.m.	6–9 a.m.	9 a.m.– 12 p.m.	12–3 p.m.	3–6 p.m.	6–9 p.m.	9 p.m.– 12 a.m.
Gen Z	NYC	1%	0%	20%	12%	21%	27%	15%	5%
	NYS other urban areas	1%	0%	19%	13%	19%	27%	16%	5%
	NYS nonurban areas	0%	0%	20%	13%	20%	29%	13%	4%
	NYS	1%	0%	19%	13%	20%	27%	15%	5%
	Rest of US	0%	1%	21%	12%	18%	29%	15%	4%
Millennials	NYC	2%	1%	13%	15%	20%	20%	19%	9%
	NYS other urban areas	1%	2%	15%	17%	19%	23%	17%	6%
	NYS nonurban areas	1%	3%	16%	16%	19%	26%	13%	6%
	NYS	2%	2%	14%	16%	20%	22%	18%	8%
	Rest of US	1%	2%	16%	16%	19%	24%	16%	6%
Gen X	NYC	1%	1%	18%	15%	19%	22%	17%	6%
	NYS other urban areas	1%	2%	17%	18%	20%	24%	15%	5%
	NYS nonurban areas	1%	2%	18%	18%	18%	27%	14%	4%
	NYS	1%	1%	17%	17%	20%	23%	16%	5%
	Rest of US	0%	2%	18%	17%	20%	24%	15%	4%
Baby boomers	NYC	1%	1%	13%	23%	24%	23%	12%	4%
	NYS other urban areas	0%	2%	14%	21%	23%	22%	14%	4%
	NYS nonurban areas	0%	2%	15%	23%	25%	21%	11%	3%
	NYS	0%	2%	14%	22%	24%	22%	13%	4%
	Rest of US	0%	2%	14%	23%	23%	22%	13%	3%

Note: Percentages may not total 100 due to rounding.

The temporal distribution of person trips by time of travel, as shown in Figure 3-14, highlights notable differences between millennials as heads of household and other household types across regions. Millennials as heads of household in NYC make 13% of their trips in the 6–9 a.m. period, which is slightly lower than in NYS nonurban areas (17%) and the rest of the US (18%). Compared with millennials living with older household members and households without millennials, the morning activity levels are similar, ranging from 15% to 17%. However, millennials as heads of household show a strong tendency to travel during the afternoon and evening, particularly in NYS nonurban areas, where 28% of their trips occur between 3 and 6 p.m. and 14% between 6 and 9 p.m. This suggests a more active late-afternoon schedule in nonurban settings, whereas in NYC, these households make 20% of their trips during both 3–6 p.m. and 6–9 p.m., indicating a balance between work and social activities. Additionally, millennials as heads of household in NYC make 10% of their trips between 9 p.m. and 12 a.m., a higher share compared with 5% or less in nonurban areas, reflecting a stronger engagement in nighttime activities, which is typical in urban settings. In contrast, households without millennials tend to reduce travel after 9 p.m., with 5% or fewer trips during this period. Millennials living with older household members show a pattern similar to that of millennials as heads of household but with slightly fewer late-night trips, particularly outside of urban regions. Overall, millennials as heads of household exhibit a more balanced and active travel schedule across the day, especially during late afternoons and evenings, compared with nonmillennial households, whose travel tapers off earlier in the evening.

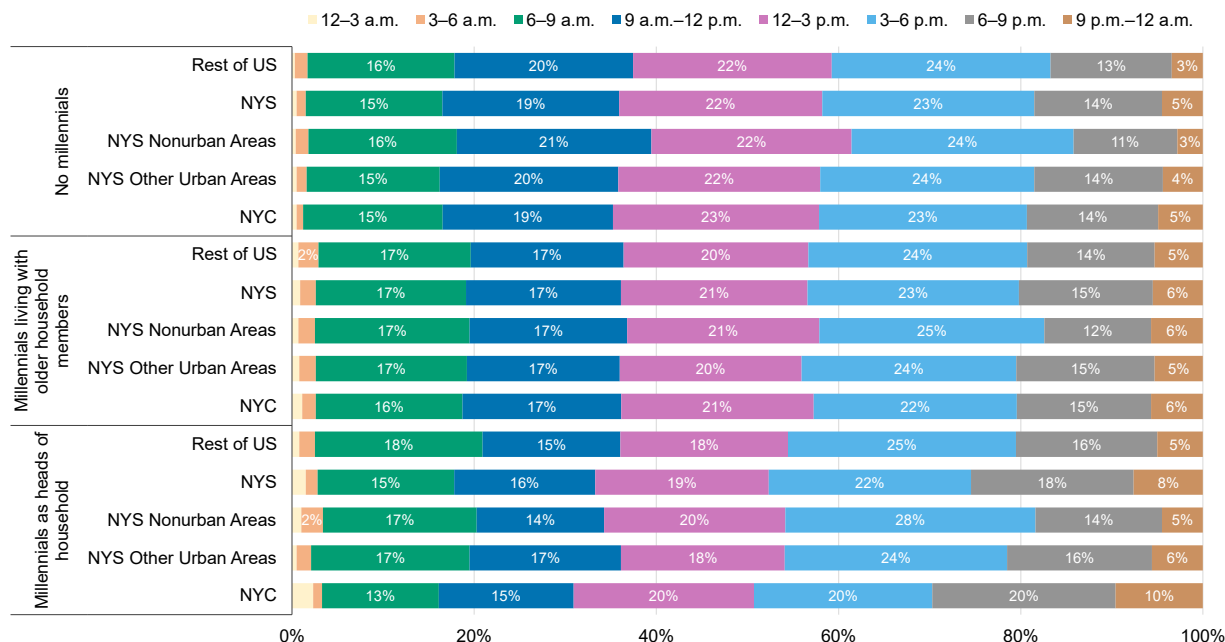


Figure 3-14. Temporal distribution of person trips by time of travel, millennial household status, and region (2017 NHTS data).

3.5 AVAILABILITY OF DRIVERS

Considering the effect of having other drivers in the household on travel frequency, Figure 3-15 illustrates that millennials tend to make slightly more daily person trips when other drivers are absent in their household, whereas the opposite is observed for baby boomers, who typically make fewer trips when other drivers are absent. This aligns with previous studies on elderly populations, such as one by Liu et al. (2022), and findings by Sadeghzaziri and Tawfik (2020), who observed that the number of trips made by baby boomers decreases when other household members are present, regardless of whether they are drivers. Millennials show an opposite tendency, albeit one that is not always statistically significant. The 2009 NHTS data indicate that millennials in NYC without other drivers made 3.2 daily trips, slightly higher than the 2.9 trips made by those with other drivers. However, this difference is not statistically significant owing to the large MOEs for the group without other drivers (0.92 trips). This trend holds across other regions, where millennials without other drivers make more trips, but the MOEs indicate that these differences are often not statistically meaningful. Even in the 2017 NHTS data, this pattern persists, with the exception of in the NYS area, where the differences showed statistical significance, indicating that regional factors might affect millennials' travel behavior.

In contrast, baby boomers demonstrate a more pronounced pattern. For example, in NYS other urban areas, baby boomers with other drivers made an average of 4.4 trips per day, significantly more than the 3.9 trips made by those without other drivers. The statistical significance of this difference is reinforced by the smaller MOEs for both groups. However, baby boomers in NYC showed no significant difference in travel behavior based on the availability of other drivers, with both groups averaging 3.1–3.3 trips per day in 2017.

Overall, whereas millennials tend to make slightly more trips when no other drivers are present in their household, these differences are typically small and often statistically insignificant, with the exception of NYS, where the trend is more pronounced and statistically significant. This suggests that the presence or absence of other drivers does not significantly affect millennials' travel behavior in most cases. This may

be a result of their greater reliance on alternative transportation options like public transit or ridesharing, particularly in urban areas, which reduce their dependence on the availability of household drivers. In contrast, baby boomers show a clearer and statistically significant pattern: they tend to make fewer trips when there are no other drivers in the household. This trend is particularly evident in nonurban regions, where baby boomers rely more heavily on personal vehicles for their daily travel needs. The data underscore that driver availability has a stronger influence on baby boomers' travel patterns, reflecting their greater dependence on private cars for mobility compared with the more diverse transportation options often used by millennials.

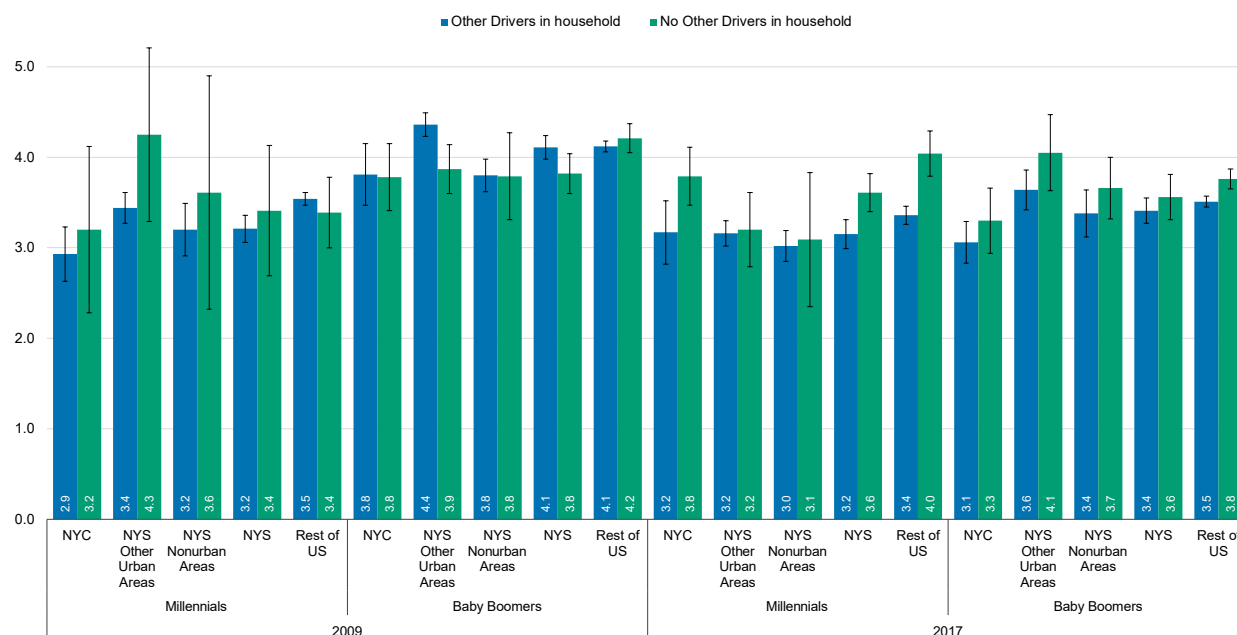


Figure 3-15. Average daily person trips per person by generation, region, and availability of other drivers in the household (NHTS data).

When considering the average daily vehicle trips per driver for millennials and baby boomers, several notable trends emerge, as highlighted in Table 3-6 and Figure 3-16. First, a clear regional difference exists, with drivers in NYC making significantly fewer vehicle trips compared with those living outside the city, reflecting NYC's extensive public transit system and lower reliance on personal vehicles. This pattern holds across both generations, showing the impact of urban infrastructure on travel behavior. The most pronounced effect of driver availability is seen among females living outside NYC, regardless of their generation. Based on 2017 NHTS data, females without other drivers in their households made more vehicle trips than those with additional drivers. For example, among millennials, women living outside NYC and without other drivers averaged 3.1 vehicle trips per day in NYS nonurban areas, compared with 2.5 trips for those with other drivers. Similarly, for baby boomers, the absence of other drivers led to higher vehicle-trip rates across various regions. This suggests that in households where women are the sole driver, they are responsible for a greater share of driving duties, leading to an increase in vehicle trips. Note that the wider MOEs associated with these estimates, particularly for females, unfortunately make drawing definitive conclusions about the impact of driver availability difficult. Although the data suggest a general trend, the statistical uncertainty in some regions limits the strength of these findings. A similar pattern is observed among male millennials; those without other drivers in the household tend to make more vehicle trips, having no statistical significance.

Table 3-6. Average daily vehicle trips per driver by generation (millennials and baby boomers), gender, region, and availability of other drivers in the household (NHTS data).

Year	Driver availability	Gender	Region	Estimates		Margin of error	
				Millennials	Baby boomers	Millennials	Baby boomers
2009	Other drivers in household	Male	NYC	0.93	1.61	0.43	0.41
			NYS other urban areas	2.46	3.50	0.26	0.16
			NYS nonurban areas	2.42	3.28	0.36	0.26
			NYS	2.02	2.85	0.21	0.18
			Rest of US	2.52	3.37	0.12	0.08
		Female	NYC	0.33	0.98	0.19	0.40
			NYS other urban areas	2.40	3.27	0.26	0.18
			NYS nonurban areas	2.29	2.57	0.50	0.22
			NYS	1.69	2.51	0.22	0.18
			Rest of US	2.59	3.03	0.14	0.07
	No other drivers in household	Male	NYC	0.28	1.26	0.49	0.52
			NYS other urban areas	2.24	3.37	0.52	0.40
			NYS nonurban areas	4.38	3.35	1.30	0.47
			NYS	0.62	2.32	0.55	0.38
			Rest of US	2.77	3.43	0.65	0.22
		Female	NYC	0.66	0.54	0.89	0.33
			NYS other urban areas	3.43	3.14	1.22	0.36
			NYS nonurban areas	4.69	3.20	1.85	0.68
			NYS	1.74	1.87	0.89	0.26
			Rest of US	2.75	3.60	0.62	0.17
2017	Other drivers in household	Male	NYC	1.00	1.20	0.37	0.23
			NYS other urban areas	2.40	3.10	0.27	0.25
			NYS nonurban areas	2.40	2.90	0.28	0.27
			NYS	1.90	2.50	0.17	0.18
			Rest of US	2.50	3.00	0.10	0.08
		Female	NYC	0.40	0.80	0.13	0.24
			NYS other urban areas	2.40	2.60	0.24	0.18
			NYS nonurban areas	2.40	2.20	0.32	0.23
			NYS	1.80	2.10	0.18	0.21
			Rest of US	2.50	2.40	0.09	0.10
	No other drivers in household	Male	NYC	1.30	1.60	0.44	0.56
			NYS other urban areas	2.80	3.20	0.42	0.79
			NYS nonurban areas	2.50	3.10	1.05	0.29
			NYS	1.80	2.30	0.34	0.42
			Rest of US	2.90	3.20	0.14	0.15

Table 3-6. Average daily vehicle trips per driver by generation (millennials and baby boomers), gender, region, and availability of other drivers in the household (NHTS data) (continued).

Year	Driver availability	Gender	Region	Estimates		Margin of error	
				Millennials	Baby boomers	Millennials	Baby boomers
		Female	NYC	1.0	1.1	0.61	0.36
			NYS other urban areas	2.9	3.7	0.70	0.56
			NYS nonurban areas	3.1	3.4	1.09	0.37
			NYS	1.8	2.5	0.44	0.53
			Rest of US	3.6	3.4	0.18	0.16

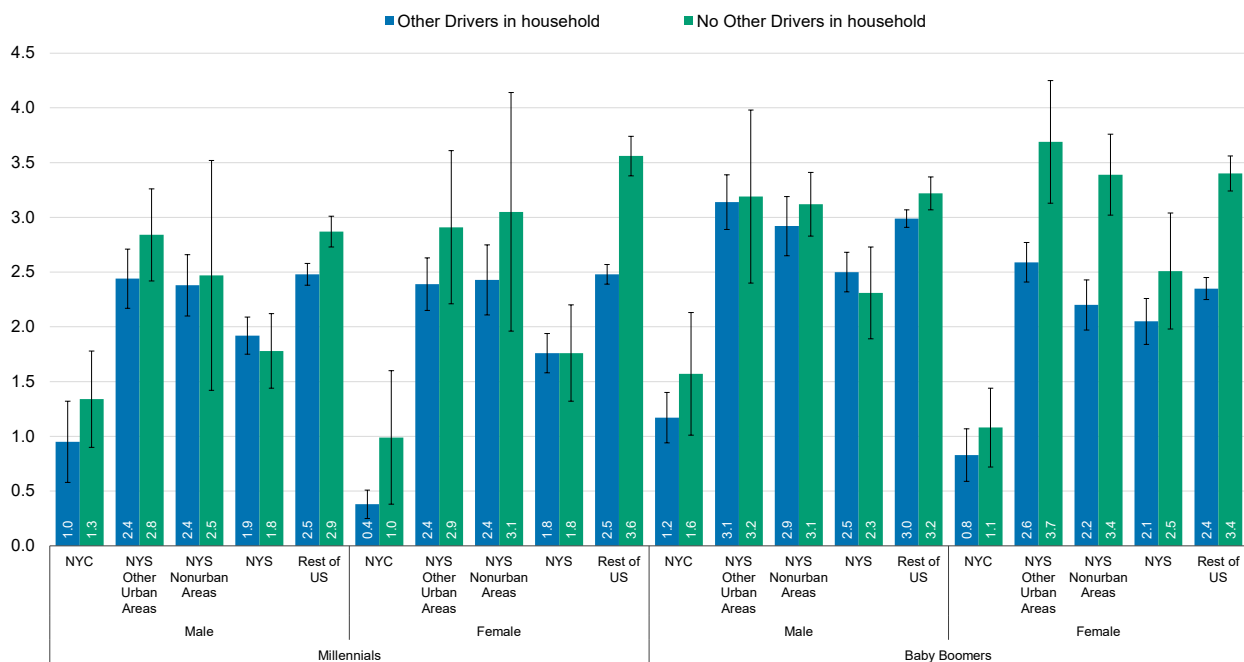


Figure 3-16. Average daily vehicle trips by generation (millennials and baby boomers), gender, and availability of other drivers in the household (2017 NHTS data).

3.6 ZERO-VEHICLE HOUSEHOLD EFFECT ON TRAVEL

Millennial households with zero vehicles are more likely to be in the low-income category, defined as those earning less than \$25,000 annually. This is clearly shown in the 2009 and 2017 NHTS data, illustrated in Figure 3-17 and Figure 3-18. It's important to note that the methodologies used to collect the 2009 and 2017 income data differ, making direct comparisons across years challenging. However, the trends for 2017 show that the share of low-income, zero-vehicle millennial households is much higher outside NYC, especially in NYS other urban and nonurban areas. Notably, in NYS nonurban areas, an overwhelming 88% of zero-vehicle millennial households are classified as low-income.

Conversely, NYC has a much smaller proportion of zero-vehicle millennial households in the low-income bracket, with only 21% in 2017. This is a significant drop compared with other regions in the state. The

lower share of low-income, zero-vehicle households in NYC can likely be attributed to the city's high cost of living and its extensive public transit system. In NYC, zero-vehicle status can apply to households with middle- or even higher-income levels; they may opt for public transit over personal vehicles because of cost, convenience, or space constraints. This stands in contrast to other areas where vehicle ownership is more strongly linked to economic capability.

In 2009, NYC had a much higher share of low-income, zero-vehicle households, with 76% falling into this income category. At the same time, NYS other urban areas and NYS nonurban areas showed much lower proportions of low-income, zero-vehicle households at 44% and 29%, respectively. Over time, this dynamic has shifted, with low-income, zero-vehicle households becoming more concentrated in nonurban regions, whereas NYC has seen a substantial decrease in the proportion of low-income, zero-vehicle households.

Looking at NYS as a whole, NYC's trends strongly influence the overall numbers. In 2009, 68% of NYS's zero-vehicle millennial households were low-income, a number largely driven by NYC's high percentage of low-income households. However, by 2017, the share of low-income, zero-vehicle households in NYS had dropped to 30%, primarily owing to the decline in NYC's low-income, zero-vehicle households. Outside of NYC, the picture is quite different, with other regions, particularly NYS nonurban areas, experiencing a significant rise in low-income, zero-vehicle households. By 2017, 88% of zero-vehicle households in NYS nonurban areas were classified as low-income, and NYS other urban areas saw an increase to 70%. These figures reflect a growing economic divide between NYC and other regions, where zero-vehicle status outside of NYC is increasingly tied to economic hardship.

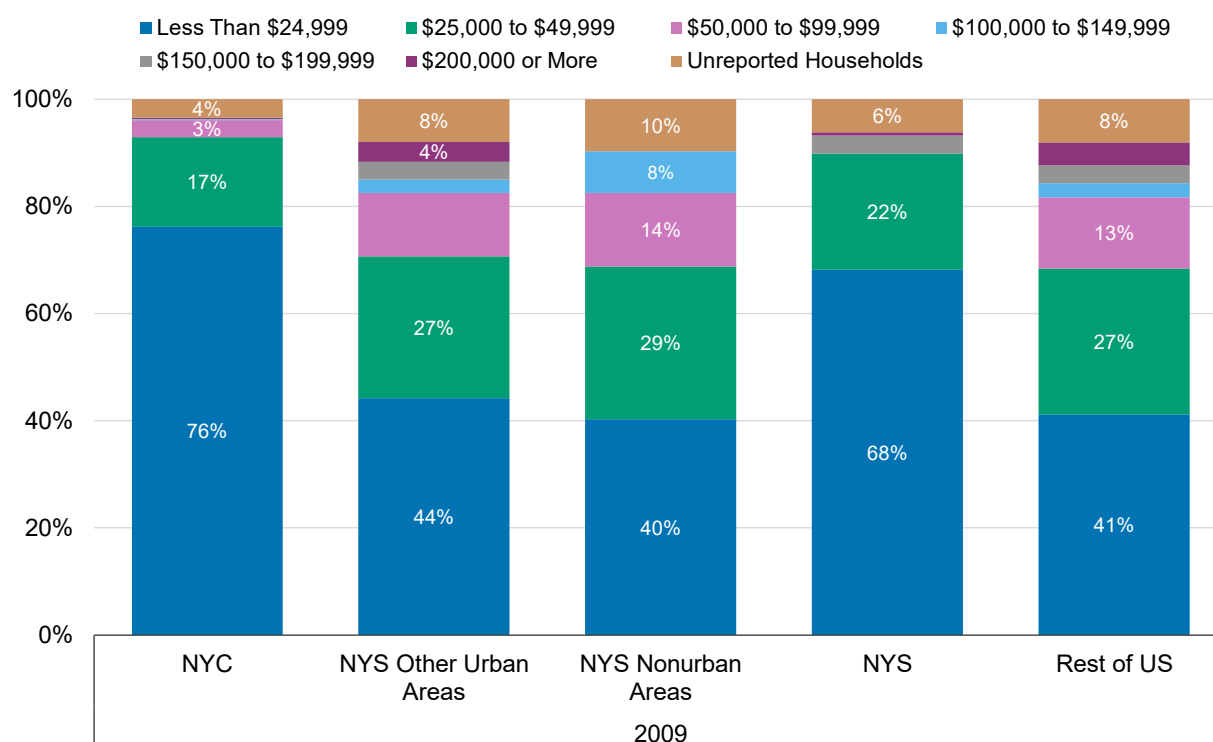


Figure 3-17. Distribution of zero-vehicle millennial households by income level and region (2009 NHTS data).

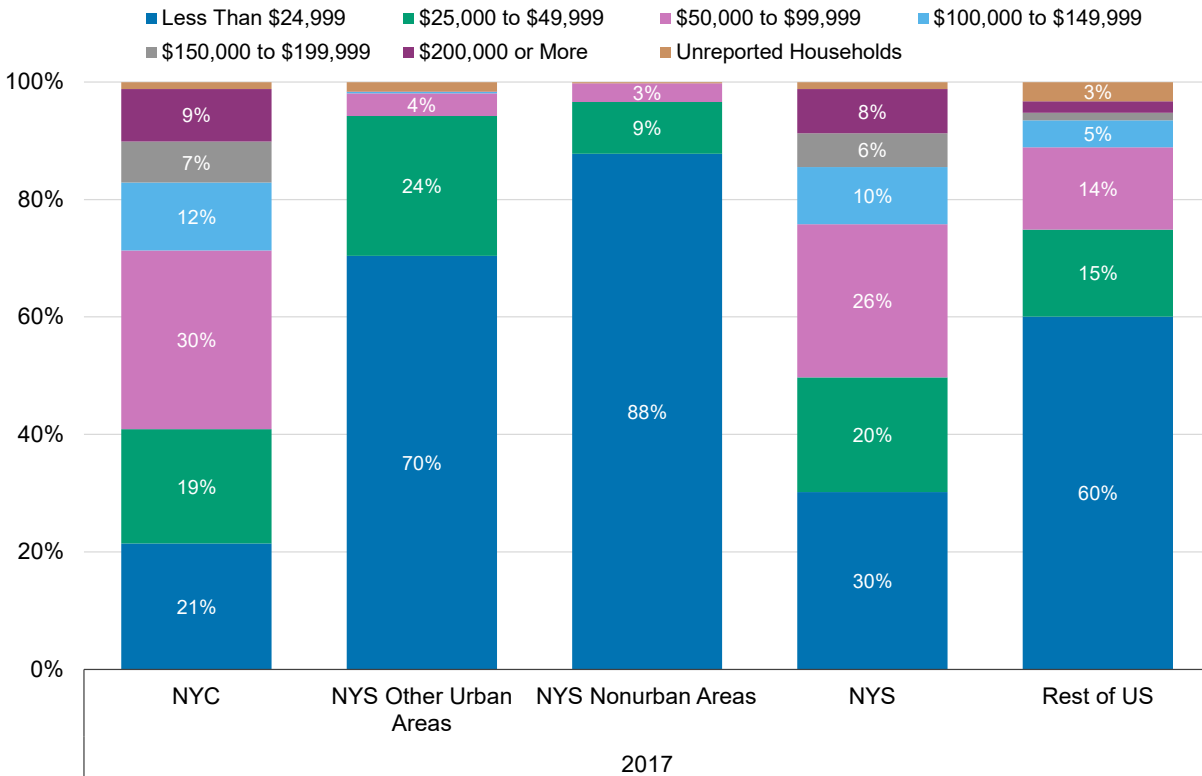


Figure 3-18. Distribution of zero-vehicle millennial households by income level and region (2017 NHTS data).

3.7 VEHICLE TRAVEL

3.7.1 Average Daily Vehicle-Trip Rate

Regarding vehicle travel, several key trends emerge in millennials' travel behavior compared with that of baby boomers, as shown in Figure 3-19. In 2009, millennials in NYC had an average vehicle-trip rate of 0.57, significantly lower than that of baby boomers, who averaged 1.19 vehicle trips per day. The MOE suggests this gap is statistically significant. By 2017, millennials' vehicle-trip rate in NYC increased to 0.87, narrowing the gap with baby boomers, whose vehicle-trip rate slightly declined to 1.15 trips per day. However, despite this increase, millennials still made fewer trips than baby boomers made, and the overlap in MOEs (0.16 for millennials and 0.15 for baby boomers) indicates that this difference was not statistically significant in 2017. In all years and regions, millennials consistently had fewer vehicle trips than those of baby boomers. This trend may be attributed to millennials' lower reliance on personal vehicles, which can be explained by their greater access to alternative modes of transportation, such as public transit, particularly in urban settings like NYC. Baby boomers, in contrast, remained more car-dependent, especially in nonurban areas where alternative transportation options are limited.

Whereas millennials saw small increases in vehicle trips over time, baby boomers experienced a decline in car use, particularly in the rest of the US and NYS regions. Despite this decrease, baby boomers still outpaced millennials in most regions. Outside of NYC, millennials made significantly more vehicle trips both in 2009 and 2017. For example, in 2017, millennials in NYS other urban areas made 2.47 vehicle trips per day on average, nearly double the rate of their counterparts in NYC. This highlights the regional variation in travel behavior, with vehicle dependence increasing in less dense areas where public transit options are fewer and personal vehicle use is more essential.

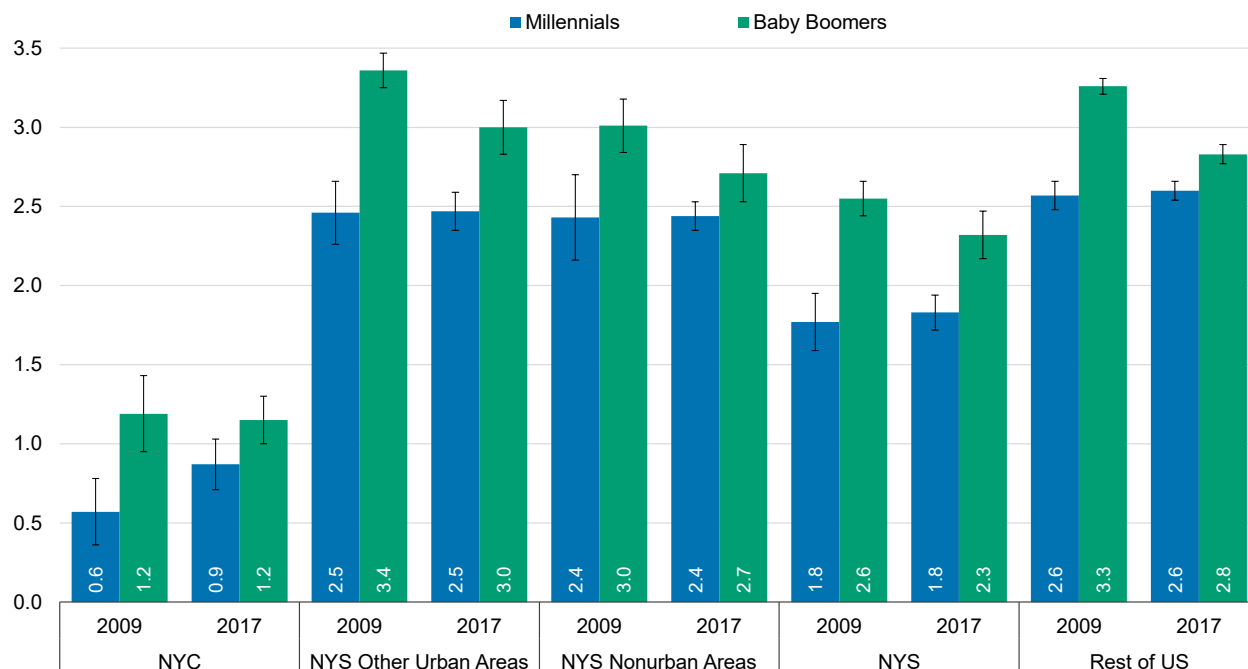


Figure 3-19. Average daily vehicle trips by generation (millennials and baby boomers) and region over time (NHTS data).

Moreover, the analysis of millennials' travel behavior by household status reveals clear distinctions, especially when comparing millennials as heads of household to other household types across regions. As shown in Table 3-7 and Figure 3-20, millennials as heads of household consistently made fewer vehicle trips than those of other household types in NYC in both 2009 and 2017. For instance, in 2009, millennials as heads of household in NYC recorded the lowest vehicle-trip rate at 0.9 trips per day, significantly lower than households without millennials (1.2 trips per day) and even slightly below millennials living with older household members (1.1 trips per day). The MOE for millennials as heads of household in NYC suggests that the difference is notable but may not be statistically significant. Nonetheless, this pattern persisted into 2017, where millennials as heads of household continued to make only 0.9 trips per day, unchanged from 2009, whereas other household types, such as those without millennials, remained higher. This stability over time highlights a consistent trend of lower vehicle dependency among millennials in NYC, likely driven by the availability of robust public transportation options.

In contrast, outside of NYC, particularly in NYS nonurban areas, millennials as heads of household had the highest vehicle-trip rate in 2009, averaging 3.5 trips per day, surpassing both households without millennials (2.9 trips per day) and millennials living with older household members (2.8 trips per day). This suggests that in nonurban areas, millennials who are heads of household are more car-dependent, possibly because of fewer transportation alternatives. By 2017, the vehicle-trip rate for this group declined slightly to 2.7 trips per day but remained higher than other household types, showing that even though car dependence slightly reduced, it was still significant in nonurban regions.

When gender differences are considered, male millennials as heads of household generally made more vehicle trips than their female counterparts in most regions. In NYC, female millennials, regardless of household status, made considerably fewer vehicle trips compared with males. For instance, females who were heads of household in NYC made only 0.7 vehicle trips per day, compared with 1.0 trips per day for males. This lower reliance on vehicles among females in urban settings reflects a greater likelihood of

using public transportation or other nonvehicle travel modes. Interestingly, in NYS nonurban areas, male and female millennials as heads of household had similar vehicle-trip rates in 2017, both averaging 2.7 trips per day, surpassing the vehicle-trip rates of other millennials household types in the same region. This suggests that gender differences in vehicle use are more pronounced in urban areas like NYC but tend to level out in nonurban regions, where reliance on vehicles is higher across the board owing to fewer alternative transportation options.

Table 3-7. Average daily vehicle trips by millennial household status and region (NHTS data).

Year	Region	Estimate			Margin of error		
		No millennials	Millennials living with older household members	Millennials as heads of household	No millennials	Millennials living with older household members	Millennials as heads of household
2009	NYC	1.20	1.10	0.90	0.18	0.26	0.54
	NYS other urban areas	3.10	3.00	2.70	0.07	0.15	0.51
	NYS nonurban areas	2.90	2.80	3.50	0.14	0.21	0.64
	NYS	2.40	2.30	1.90	0.09	0.15	0.49
	Rest of US	3.10	3.10	2.90	0.04	0.06	0.22
2017	NYC	1.20	1.10	0.90	0.21	0.22	0.16
	NYS other urban areas	2.90	2.50	2.70	0.15	0.20	0.24
	NYS nonurban areas	2.60	2.40	2.70	0.13	0.24	0.18
	NYS	2.30	2.10	1.90	0.11	0.16	0.19
	Rest of US	2.80	2.50	2.70	0.03	0.08	0.08

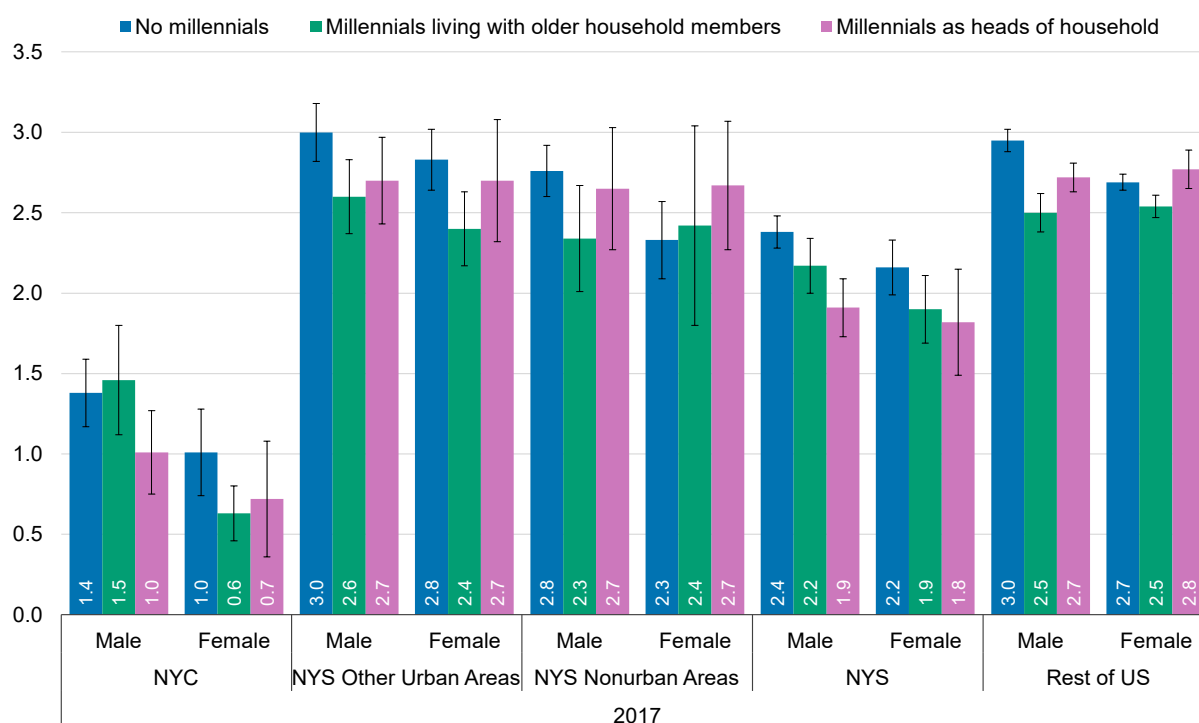


Figure 3-20. Average daily vehicle trips by millennial household status, gender, and region (2017 NHTS data).

3.7.2 Average Vehicle-Trip Length

In Figure 3-21, the 2017 NHTS data on average vehicle-trip length by generation and region reveal some interesting patterns when comparing millennials with other generations. In NYC, millennials had an average vehicle-trip length of 11.8 miles, which was significantly longer than that of baby boomers (7.4 miles) and Gen Xers (9.8 miles). This suggests that when millennials in NYC do use vehicles, they tend to travel longer distances compared with older generations. However, MOE for millennials in NYC is notably high (3.88 miles), making difficult a determination of whether this difference is statistically significant when compared with baby boomers or Gen Xers, whose MOEs are lower. For Gen Zers, the average trip length was much shorter at 3.5 miles, but the large MOE for this group (4.10 miles) makes this estimate less reliable.

In other urban areas outside NYC, millennials continue to exhibit vehicle-trip lengths longer than those of other generations across all regions. Millennials' average vehicle-trip length was 11.0 miles, compared with 9.7 miles for Gen Xers and 8.8 miles for baby boomers. The MOE integration suggests a significant difference between millennials and baby boomers, emphasizing that millennials in these areas tend to travel farther by car than baby boomers travel.

In NYS nonurban areas, millennials had an average vehicle-trip length of 12.5 miles, slightly higher than that of baby boomers (11.2 miles) but slightly lower than that of Gen Xers (13.3 miles). Whereas the difference between millennials and Gen Xers is not statistically significant because of the overlapping MOEs (1.68 miles for millennials and 2.27 miles for Gen Xers), this still indicates that millennials in nonurban areas are taking vehicle trips longer than those of baby boomers, likely reflecting their reliance on private vehicles in less densely populated regions with fewer transportation alternatives.

Looking at NYS overall, millennials had an average trip length of 11.3 miles, once again outpacing both baby boomers (8.9 miles) and Gen Xers (10.1 miles). The smaller MOE for millennials (0.71 miles) compared with that of baby boomers (0.60 miles) indicates that the difference in vehicle-trip lengths is likely statistically significant, with millennials traveling farther than baby boomers travel. In the rest of the US, millennials followed a similar pattern, with an average vehicle-trip length of 11.9 miles, again higher than that of baby boomers (10.0 miles) and Gen Xers (10.8 miles). The difference in trip length between millennials and baby boomers in this region is statistically significant, given the respective MOEs (1.53 miles for millennials and 0.40 miles for baby boomers). This further emphasizes millennials' inclination to take longer vehicle trips compared with those of older generations across the country. Overall, millennials tend to make longer vehicle trips than baby boomers make across all regions, a trend that is statistically significant in many areas because of the lower MOEs.

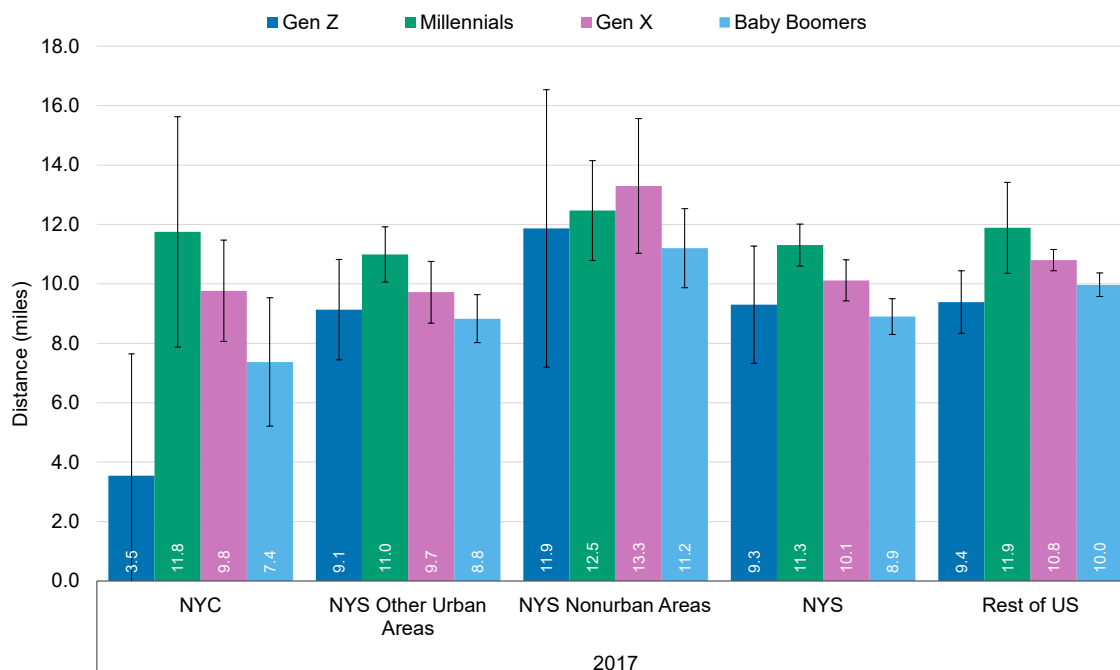


Figure 3-21. Average daily vehicle-trip length by generation and region (2017 NHTS data).

3.7.3 Vehicle Trips by Time of Travel

Figure 3-22 illustrates that millennials, regardless of their region, were less likely than baby boomers were to make vehicle trips during the period from 9 a.m. to 6 p.m. Instead, millennials' vehicle travel was more evenly distributed across the day, particularly between 6 a.m. and 9 p.m. One key finding is that, in every region, millennials consistently made more vehicle trips during the evening hours (6 p.m. to midnight) than baby boomers made in both 2009 and 2017. This pattern indicates that millennials tend to shift their vehicle travel to later hours of the day, suggesting a lifestyle that may be more oriented toward evening activities, whether social or work-related, compared with the lifestyles of their older counterparts.

Like their person trips, millennials' vehicle trips have evolved over time, with a clear concentration of trips during the afternoon and early evening hours. For example, in 2009, millennials in NYS nonurban areas had a notable concentration of vehicle trips in the afternoon, with 52% of trips occurring between 12 and 3 p.m. and 60% occurring between 3 and 6 p.m. These patterns reflect millennials' reliance on vehicles during traditional work hours or school commute times. However, by 2017, their vehicle trips became more evenly distributed across time slots, with only 28% of trips happening between 3 and 6 p.m. This decline suggests that as millennials aged, their travel behavior diversified, spreading their trips more evenly across the day.

A similar pattern emerged in the rest of the US. In 2009, 60% of millennials' vehicle trips occurred between 3 and 6 p.m., underscoring their significant dependence on afternoon travel. By 2017, only 25% of trips occurred during this period, showing a notable reduction in reliance on peak-hour vehicle travel. This change implies that millennials are shifting away from the traditional afternoon commute window, possibly owing to changes in work patterns, the adoption of flexible hours, or alternative transportation methods.

In NYC, millennials also demonstrated an increase in vehicle travel during the afternoon. In 2009, only 15% of their trips occurred between 12 and 3 p.m., but this figure rose to 25% by 2017. This increase

suggests that millennials are making more midday trips, possibly reflecting flexible work arrangements or errand-running during off-peak hours, a trend less evident in baby boomers. Furthermore, millennials in NYC also made a higher proportion of vehicle trips during the evening hours. In 2017, 8% of their trips occurred between 9 p.m. and 12 a.m., compared with only 5% of baby boomers' trips. This reinforces the idea that millennials engage more frequently in late-night activities, likely involving social or recreational outings, compared with older generations, who appear to taper off their vehicle trips earlier in the evening.

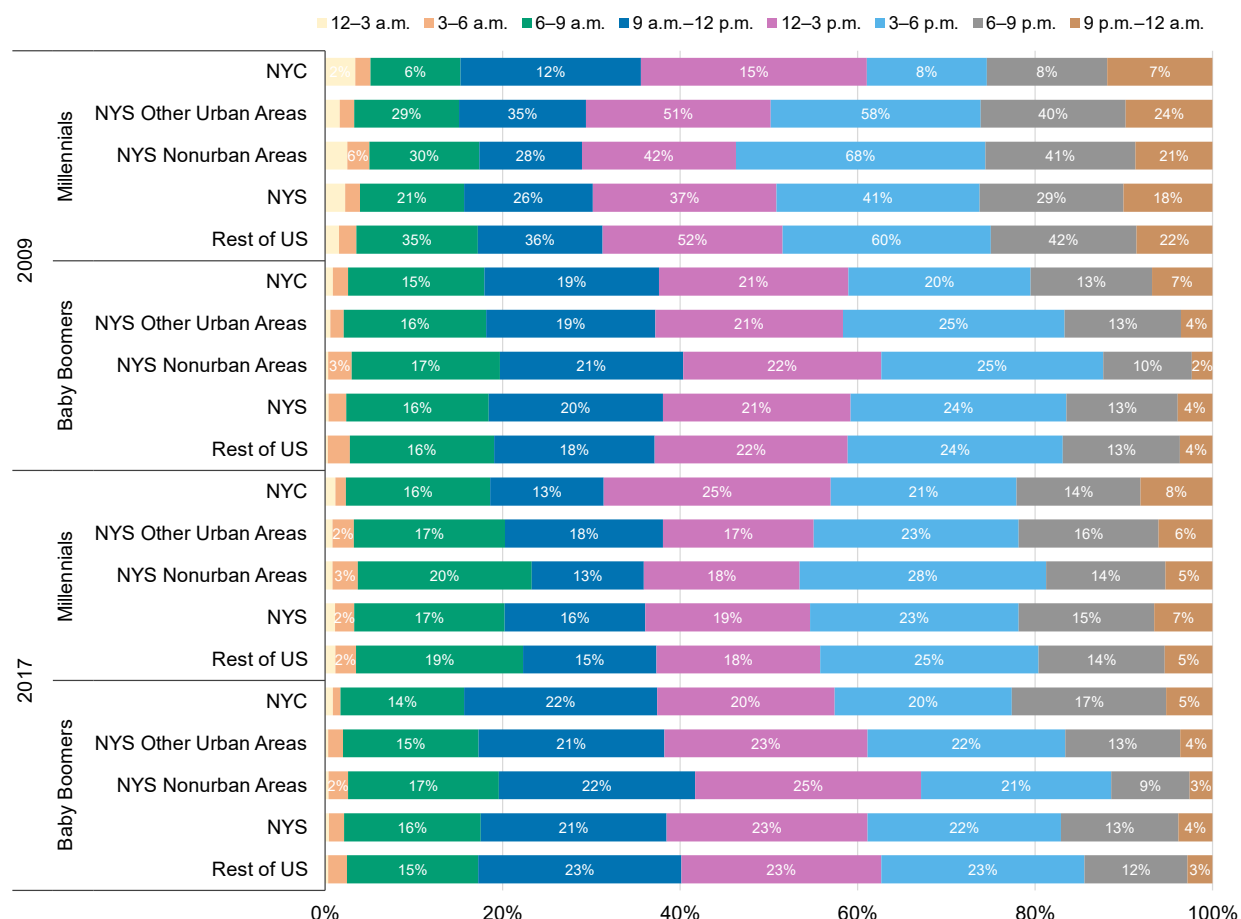


Figure 3-22. Comparison of the temporal distribution of vehicle trips by generation (millennials and baby boomers), time of travel, and region (NHTS data).

The temporal distribution of vehicle trips for millennials based on their household status reveals distinct patterns in the 2017 NHTS data (Table 3-8). Millennials as heads of household across most regions display a marked concentration of vehicle trips during midday and early evening, with less emphasis on early morning travel compared with that of other household types. In contrast, households without millennials tend to spread vehicle trips more evenly throughout the day, particularly in the morning hours. Millennials as heads of household are also more likely to make late-evening trips, indicating a more active lifestyle during the evening, particularly in urban areas like NYC. For example, in NYC, 25% of trips made by millennials as heads of household occur between 12 and 3 p.m., and another 23% between 3 and 6 p.m. This reflects a strong reliance on vehicle travel during midday and early evening, suggesting work or other activities concentrated in those periods. When compared with millennials living with older household members, millennials as heads of household in NYC make slightly fewer vehicle trips during the morning (15% between 6 and 9 a.m.), whereas those living with older household members have a

more morning-focused pattern (20% between 6 and 9 a.m.). In contrast, households without millennials have a broader distribution of trips, with 17% occurring during the morning rush (6 to 9 a.m.) and 19% between 9 a.m. and 12 p.m. This suggests that millennials as heads of household in NYC tend to have a later, more spread-out travel pattern, potentially because of more flexible work schedules or a greater reliance on public transportation for certain trips.

Table 3-8. Temporal distribution of vehicle trips by millennial household status (2017 NHTS data).

Household status	Region	12–3 a.m.	3–6 a.m.	6–9 a.m.	9 a.m.–12 p.m.	12–3 p.m.	3–6p.m.	6–9 p.m.	9 p.m.–12 a.m.
Millennials as heads of household	NYC	2%	1%	15%	15%	25%	23%	13%	7%
	NYS other urban areas	0%	2%	17%	19%	17%	23%	15%	6%
	NYS nonurban areas	0%	3%	19%	12%	18%	28%	14%	6%
	NYS	1%	2%	17%	17%	19%	24%	14%	6%
	Rest of US	1%	2%	20%	15%	18%	25%	14%	5%
Millennials living with older household members	NYC	1%	2%	20%	14%	21%	21%	14%	6%
	NYS other urban areas	1%	2%	17%	16%	19%	24%	15%	5%
	NYS nonurban areas	1%	3%	18%	16%	20%	26%	11%	5%
	NYS	1%	2%	18%	16%	20%	24%	14%	5%
	Rest of US	1%	3%	17%	17%	20%	24%	13%	5%
No millennials	NYC	1%	1%	17%	19%	20%	20%	17%	6%
	NYS other urban areas	1%	1%	15%	21%	22%	23%	13%	4%
	NYS nonurban areas	0%	2%	17%	22%	23%	24%	10%	3%
	NYS	0%	1%	16%	21%	22%	23%	13%	4%
	Rest of US	0%	2%	16%	21%	22%	23%	12%	3%

Note: Percentages may not total 100 due to rounding.

3.7.4 Average Vehicle Trips per Driver by Purpose

In Figure 3-23, the distribution of daily average vehicle trips by millennials and baby boomers shows notable differences in the purpose of travel. Millennials have a higher share of trips dedicated to earning a living across all regions compared with those of baby boomers. For example, in the NYS region, 31% of millennials' vehicle trips were for earning a living, significantly higher than the 22% of trips for baby boomers. Similarly, in NYC, 29% of millennials' vehicle trips were for earning a living, higher than the 17% of baby boomers. This suggests that millennials are more likely to use vehicles for commuting or work-related purposes, potentially reflecting their higher participation in the workforce or more dispersed work locations requiring vehicle travel.

Compared with millennials, baby boomers have a greater proportion of trips dedicated to family and personal business. In NYS nonurban areas, 49% of baby boomers' vehicle trips were for family and personal business, compared with 39% for millennials. This trend is consistent across regions, highlighting that baby boomers are more likely to use vehicles for errands or household responsibilities, possibly owing to different life stages, such as retirement or caregiving.

When comparing social and recreational trips, millennials in NYC have a higher share of such trips (25%) than baby boomers have (20%), which may reflect millennials' greater focus on social and leisure activities. This pattern is less pronounced in nonurban areas, where baby boomers and millennials have closer percentages of social and recreational trips. Additionally, baby boomers consistently have fewer vehicle trips for purposes like civic, educational, and religious activities compared with those of millennials, though the differences are minimal (e.g., 5% for millennials in the rest of the US vs. 4% for baby boomers).

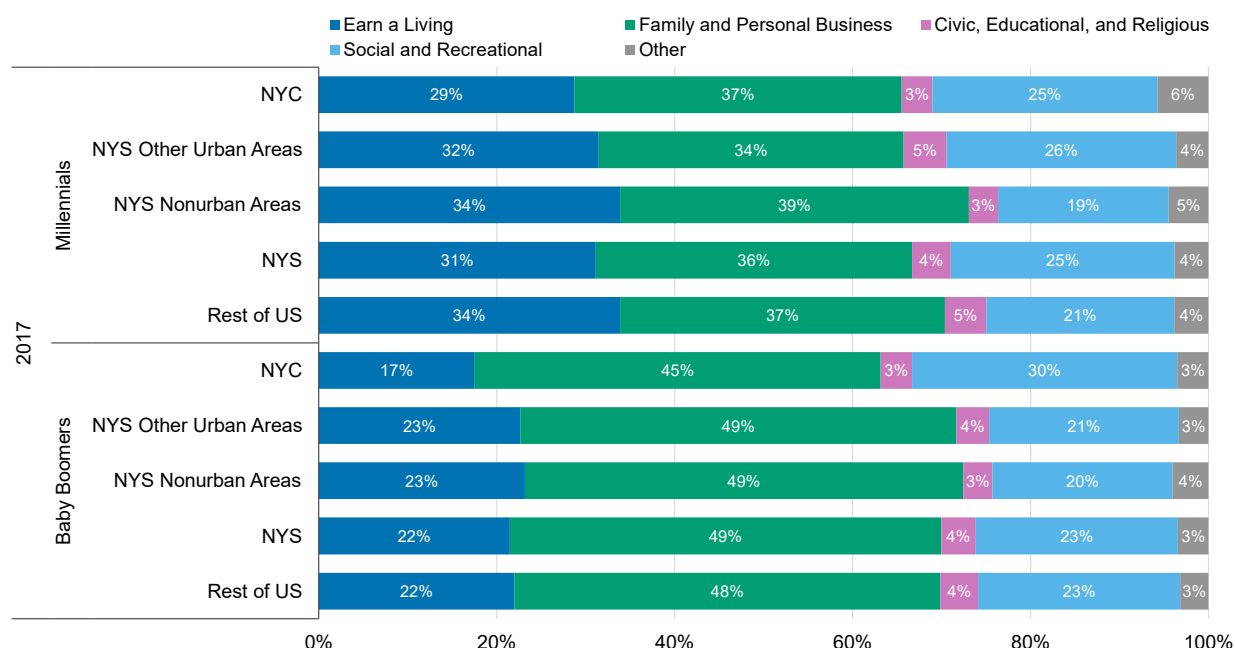


Figure 3-23. Distribution of daily average vehicle trips by generation, trip purpose, and region (2017 NHTS data).

In Table 3-9, gender differences among millennials are clear. Male millennials have a higher share of trips for earning a living compared with females across all regions. For instance, in NYS, 34% of male millennials' vehicle trips were for earning a living, compared with 28% of trips for females. Similarly, in NYS nonurban areas, 40% of male millennials' trips were for work, significantly higher than the 27% for females. This suggests that male millennials may be more likely to use vehicles for commuting to work, reflecting potential differences in employment patterns or the distribution of household responsibilities.

In contrast, female millennials have a higher proportion of trips for family and personal business. For example, in NYS nonurban areas, 46% of female millennials' vehicle trips were for family and personal business, compared with 33% for males. This pattern suggests that females may take on more household-related responsibilities, which is consistent across other regions.

For social and recreational purposes, male millennials have a slightly higher share of trips in NYC (28% for males vs. 19% for females), whereas in NYS other urban areas, females take more social and recreational trips (26% vs. 25% for males). This indicates that although men are slightly more engaged in recreational activities in urban areas like NYC, women have a more balanced distribution across regions. Overall, these gender differences reflect the varied roles and responsibilities between male and female millennials, influencing how they allocate vehicle trips across different purposes.

Table 3-9. Distribution of daily average vehicle trips by trip purpose and gender of millennials across regions (2017 NHTS data).

Gender	Region	Earn a living	Family and personal business	Civic, educational, and religious	Social and recreational	Other
Male	NYC	30%	35%	3%	28%	6%
	NYS other urban areas	34%	31%	5%	25%	4%
	NYS nonurban areas	40%	33%	3%	19%	5%
	NYS	34%	32%	4%	25%	5%
	Rest of US	38%	31%	5%	22%	4%
Female	NYC	29%	41%	7%	19%	5%
	NYS other urban areas	28%	38%	4%	26%	3%
	NYS nonurban areas	27%	46%	3%	19%	4%
	NYS	28%	40%	5%	24%	3%
	Rest of US	30%	42%	4%	21%	3%

Note: Percentages may not total 100 due to rounding.

Additionally, we explored the distribution of daily vehicle trips by trip purpose for millennials based on their household status. Figure 3-24 shows important trends in how millennials as heads of households allocate their vehicle trips, compared with other household types in 2017. Across most regions, millennials as heads of household prioritize vehicle trips for work-related purposes, with the higher proportion of trips allocated to “earn a living” than those of non-millennials households. For instance, in the NYS region, 28% of vehicle trips for this group are work-related, which is notably higher than the 21% for households without millennials. This indicates that millennials as heads of household have a stronger reliance on vehicle trips for employment purposes, compared with households without millennials. In NYC, however, the share of work-related trips is lower for millennials as heads of household (23%), possibly reflecting the availability of public transportation options and reduced reliance on cars for commuting.

Millennials as heads of household in NYC also stand out for making more social and recreational trips (27%) compared with their counterparts living with older household members (24%) and households without millennials (24%). This suggests that millennials heading their own households in urban environments like NYC might have more flexibility or desire to engage in social activities, perhaps reflecting different lifestyle priorities. However, they allocate fewer trips to family and personal business compared with households without millennials. In NYS nonurban areas, for instance, millennials as heads of households spend 44% of their vehicle trips on family and personal business, which is higher than other millennials households but lower than the 47% for households without millennials.

In Figure 3-25, we see gender differences in vehicle trip purposes for millennials as heads of households across regions. Generally, male millennials as heads of households make a higher percentage than females of vehicle trips for earning a living. For example, in the NYS other urban areas, 33% of male millennials’ vehicle trips are for work-related purposes, compared with 25% of trips for female millennials. This trend is consistent across other regions such as NYS nonurban areas, where 37% of male millennials’ trips are for earning a living compared with only 25% for females. This indicates a stronger focus on work-related travel for men.

Females, on the other hand, show a higher proportion of vehicle trips for family and personal business. In NYS nonurban areas, 51% of trips by female millennials are for family and personal business, significantly higher than the 37% for males. This suggests that women, particularly in nonurban areas,

may bear more household responsibilities, which could explain their greater reliance on vehicle trips for personal errands and family duties.

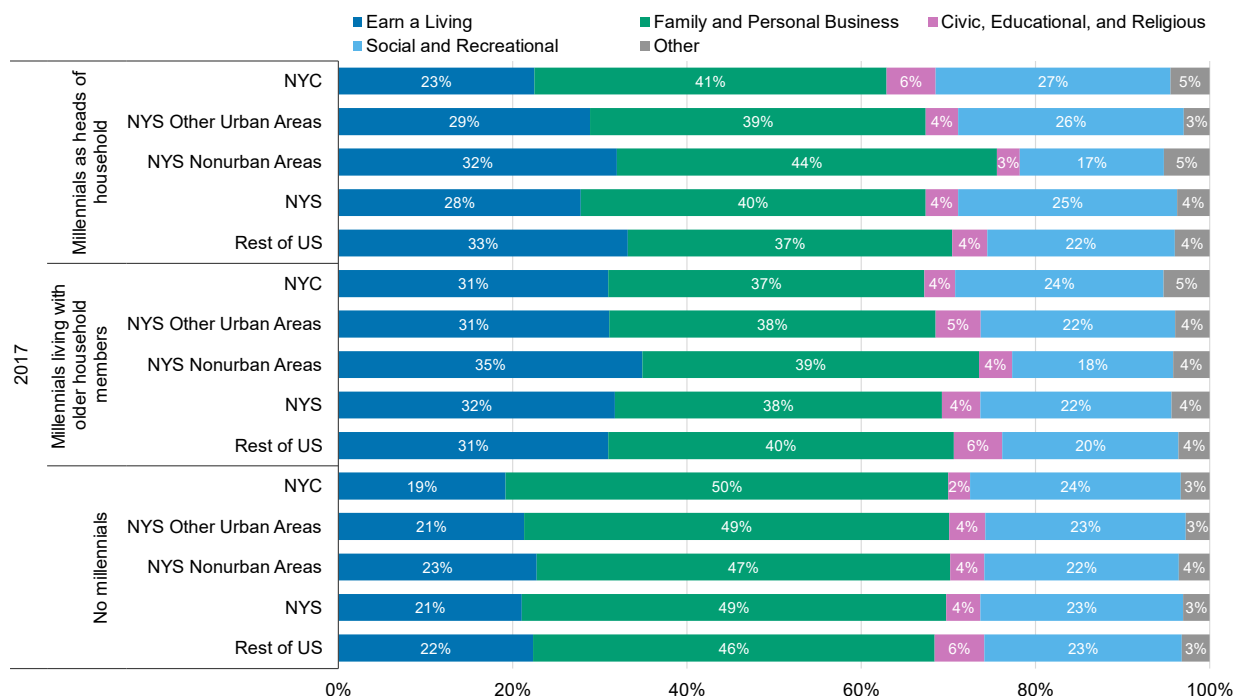


Figure 3-24. Distribution of daily average vehicle trips by trip purpose and millennial household status across regions (2017 NHTS data).

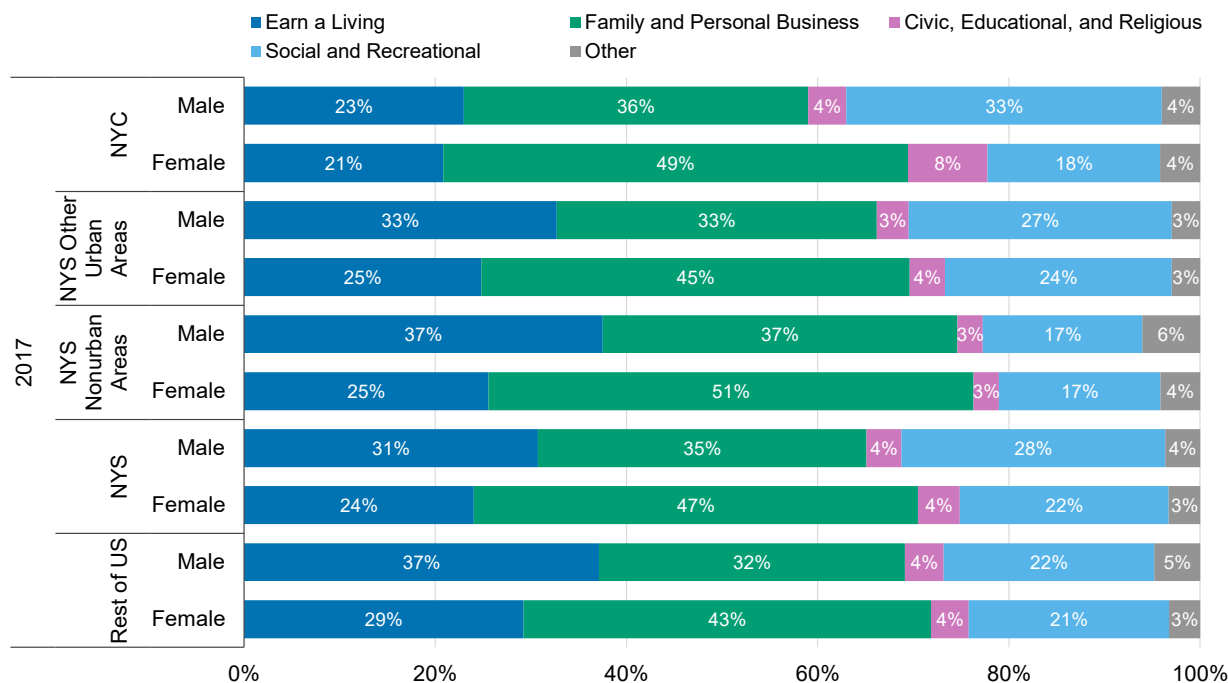


Figure 3-25. Distribution of daily average vehicle trips by trip purpose and gender of millennials as heads of household across regions (2017 NHTS data).

In NYC, the gender differences are particularly big in terms of social and recreational trips. Male millennials allocate 33% of their vehicle trips for social and recreational purposes, compared with only 18% for females. This large discrepancy could reflect differing social habits or levels of access to recreational opportunities. Additionally, female millennials in NYC allocate a higher share of trips to family and personal business (49%) compared with males (36%), further emphasizing the role of gender in shaping vehicle travel behavior, especially in urban areas.

3.8 SIZE OF TRAVEL PARTY ON TRIPS INVOLVING MILLENNIALS

To determine whether millennials traveled alone, in pairs, or in larger groups in 2017, data on travel party size from the NHTS, which capture the number of people traveling together on the same trip, were analyzed. As shown in Table 3-10, across most regions, millennials generally travel longer distances when alone or with smaller groups, especially when compared with baby boomers. For instance, in NYC, millennials traveling alone (travel party of one) recorded an average person miles of travel (PMT) of 14.08 miles, which was higher than that of baby boomers (9.26 miles). However, the MOE for millennials' PMT in NYC is quite large (8.64 miles), indicating that this difference might not be statistically significant. Regardless, this could suggest that when millennials in NYC use personal vehicles for solo trips, they tend to travel longer distances. However, the data show a sharp drop in travel distances for larger groups; when traveling with a party of four or more, millennials' PMT in NYC was just 1.67 miles.

In nonurban areas and the rest of the US, millennials consistently exhibited higher PMT across various travel party sizes compared with both baby boomers and younger generations. For instance, in the rest of the US, millennials traveling alone recorded a PMT of 21.23 miles, higher than that of baby boomers (20.00 miles) and significantly higher than that of Gen Zers (4.53 miles). A similar trend is observed in NYS nonurban areas, where millennials traveling alone recorded a PMT of 19.12 miles, indicating longer travel distances in rural areas. In comparison, baby boomers had a PMT of 22.10 miles, but Gen Zers recorded much shorter solo trips (5.23 miles), reinforcing the idea that millennials in nonurban areas tend to travel farther distances. In contrast, Gen Zers tend to travel more with larger groups, particularly with a travel party of four or more. This is especially evident in the rest of the US, where millennials with travel parties of four or more have an average PMT of 6, significantly lower than that of Gen Zers. For travel parties of two or three, we observe no significant difference between millennials and Gen Zers across most regions, with the exception of the rest of the US, where Gen Zers again shows higher PMT.

When examining vehicle miles traveled (VMT) across travel party sizes, millennials also exhibited interesting trends, as shown in Table 3-11. In NYC, millennials traveling alone had a VMT of 3.94 miles, which was lower than that of Gen Xers (5.28 miles) and baby boomers (4.61 miles). This indicates that millennials in NYC tend to travel shorter distances by car when driving solo, likely because of a combination of dense urban infrastructure, shorter distances between destinations, and the availability of public transit alternatives. However, the MOE for this group suggests that the difference is not statistically significant.

Outside of NYC, the data reveal that millennials traveled significantly longer distances when driving in smaller groups. In the rest of the US, millennials recorded a VMT of 7.57 miles for travel parties of two, which is higher than that for both Gen Zers (3.45 miles) and Gen Xers (7.54 miles). This suggests that millennials in rural areas depend more heavily on vehicles and travel long distances. However, an opposite trend is evident in NYS nonurban areas, where millennials in a travel party of two had a VMT of 5.01 miles, compared with baby boomers' 8.63 miles and Gen Xers' 9.16 miles.

Table 3-10. Average daily PMT per person by generation and travel party size across regions (2017 NHTS data).

Generation	Region	Estimate				Margin of error			
		Travel party of one	Travel party of two	Travel party of three	Travel party of four or more	Travel party of one	Travel party of two	Travel party of three	Travel party of four or more
Gen Z	NYC	3.60	1.98	6.71	4.40	1.27	0.62	6.85	3.24
	NYS other urban areas	4.42	5.89	3.99	10.60	1.95	2.32	1.21	5.06
	NYS nonurban areas	5.23	5.55	4.19	9.69	2.18	2.03	0.85	2.54
	NYS	4.23	4.53	4.93	8.41	0.91	1.16	2.32	3.08
	Rest of US	4.53	5.26	6.15	10.76	0.39	0.7	1.35	1.97
Millennials	NYC	14.08	7.07	12.53	1.67	8.64	4.63	11.62	0.95
	NYS other urban areas	22.45	9.77	4.38	5.36	8.79	7.37	1.79	3.26
	NYS nonurban areas	19.12	6.49	5.35	5.46	6.10	1.87	1.17	2.46
	NYS	18.18	8.21	8.35	3.61	7.90	3.45	5.72	1.54
	Rest of US	21.23	12.09	5.78	5.92	2.65	4.39	1.00	1.96
Gen X	NYC	16.37	10.07	4.62	3.57	7.17	10.85	4.64	2.60
	NYS other urban areas	21.82	9.44	3.51	8.28	3.61	1.34	0.52	5.46
	NYS nonurban areas	19.89	11.36	4.46	6.71	3.57	4.95	1.04	4.57
	NYS	19.28	9.88	4.07	6.09	3.61	4.78	1.98	2.78
	Rest of US	22.83	11.62	5.20	7.50	1.63	1.13	0.57	1.40
Baby boomers	NYC	9.26	4.54	1.52	0.61	1.58	3.29	0.91	0.50
	NYS other urban areas	26.43	10.49	4.38	2.83	10.68	2.27	2.51	1.15
	NYS nonurban areas	22.10	16.00	3.18	4.67	3.67	4.61	0.54	2.69
	NYS	18.39	8.37	3.00	2.02	4.92	1.76	1.24	0.49
	Rest of US	20.00	16.13	3.62	2.86	1.06	1.50	0.44	0.67

Table 3-11. Average daily VMT per driver by generation and travel party size across regions (2017 NHTS data).

Generation	Region	Estimate				Margin of error			
		Travel party of one	Travel party of two	Travel party of three	Travel party of four or more	Travel party of one	Travel party of two	Travel party of three	Travel party of four or more
Gen Z	NYC	2.39	0.13	0.12	0.00	3.17	0.20	0.24	0.00
	NYS other urban areas	8.65	4.35	0.36	0.25	1.50	2.43	0.31	0.22
	NYS nonurban areas	14.48	2.38	0.05	0.04	10.26	1.80	0.06	0.08
	NYS	9.01	3.55	0.28	0.18	1.79	1.65	0.23	0.15
	Rest of US	12.01	3.45	1.31	0.49	1.67	0.92	0.51	0.26
Millennials	NYC	3.94	4.02	1.18	1.03	1.32	2.96	1.17	1.03
	NYS other urban areas	16.69	6.00	2.12	2.35	2.25	3.04	0.76	2.11
	NYS nonurban areas	19.72	5.01	3.38	2.33	5.12	1.97	1.30	1.16
	NYS	11.91	5.12	1.86	1.82	1.52	1.90	0.49	1.08
	Rest of US	18.12	7.57	2.76	2.46	1.08	3.71	0.34	0.71
Gen X	NYC	5.28	2.65	2.16	2.28	1.93	0.96	1.77	2.21
	NYS other urban areas	17.61	6.48	2.65	3.48	2.40	0.84	0.62	1.13
	NYS nonurban areas	18.59	9.16	3.08	4.87	1.45	4.78	0.91	2.75
	NYS	12.99	5.27	2.50	3.15	1.70	0.82	0.66	0.97
	Rest of US	19.33	7.54	3.04	3.55	0.88	0.72	0.30	0.61
Baby boomers	NYC	4.61	2.93	0.68	0.22	1.78	2.08	0.59	0.22
	NYS other urban areas	16.74	5.9	2.58	1.27	1.75	0.67	1.40	0.78
	NYS nonurban areas	18.76	8.63	1.94	1.07	1.49	2.39	0.30	0.49
	NYS	12.76	5.19	1.85	0.88	1.47	0.74	0.78	0.39
	Rest of US	17.48	7.66	2.01	1.05	0.48	0.53	0.36	0.13

The 2017 NHTS data provide insight into the gender differences in VMT among millennials based on the size of the travel party across various regions. As illustrated in Figure 3-26, the VMT patterns for male and female millennials display notable differences depending on both the travel party size and region. Male millennials in NYC recorded an average VMT of 5 miles when traveling alone, compared with only 2 miles for female millennials. The large MOE for male millennials (2.30 miles) suggests that the difference between males and females for solo trips is not statistically significant. However, the lower VMT for females suggests a trend in which female millennials are more likely to make shorter trips.

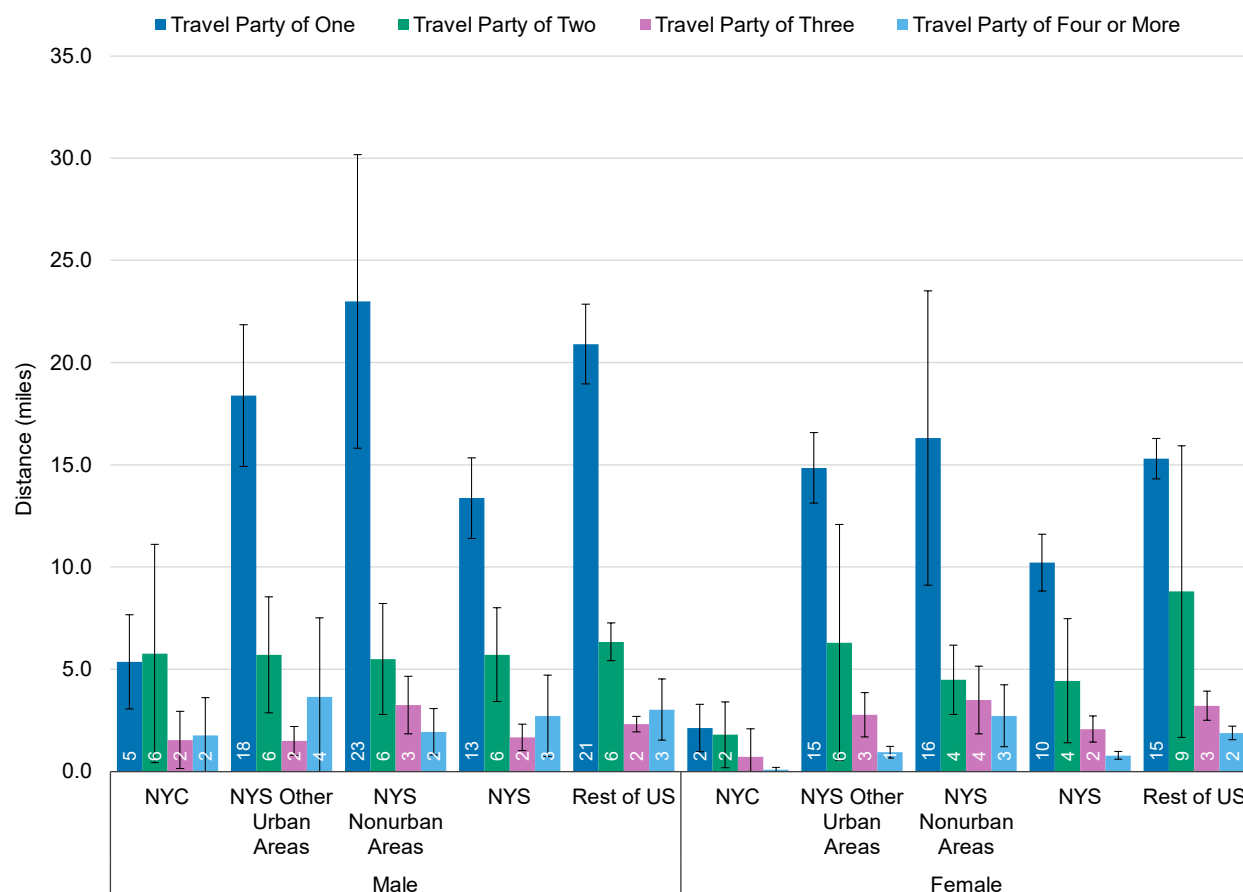


Figure 3-26. Average daily VMT per driver by millennials' gender and travel party size across regions (2017 NHTS data).

4. VIEWS OF TRANSPORTATION SERVICES

In the 2017 NHTS recruitment survey, one adult from each household was asked questions regarding their perceptions on the effect of travel costs on their travel. Specifically, the survey included the following question for each respondent: “How much do you agree or disagree with each of the following statements?” The statements were as follows:

- The price of gas affects the number of places I go
- Getting from place to place costs too much
- I walk to places to save money
- I bike to places to save money
- I use public transportation to save money

For each part listed above, the survey respondent would select one option from the provided list of strongly agree, agree, neutral (neither agree or disagree), disagree, and strongly disagree.

The discussions are provided for respondents living in NYC; New York Metropolitan Transportation Council (NYMTC); other urban areas in NYS (excluding NYMTC), NYS as a whole, and the rest of the United States; and rural areas in NYS as a whole and the rest of the United States. Categories are further refined to obtain responses based on whether millennials are present in the household.

4.1 PRICE OF GAS AFFECTS AMOUNT OF TRAVEL

Households with millennials as heads have a higher share of disagree (23%) and strongly disagree (30%) than those of other households in NYC (Figure 4-1). Similar pattern is found in NYMTC, NYS statewide, and the rest of the United States. In nonurban areas, a slightly lower share of households with millennials as heads disagree or strongly disagree (20%) than households with no millennials (23%). Households with millennials living with older household members have the highest share of strongly agree or agree of all household types. For example, in nonurban areas in NYS, 67% of the households with millennials living with older household members agree and strongly agree with this statement vs. only 39% for this household type in NYC.

4.2 TRAVEL COSTS TOO MUCH

Figure 4-2 shows that households with millennials living with older household members had the highest share of agreeing and strongly agreeing with the statement about travel costing too much in all regions except for one: other urban areas in NYS. The highest share was observed in NYC. In particular, 64% of the households with millennials living with older household members in NYC agree or strongly agree with that travel costs too much. In households where millennials are heads, the rest of the United States household type saw the highest share of households that disagree or strongly disagree with this statement.

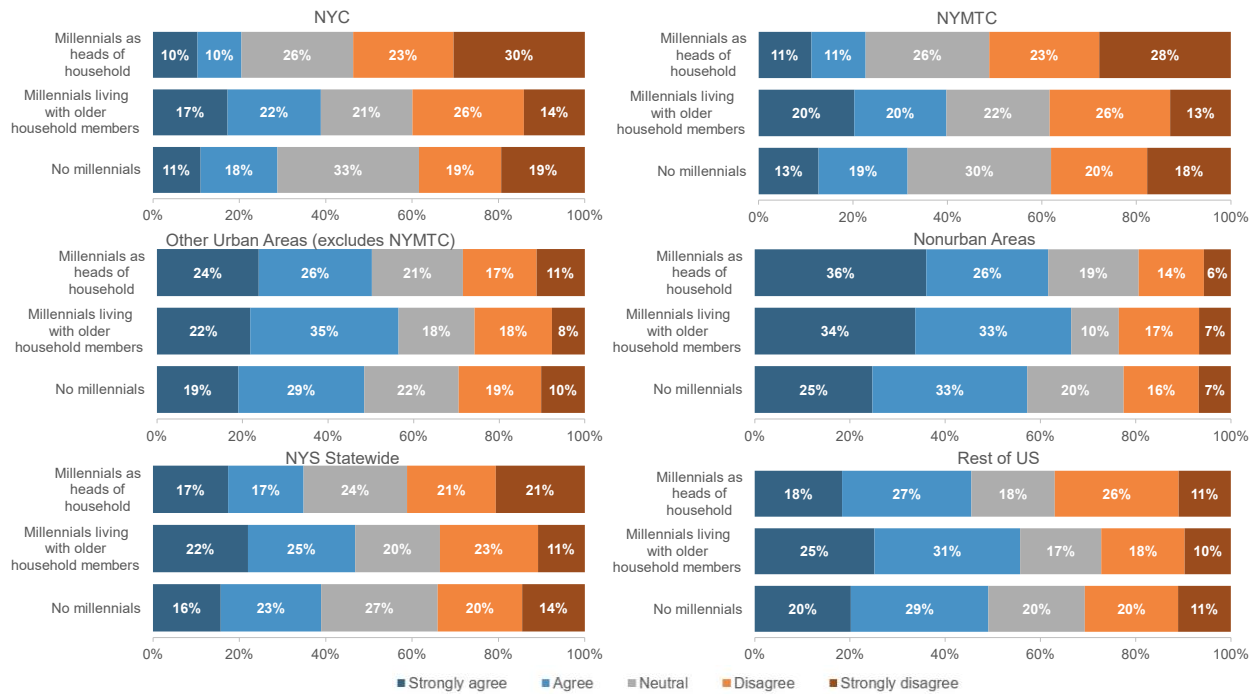


Figure 4-1. Share of ratings by region and millennial household status in response to the survey statement, "The price of gas affects the number of places I go" (2017 NHTS data).

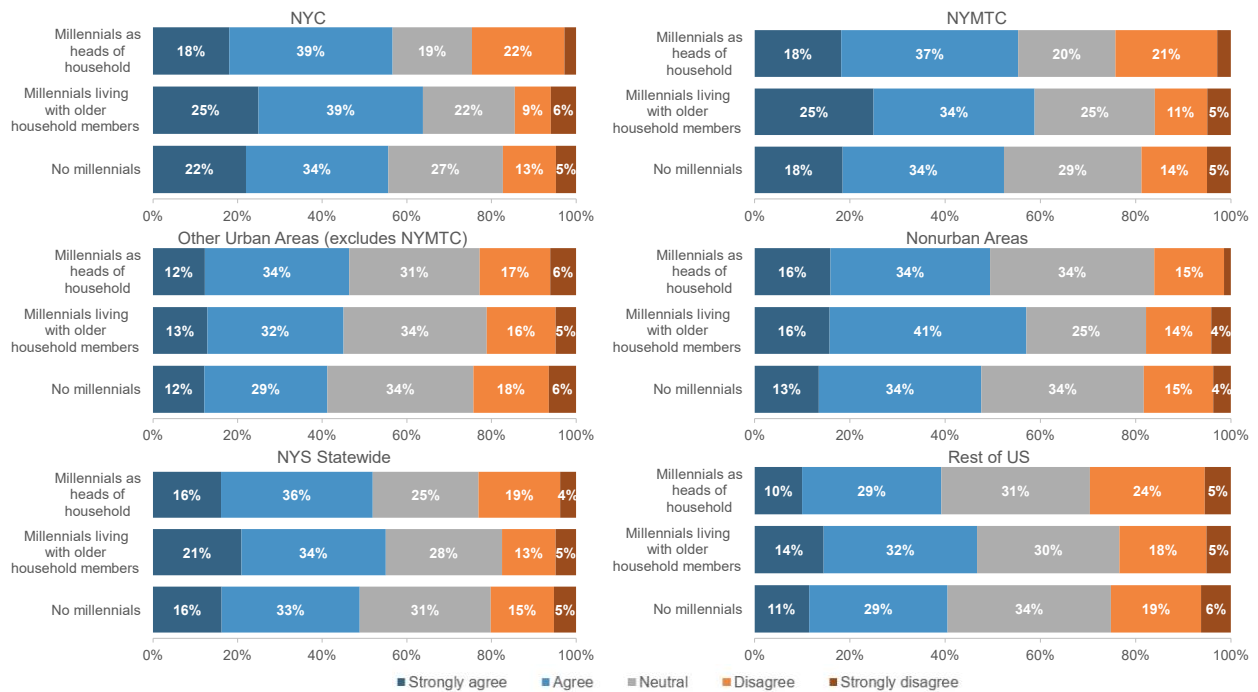


Figure 4-2. Share of ratings by region and millennial household status in response to the survey statement, "Getting from place to place costs too much" (2017 NHTS data).

4.3 WALK TO PLACES TO SAVE MONEY

In NYC and NYMTC, households headed by millennials have the highest share of those agreeing or strongly agreeing that they walked to places to save money (Figure 4-3). This share reaches 66% in NYC and 61% in NYMTC, a pattern that differs in other areas of NYS and the rest of the United States. In other urban areas of NYS outside of NYMTC, nearly half of millennial-headed households (49%) either disagree or strongly disagree with this statement, whereas in the rest of the United States, this share is 52%. The pattern of a higher share agreeing with the statement in NYC and NYMTC, with a lower share agreeing in other regions, is consistent across other types of households, as well.

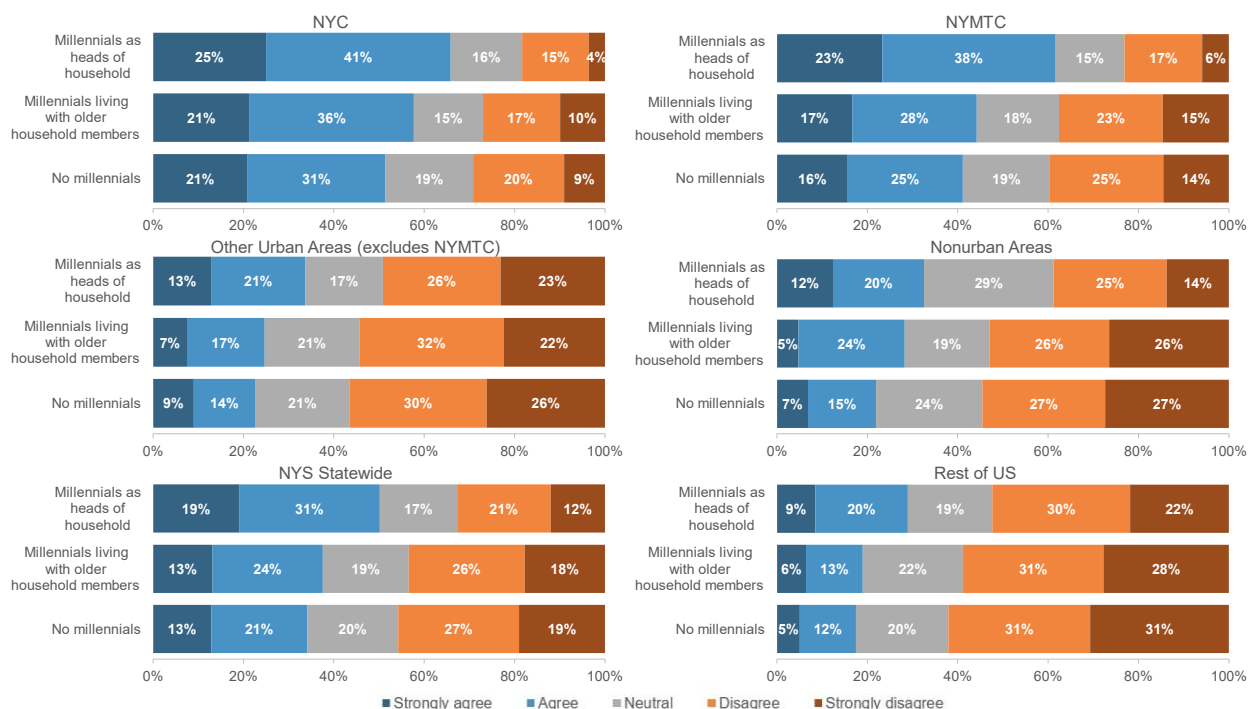


Figure 4-3. Share of ratings by region and millennial household status in response to the survey statement, “I walk to places to save money” (2017 NHTS data).

4.4 BIKE TO PLACES TO SAVE MONEY

Households without millennials have the highest share of disagreement with the statement that they biked to places to save money (Figure 4-4). In NYC, 55% of these households disagree or strongly disagree, surpassing the share among households with millennials. This percentage is even higher in the rest of NYS and the rest of the United States. In other urban areas outside of NYMTC, 72% of households without millennials disagree or strongly disagree with this statement. Compared with the rest of the United States, a higher share of households across all types in these regions held a neutral attitude toward this statement. For instance, 29% of households with millennials living with older household members were neutral about biking to save money in NYS, compared with 21% in the rest of the United States.

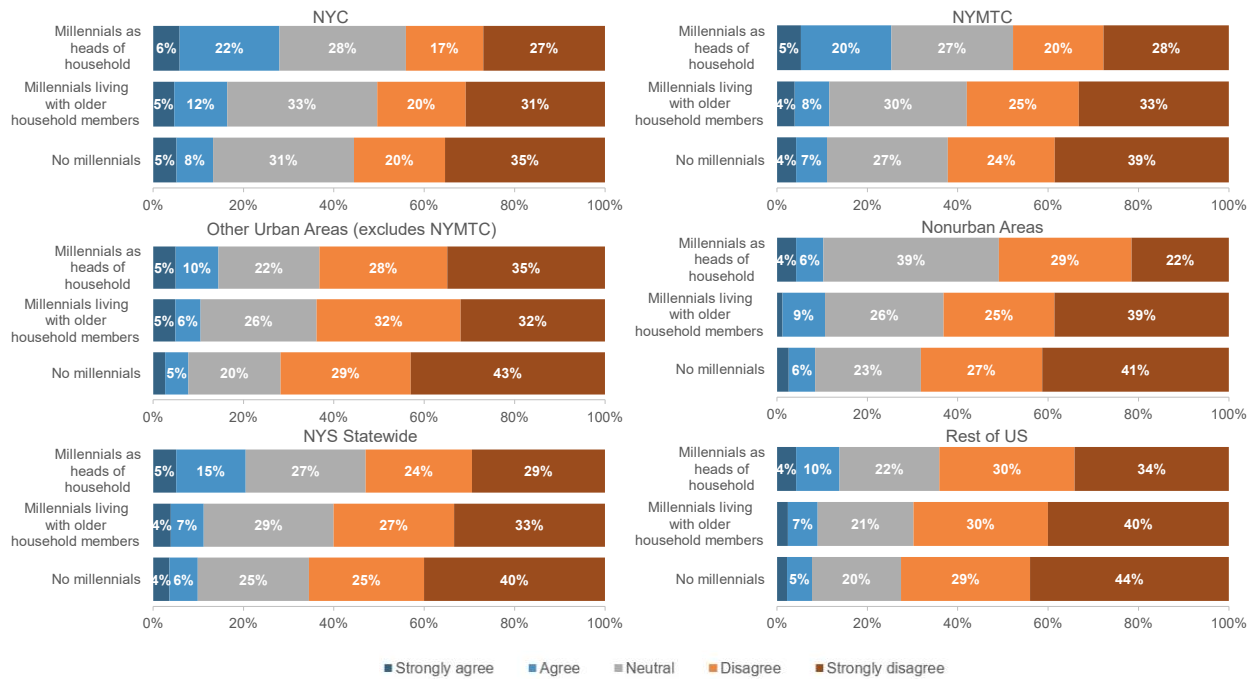


Figure 4-4. Share of ratings by income status, region, and millennial household status in response to the survey statement, "I bike to places to save money" (2017 NHTS data).

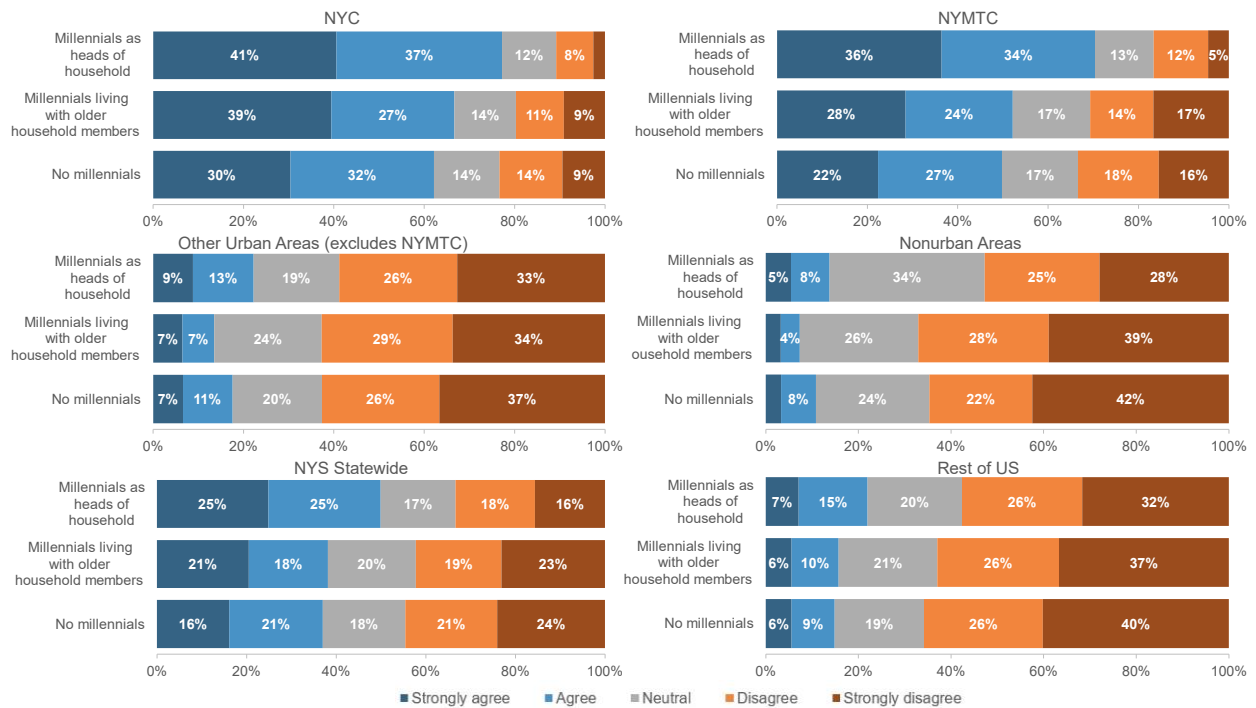


Figure 4-5. Share of ratings by region and millennial household status in response to the survey statement, "I use public transportation to save money" (2017 NHTS data).

4.5 USE PUBLIC TRANSPORTATION TO SAVE MONEY

Compared with other types of households, a higher percentage of households headed by millennials in NYC and NYMTC agree or strongly agree that they used public transportation to save money. In NYC, 78% of millennial-headed households agree or strongly agree with this statement, which is significantly higher than the 62% of nonmillennial-headed households. In general, urban areas in NYS have a higher share of millennial-headed households that agree or strongly agree with this statement, compared with the survey responses from nonurban areas of NYS. In nonurban areas of NYS, only 13% of millennial-headed households agree or strongly agree, a figure lower than the 22% seen across the rest of the United States.

4.6 OVERALL COST CONCERNS

Figure 4-6 compares overall travel cost concerns among households with millennials as head of the household, showing the percentage of respondents who agree or strongly agree with each statement. Outside of NYC and the NYMTC region, the most common concern was the impact of gas prices on travel, with 45% of such households across the rest of the United States expressing this worry. In nonurban areas of NYS, this figure rises to 62%. However, in NYC and the NYMTC region, the top concerns among households without millennials were the high costs of getting around and using public transportation to save money. In NYC, 77% of these households agree or strongly agree with this statement.

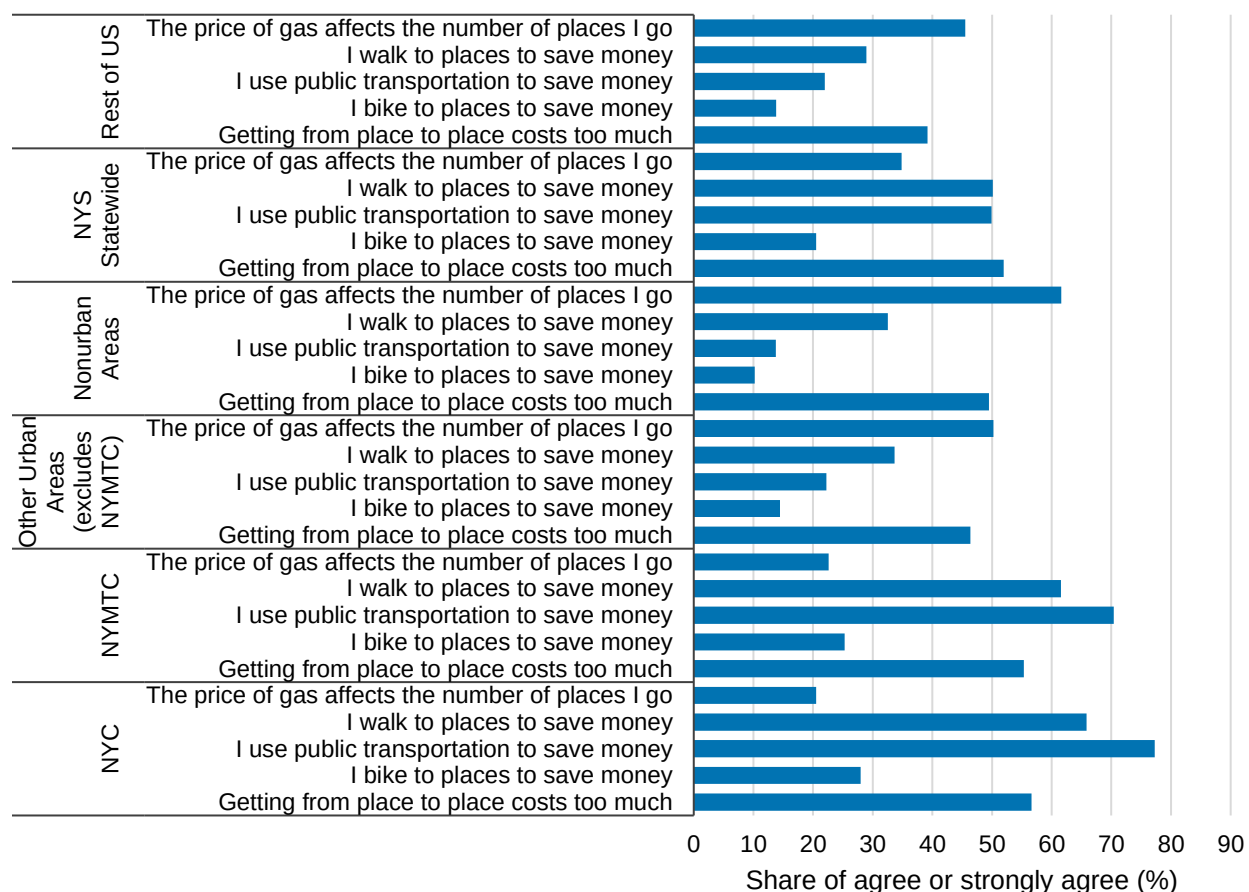


Figure 4-6. Travel cost concerns among millennials as heads of household by region (2017 NHTS data).

5. USE OF TRANSPORTATION TECHNOLOGIES

The 2017 NHTS data provide insight into millennials' behavior in terms of their use of transportation technologies that have the potential to reduce reliance on traditional car travel. Such technologies include online shopping, work from home, rideshares, carshares, and bikeshares. This chapter explores each of these, focusing on millennials' engagement with these technologies and comparing their behaviors with those of other generations across different regions. The analysis in this chapter is based on count responses, where respondents indicated how many times they used these technologies in the past 30 days.

5.1 ONLINE SHOPPING

Millennials display a greater tendency toward online shopping compared with baby boomers and Gen Xers across all regions, as shown in Figure 5-1. In NYC, for instance, only 18% of millennials reported not making online purchases, compared with 55% of baby boomers and 30% of Gen Xers. This lower percentage of nonpurchasers among millennials highlights their stronger adoption of e-commerce. Millennials also stand out in the 6–10 times and 11–30 times categories. In NYC, 17% of millennials made online purchases 6–10 times, whereas 9% made purchases 11–30 times. In comparison, baby boomers show only 8% making 6–10 purchases and 2% making 11–30 purchases in the same region. These figures underscore millennials' greater reliance on online shopping as part of their lifestyle.

NYC shows the highest rate of frequent online shopping among millennials, with a large proportion (56%) making 1–5 purchases. The higher urban density and accessibility to delivery services may make online shopping more convenient in NYC, contributing to these higher engagement rates. In contrast, millennials in NYS nonurban areas have a higher percentage of nonpurchasers (46%) and lower frequent purchase rates (6% for 11–30 purchases). The rest of the US region shows a similar trend to that of NYS nonurban areas, with 35% of millennials reporting no online purchases. However, this region still shows moderate engagement, with 48% making 1–5 purchases and 11% making 6–10 purchases.

Compared with millennials, younger generations (i.e., Gen Z), especially in nonurban areas, show the highest percentage of individuals who did not shop online, such as 63% in NYS nonurban areas. This could reflect limited income or reliance on family-based shopping habits. Across all regions, baby boomers show lower engagement with online shopping, with over 50% in many areas, like NYC and the rest of the US, not making any online purchases. Their frequency of purchases also remains low, with minimal representation in the higher frequency categories (11–30 times). Overall, millennials lead in online shopping frequency, especially in urban areas. Whereas Gen Xers participate in online shopping, millennials show higher engagement in mid-to-high frequency purchase categories, suggesting that e-commerce is more central to their lifestyle. Regional differences indicate that urban millennials are more active online shoppers than their rural counterparts.

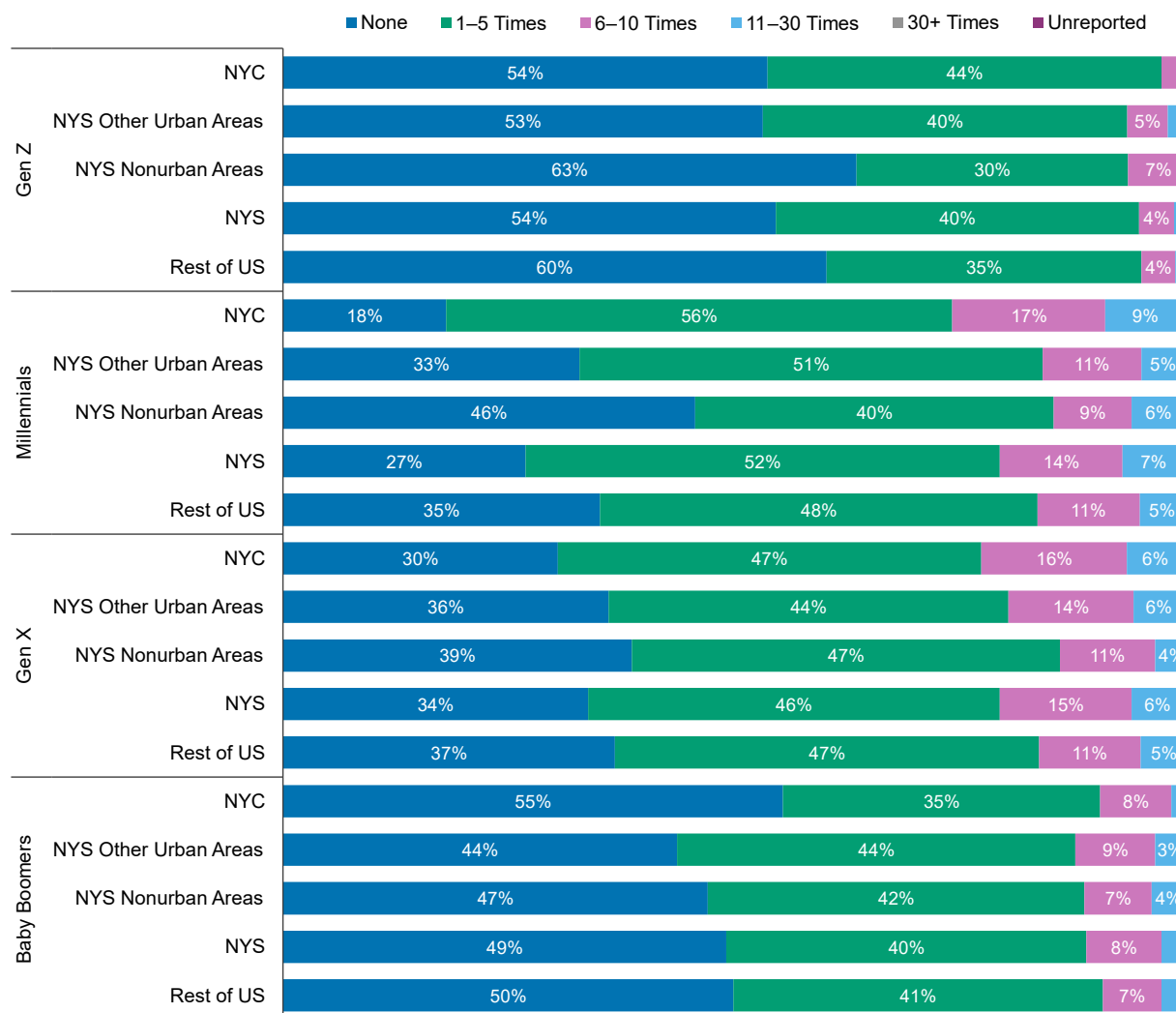


Figure 5-1. Distribution of online purchases by generation and region (2017 NHTS data).

An examination of the gender differences of online purchases between millennials and baby boomers shows that female millennials appear to engage in online shopping slightly more frequently than male millennials, particularly in other urban areas and nonurban areas (Figure 5-2). In the NYS nonurban areas, only 36% of female millennials reported making no online purchases, compared with 54% of males. Additionally, 13% of females shopped online 6–10 times, versus just 5% of males, and 8% of females made 11–30 purchases, compared with 4% of males. In NYC, male and female millennials report similar frequencies of 1–5 and 6–10 online purchases (56% and 17% for both genders), respectively. However, females have a slight edge in the higher frequency (11–30 times) category, with 10% compared with 8% for males. In NYS other urban areas, 27% of female millennials report no online purchases, compared with 38% of males. Females are also more likely to make 6–10 purchases, with 13% compared with 9% for males, indicating that female millennials in these regions might use online shopping more consistently than males.

Both male and female baby boomers show a much higher percentage of nonpurchasers across all regions compared with millennials. In NYC, 58% of male and 53% of female baby boomers did not shop online, a huge contrast to the 18% of millennials who reported no online purchases.

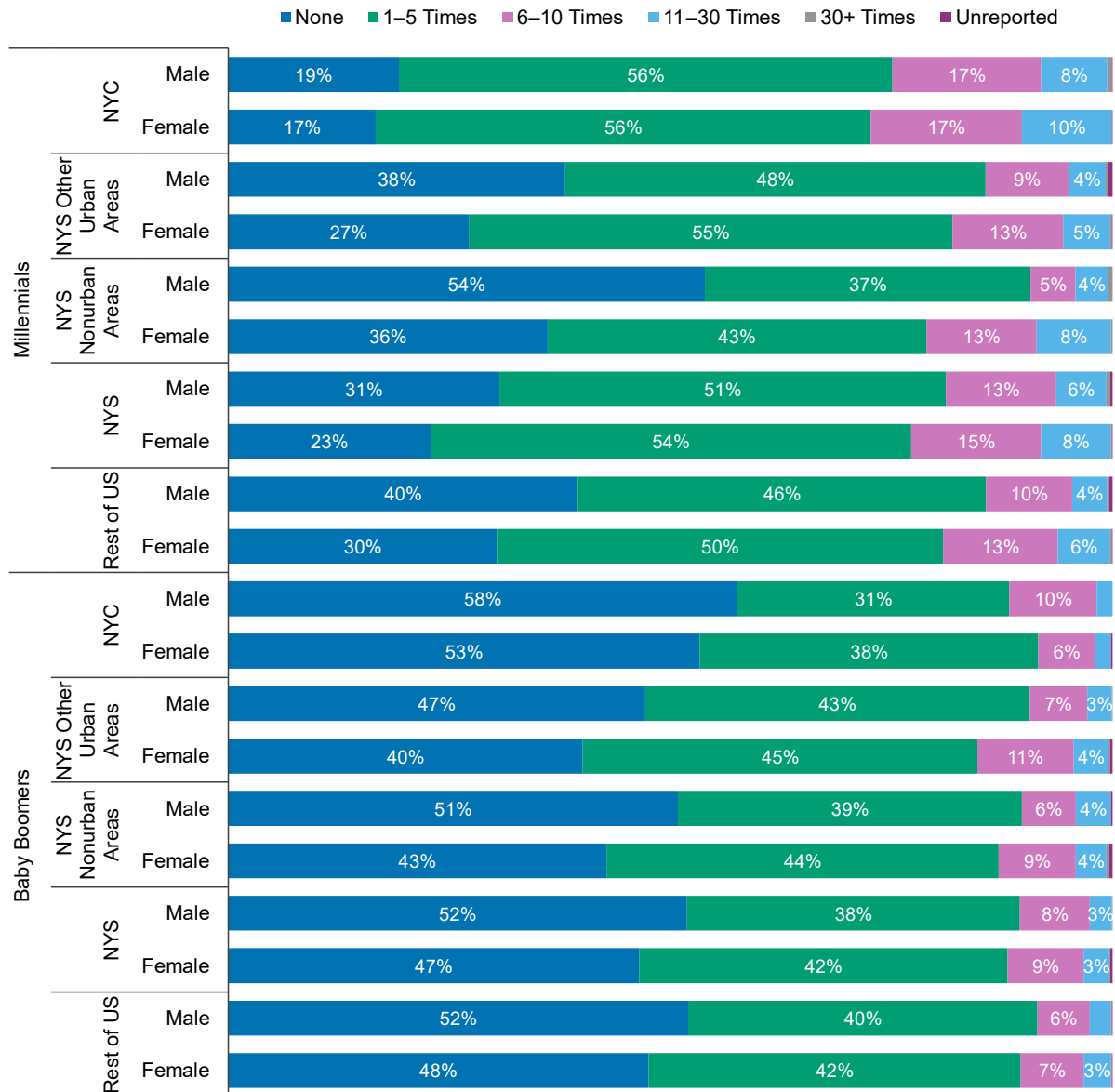


Figure 5-2. Distribution of online purchases by generation, region, and gender (2017 NHTS data).

Further diving into the millennials' online shopping behavior by their household status reveals notable differences in purchasing frequency, especially for millennials who are heads of households compared with those in other household arrangements (Table 5-1). Millennials as heads of household in NYC show a markedly higher engagement in online shopping compared with other household types. Only 13% reported no online purchases, significantly lower than the 48% for households without millennials. Moreover, 56% of millennials as heads of household in NYC made 1–5 purchases, and 19% made 6–10 purchases, indicating a higher frequency of online shopping in this group. Similarly, in NYS other urban areas, only 33% of millennials as heads of household did not engage in online shopping, compared with 45% in households without millennials. In NYS overall, only 23% of millennials as heads of household reported no online purchases, whereas the figure was 47% for households without millennials. This suggests that millennials heading households in urban and suburban settings are more likely to engage in online shopping than households without millennials. In NYS nonurban areas, 43% of millennials as

heads of household did not make online purchases, showing a less pronounced difference from other household types (where around 49%–50% reported no online purchases). However, a notable 10% made 6–10 purchases, and 7% made 11–30 purchases.

Table 5-1. Distribution of online purchases by millennial household status and region (2017 NHTS data).

Millennial household status	Region	None	1–5 times	6–10 times	11–30 times	30+ times	Unreported
No millennials	NYC	48.3%	37.9%	10.1%	3.4%	0.2%	0.2%
	NYS other urban areas	45.4%	40.9%	9.5%	3.5%	0.1%	0.5%
	NYS nonurban areas	50.2%	39.8%	7.2%	2.7%	0.0%	0.1%
	NYS	47.1%	39.5%	9.6%	3.4%	0.1%	0.3%
	Rest of US	48.7%	40.5%	7.4%	3.2%	0.1%	0.2%
Millennials living with older household members	NYC	40.2%	45.2%	10.8%	3.7%	0.0%	0.1%
	NYS other urban areas	40.0%	47.3%	8.5%	3.7%	0.2%	0.3%
	NYS nonurban areas	48.8%	38.2%	8.1%	4.3%	0.4%	0.1%
	NYS	40.8%	45.7%	9.4%	3.7%	0.2%	0.2%
	Rest of US	48.0%	40.6%	7.6%	3.3%	0.2%	0.2%
Millennials as heads of household	NYC	12.7%	56.2%	19.4%	11.4%	0.4%	0.0%
	NYS other urban areas	33.0%	47.4%	13.2%	6.0%	0.2%	0.2%
	NYS nonurban areas	43.0%	40.0%	10.2%	6.6%	0.2%	0.0%
	NYS	23.2%	51.3%	16.2%	8.9%	0.3%	0.1%
	Rest of US	29.8%	51.1%	13.0%	5.6%	0.2%	0.2%

5.2 WORK FROM HOME

5.2.1 Usually Work from Home

According to 2017 NHTS data, millennials in NYC and the rest of the US show slightly higher rates of working from home compared with other areas, as shown in Figure 5-3. In NYC, 8% of millennials reported usually working from home, similar to the 8% in the rest of the US. This is higher than Gen Zers and lower than Gen Xers and baby boomers. In NYS nonurban areas, only 5% of millennials reported working from home, which is slightly lower than the rates seen in more urbanized areas. This aligns with the broader trend seen across generations, where nonurban areas generally show lower rates of working from home.

Baby boomers and Gen Xers exhibit generally higher rates of work from home behavior across most regions. For example, baby boomers in the rest of the US show a substantial 16% working from home, the highest across all generations and regions. Similarly, Gen Xers in the rest of the US and NYC also report higher work from home rates (13% and 12%, respectively) than those of millennials. Millennials' work from home shares are relatively close to those of Gen Zers, especially in nonurban areas. For instance, in NYS nonurban areas, both millennials and Gen Zers show low work from home rates (5% and 3%, respectively).

Overall, millennials display moderate work from home engagement, particularly in urban regions like NYC. However, their rates remain lower than those of baby boomers and Gen Xers, likely because of differences in job roles and industry presence. Urban areas generally facilitate higher work from home

rates across all generations, with NYC serving as a notable center for remote work, especially for older generations.

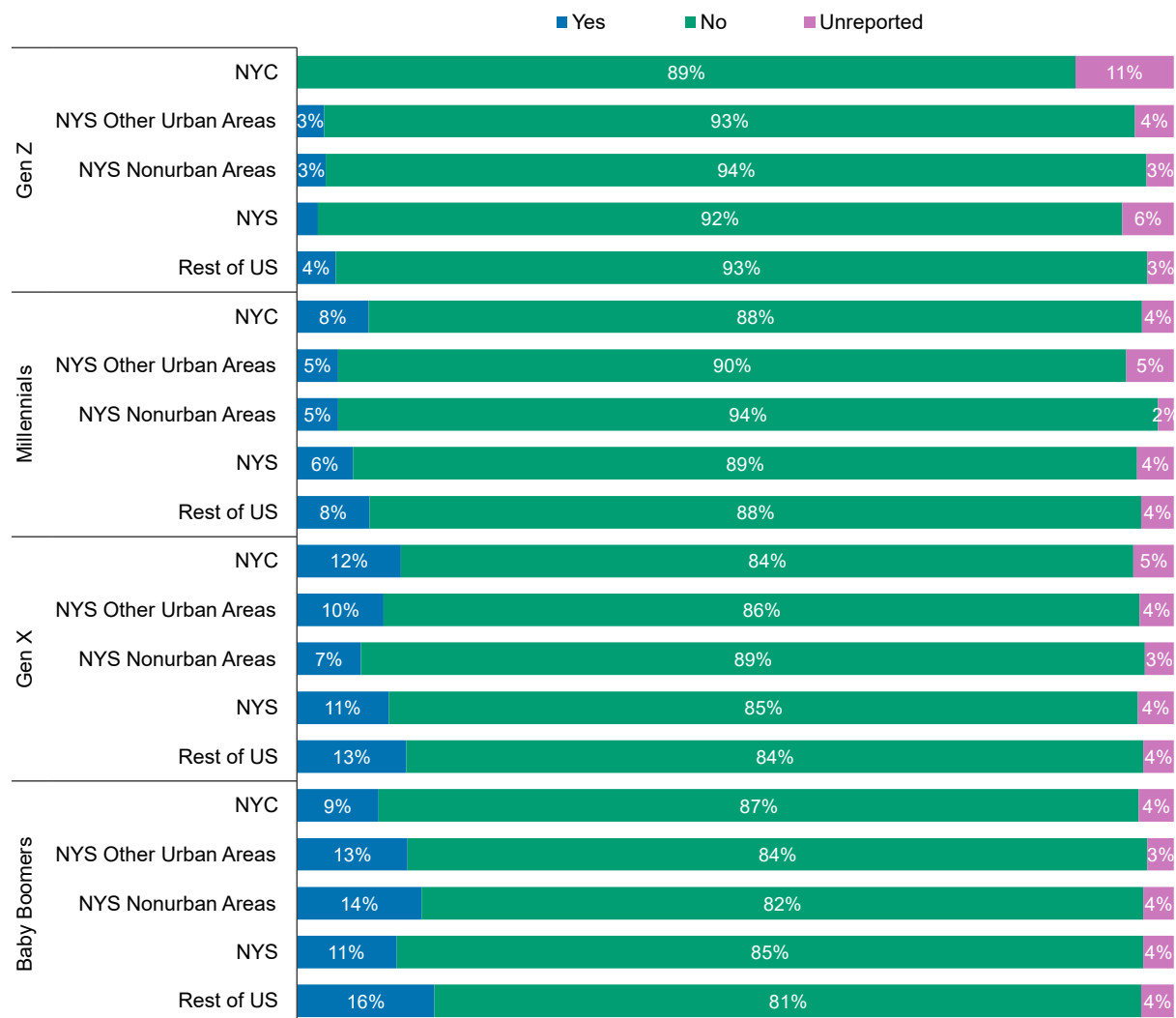


Figure 5-3. Distribution of work from home by generation and region (2017 NHTS data).

Figure 5-4 shows the distribution of work from home rates among millennials based on their household status in the 2017 NHTS data, providing insight into how household members' generations may influence the likelihood of working from home. Millennials who are heads of household in NYC show a work-from-home share of 11%, notably higher than that of millennials living with older household members (6%) in the same region. This might indicate that those heading their own households in urban areas have more flexibility in their roles, enabling them to work from home more frequently.

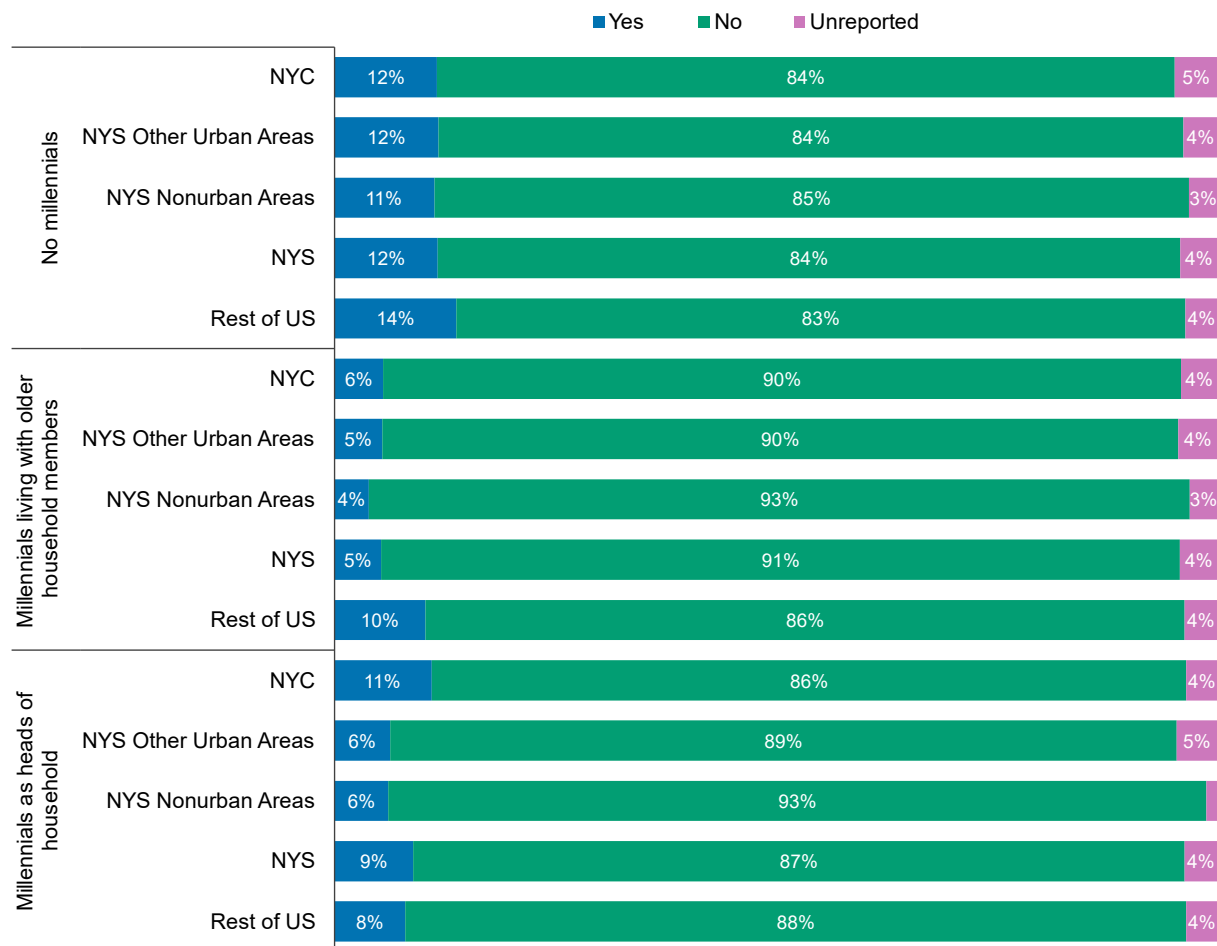


Figure 5-4. Distribution of work from home by millennial household status and region (2017 NHTS data).

5.2.2 Options for Working from Home

The data on whether workers have the option to work from home, even if they are not currently doing so, reveal several generational and regional variations, particularly among millennials (Figure 5-5). In NYC, 22% of millennials report having the option to work from home, a higher rate compared with those of both baby boomers (18%) and Gen Zers (9%) in the same region. Outside of NYC, millennials' access to work from home options declines but remains notable in NYS (17%) overall and the rest of the US (16%). These rates are lower than those for Gen Xers, who consistently report higher work from home options across regions. For instance, in NYS other urban areas, 21% of Gen Xers have this option, compared with 13% of millennials.

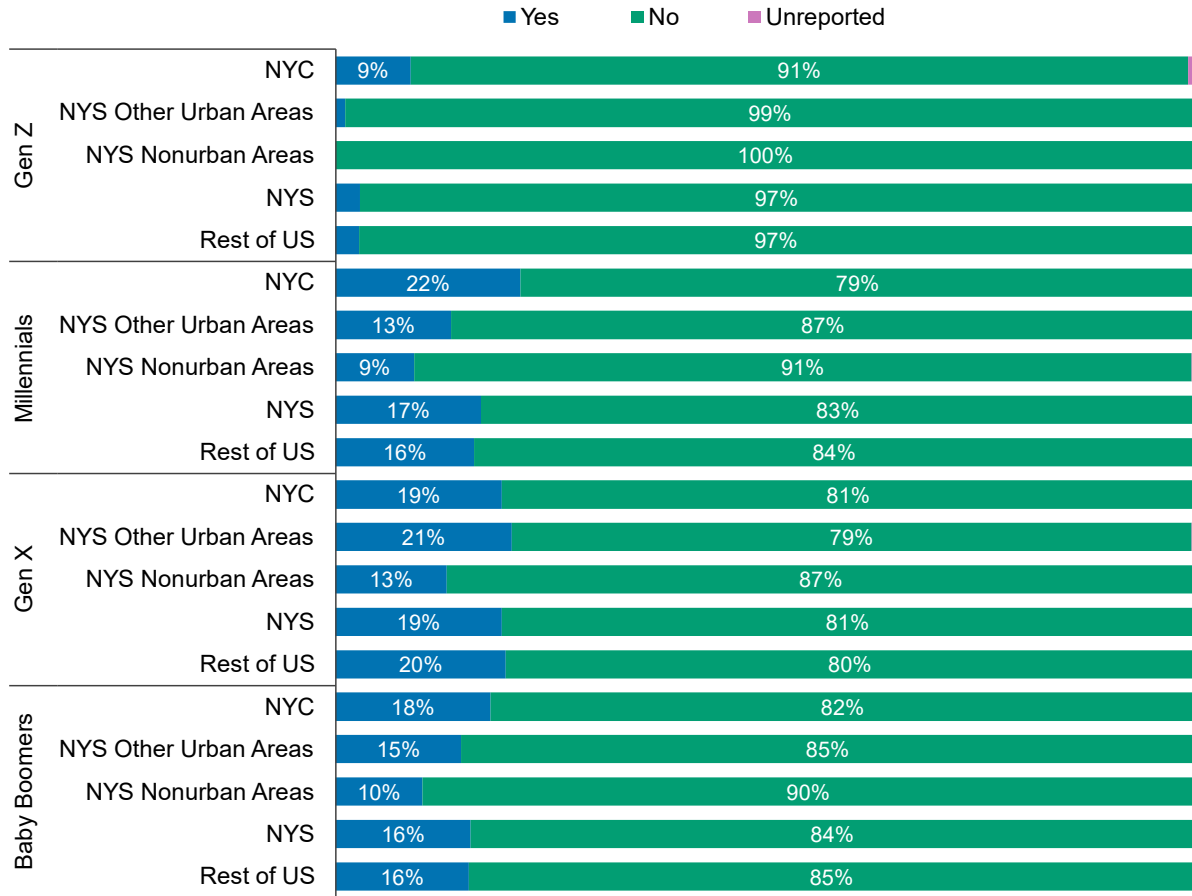


Figure 5-5. Distribution of option to work from home by generation and region (2017 NHTS data).

Table 5-2 highlights the gender-based differences in work-from-home flexibility across generations and regions. Among millennials, males in NYC report the highest percentage (23%) of having the option to work from home, slightly higher than that of their female counterparts (19%). Across regions, millennial males generally have more consistent work-from-home options than females have, though the rates vary by region. For instance, in NYS other urban areas, 17% of millennial males have this option, compared with only 10% of females. Millennial females in NYS nonurban areas have particularly limited access to work-from-home options, with only 7% reporting this option, compared with 11% of their male counterparts.

Among Gen Xers, both males and females in NYC and NYS other urban areas report relatively high work-from-home options of around 21% for males and 19% for females. For baby boomers, a notable gender difference exists in NYC, where 25% of males have work-from-home options, compared with only 12% of females.

Table 5-2. Distribution of option to work from home by generation, region, and gender (2017 NHTS data).

Generation	Region	Gender	Yes	No	Unreported
Millennials	NYC	Male	23%	77%	0%
		Female	19%	81%	0%
	NYS other urban areas	Male	17%	83%	0%
		Female	10%	90%	0%
	NYS nonurban areas	Male	11%	89%	0%
		Female	7%	93%	1%
	NYS	Male	19%	81%	0%
		Female	14%	86%	0%
	Rest of US	Male	17%	83%	0%
		Female	15%	85%	0%
Gen X	NYC	Male	21%	79%	0%
		Female	18%	82%	0%
	NYS other urban areas	Male	21%	79%	0%
		Female	19%	80%	1%
	NYS nonurban areas	Male	14%	86%	0%
		Female	11%	89%	0%
	NYS	Male	20%	80%	0%
		Female	18%	82%	0%
	Rest of US	Male	23%	77%	0%
		Female	16%	84%	0%
Baby boomers	NYC	Male	25%	75%	0%
		Female	12%	88%	0%
	NYS other urban areas	Male	18%	82%	0%
		Female	11%	89%	0%
	NYS nonurban areas	Male	12%	88%	0%
		Female	11%	89%	0%
	NYS	Male	20%	80%	0%
		Female	11%	89%	0%
	Rest of US	Male	18%	82%	0%
		Female	13%	88%	0%

Note: Percentages may not total 100 due to rounding.

5.2.3 Frequency of Work from Home

Figure 5-6 illustrates the count of days working from home when the participant is a worker and when teleworking is an option. Millennials in NYS nonurban areas have the highest frequency of working from home among their peers, with 18% working from home 10+ times and 30% working from 6–10 times in the past 30 days. In urban areas, especially in NYC, 66% of millennials report working from home 1–5 times, whereas only 4% work from home 10+ times.

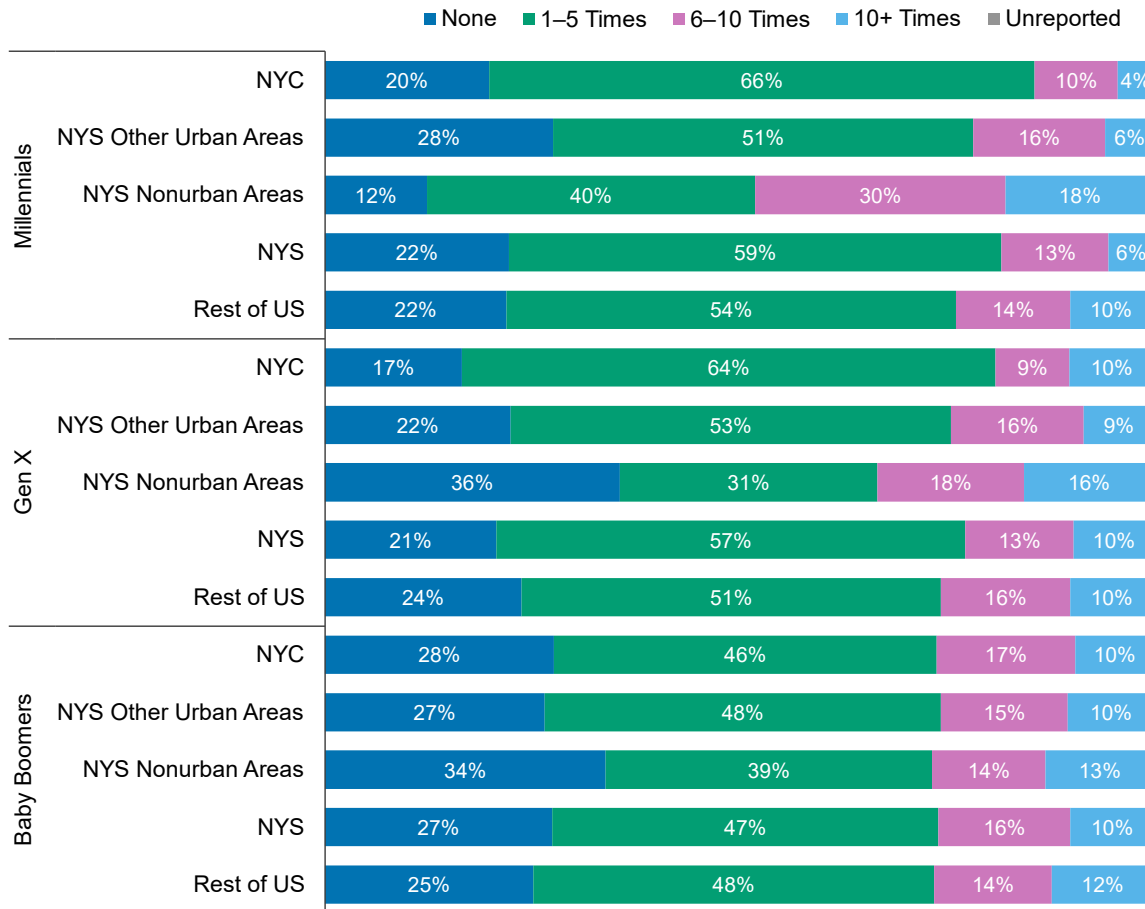


Figure 5-6. Distribution of count of days worked from home by generation and region (2017 NHTS data).

Additionally, the distribution of work from home days among millennials reveals some nuanced gender and regional differences, especially when comparing them with baby boomers (Table 5-3). Among millennials, males in nonurban areas such as NYS nonurban areas, tend to work from home at higher frequencies than that of females. Specifically, 40% of male millennials in NYS nonurban areas work from home 6–10 times, and 23% work from home 10+ times. In contrast, female millennials in the same region show a greater concentration at lower frequencies, with 71% working from home 1–5 times and only 7% working from home 10+ times. In NYC, the distribution is somewhat more balanced between genders, with both male and female millennials predominantly working from home 1–5 times per month (65% for males and 66% for females). However, the number of females working from home 10+ times in NYC (4%) is lower than that of their male counterparts (5%).

The gender difference in work-from-home frequency is more pronounced among millennials than among baby boomers, especially in nonurban areas. For example, in NYS nonurban areas, 23% of male millennials work from home 10+ times, compared with only 7% of female millennials. Among baby boomers in the same region, this difference is smaller, with 13% of males and 14% of females working from home 10+ times, showing almost no gender gap for this older generation. In urban areas like NYC, both millennials and baby boomers show smaller gender differences in work-from-home frequency. For instance, 5% of millennial males and 4% of millennial females work from home 10+ times. Among baby

boomers in NYC, 12% of males and 4% of females report working from home 10+ times, suggesting a larger gender gap for baby boomers in urban areas than millennials.

Millennials display a larger gender gap in work-from-home frequency, especially in nonurban regions, where males tend to work from home more frequently than females do. This trend contrasts with baby boomers, who show more balanced work-from-home behavior between genders, with minimal differences in nonurban areas and a slightly larger gap in urban settings.

Table 5-3. Distribution of count of days worked from home by generation, region, and gender (2017 NHTS data).

Generation	Region	Gender	None	1–5 times	6–10 times	10+ times
Millennials	NYC	Male	23%	65%	7%	5%
		Female	16%	66%	14%	4%
	NYS other urban areas	Male	32%	52%	10%	6%
		Female	19%	49%	27%	5%
	NYS nonurban areas	Male	11%	26%	40%	23%
		Female	15%	71%	7%	7%
	NYS	Male	26%	58%	10%	6%
		Female	17%	61%	18%	5%
	Rest of US	Male	24%	51%	14%	10%
		Female	19%	58%	14%	9%
Baby boomers	NYC	Male	31%	44%	13%	12%
		Female	21%	51%	24%	4%
	NYS other urban areas	Male	29%	46%	14%	10%
		Female	21%	51%	17%	11%
	NYS nonurban areas	Male	34%	40%	14%	13%
		Female	34%	39%	13%	14%
	NYS	Male	31%	45%	14%	11%
		Female	22%	50%	20%	8%
	Rest of US	Male	26%	46%	15%	12%
		Female	23%	52%	13%	13%

Note: Percentages may not total 100 due to rounding.

When examining the millennial household differences, interesting patterns emerge, as shown in Figure 5-7. Millennials as heads of household generally show higher rates of working from home compared with those of millennials living with older household members and households without millennials. For instance, in NYC, 69% of millennials as heads of household worked from home 1–5 times per month, with 9% working from home 6–10 times. This frequency is slightly higher than that of households without millennials, where 55% reported working from home 1–5 times and only 14% worked from home 6–10 times. In NYS nonurban areas, 28% of millennials as heads of household worked from home 6–10 times and 17% worked from home 10+ times per month. This contrasts with millennials living with older household members, where the majority (33%) worked from home only 1–5 times, and those working 10+ times were fewer (8%).

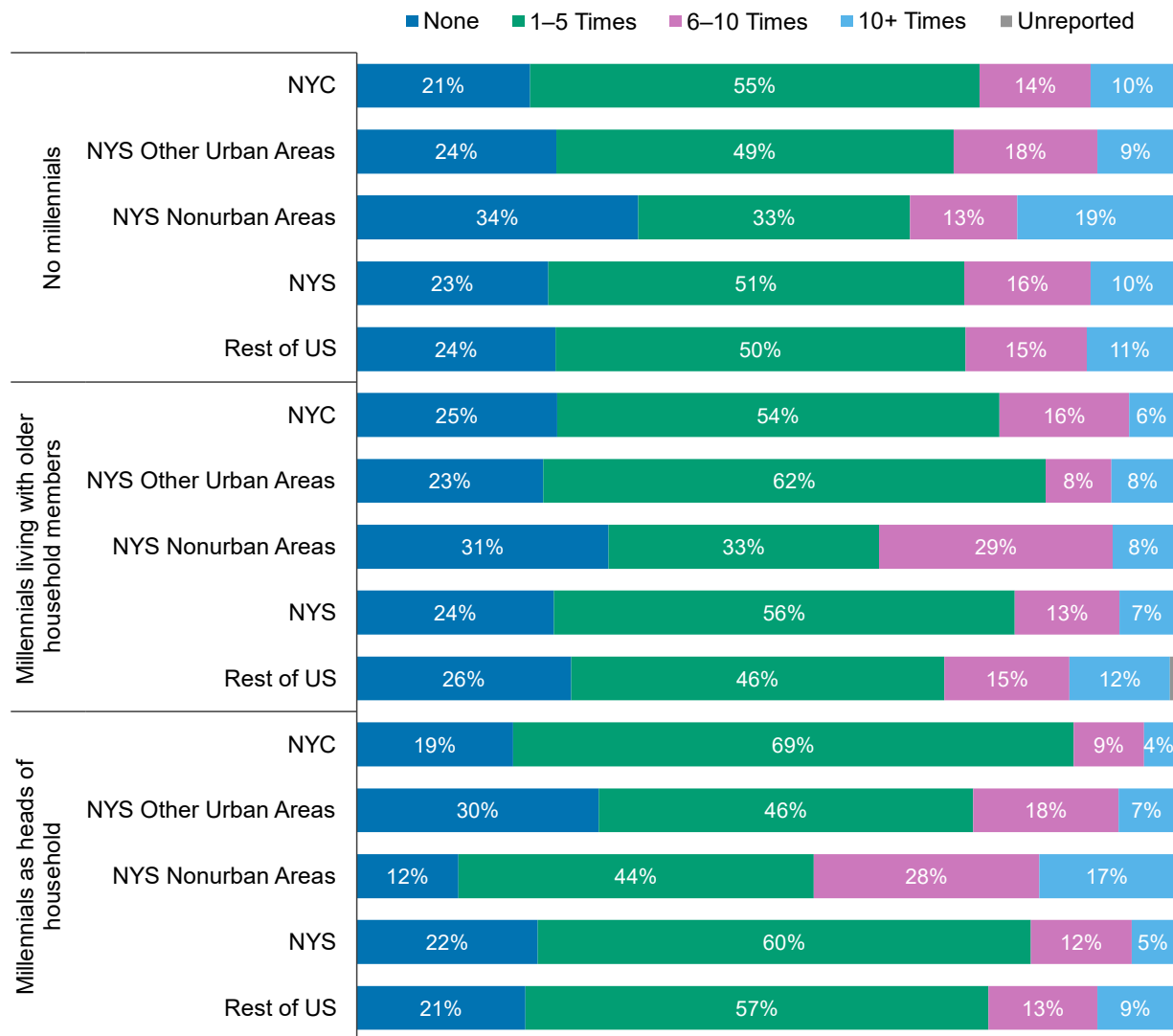


Figure 5-7. Distribution of count of days worked from home by millennial household status and region (2017 NHTS data).

However, examining gender differences in households with millennials as heads of the household reveals some interesting patterns (Table 5-4). In NYC, for example, 74% of female millennials worked from home 1–5 times per month compared with 64% of male millennials. However, a higher proportion of male millennials (9%) worked from home 6–10 times, compared with 8% for females. This pattern is consistent across several regions, with female millennials generally working from home more frequently in lower frequency categories (1–5 times) but fewer reaching the higher frequency ranges (10+ times). In NYS nonurban areas, a substantial difference emerges: 81% of female millennials as heads of household worked from home 1–5 times, whereas only 6% worked from 6–10 times and 2% 10+ times. In contrast, 36% of male millennials worked from home 6–10 times and 22% 10+ times, showing a higher frequency of remote work among males in this region.

Table 5-4. Distribution of count of days worked from home by millennials as heads of household, region, and gender (2017 NHTS data).

Household status	Region	Gender	None	1–5 times	6–10 times	10+ times
Millennials as heads of household	NYC	Male	24%	64%	9%	3%
		Female	13%	74%	8%	5%
	NYS other urban areas	Male	34%	48%	11%	7%
		Female	22%	41%	31%	6%
	NYS nonurban areas	Male	13%	30%	36%	22%
		Female	12%	81%	6%	2%
	NYS	Male	27%	57%	11%	5%
		Female	16%	65%	14%	5%
	Rest of US	Male	23%	55%	14%	9%
		Female	18%	60%	13%	10%

Note: Percentages may not total 100 due to rounding.

5.3 RIDESHARE

In Table 5-5, rideshare app usage across generations reveals notable generational and regional variations in the adoption of this technology. Millennials show a higher engagement with rideshare services compared with that of other generations, especially in urban areas like NYC. In NYC, only 52% of millennials reported never using a rideshare app, compared with significantly higher nonusage rates for other generations (82% for Gen Zers, 75% for Gen Xers, and 91% for baby boomers). Furthermore, 33% of millennials in NYC used rideshare apps 1–5 times, and a notable 11% reported using them 6–10 times, with 5% using them over 10 times. This frequency of use indicates that millennials are more likely to rely on rideshare apps regularly, reflecting a preference for flexible, on-demand transportation options in dense urban areas. In contrast, outside NYC, millennials' usage drops but still remains higher than that of other generations. In the rest of the US, 82% of millennials reported no usage, which is still lower than the 93% for Gen Zers and 90% for Gen Xers. In NYS overall, 20% of millennials reported using rideshare apps 1–5 times, with 6% using them 6–10 times and 2% using them 10+ times.

Millennials stand out as the primary adopters of rideshare technology, particularly in urban settings where services are readily available and convenient. In contrast, older generations, especially baby boomers, display limited engagement, with usage concentrated in major urban centers and almost nonexistent in rural areas. This pattern highlights the role of both generational preferences and geographic factors in the adoption of transportation technologies like rideshare services.

Millennials by household status reveals distinct patterns based on household composition, with millennials as heads of household showing significantly higher usage compared with those living with older household members or in households without millennials, as shown in Figure 5-8. Millennials as heads of household show a strong tendency to use rideshare services, particularly in urban areas like NYC. Only 37% of millennials in this group reported no rideshare usage in NYC, whereas a substantial 41% used it 1–5 times, 15% used it 6–10 times, and 8% used it 10+ times. In other areas, whereas the overall usage drops, millennials as heads of household still report higher adoption rates compared with those of other household types. For instance, in NYS overall, 63% reported no rideshare use, but 25% used it 1–5 times, 8% used it 6–10 times, and 4% used it 10+ times. In the rest of the US, 78% reported no rideshare usage, leaving 22% who use rideshare services at varying frequencies.

Table 5-5. Distribution of rideshare app usage by generation and region (2017 NHTS data).

Generation	Region	None	1–5 times	6–10 times	10+ times
Gen Z	NYC	82%	14%	2%	2%
	NYS other urban areas	94%	4%	1%	0%
	NYS nonurban areas	99%	1%	0%	0%
	NYS	91%	7%	1%	1%
	Rest of US	93%	6%	1%	1%
Millennials	NYC	52%	33%	11%	5%
	NYS other urban areas	90%	9%	1%	0%
	NYS nonurban areas	97%	2%	0%	0%
	NYS	72%	20%	6%	2%
	Rest of US	82%	14%	3%	1%
Gen X	NYC	75%	19%	4%	2%
	NYS other urban areas	93%	6%	1%	0%
	NYS nonurban areas	98%	2%	0%	0%
	NYS	85%	11%	2%	1%
	Rest of US	90%	8%	1%	1%
Baby boomers	NYC	91%	7%	1%	0%
	NYS other urban areas	97%	3%	0%	0%
	NYS nonurban areas	99%	1%	0%	0%
	NYS	95%	4%	1%	0%
	Rest of US	96%	4%	0%	0%

Note: Percentages may not total 100 due to rounding.

In contrast, millennials living with older household members and households without millennials show a far lower reliance on rideshares. For example, in NYC, 82% of millennials living with older household members reported no rideshare usage, with only 15% using it 1–5 times, and a minimal 3% and 1% using it 6–10 and 10+ times, respectively. Similarly, in NYS other urban areas, 94% of millennials in multigenerational households reported no rideshare usage, whereas 5% used it 1–5 times.

Additionally, some notable gender differences are present among millennials as heads of household across urban and nonurban regions, as shown in Figure 5-9. Millennials as heads of households in urban areas like NYC are much more frequent users of rideshare services, with females in NYC showing a higher propensity for frequent rideshare use (6–10 times and more) compared with males. In contrast, in less urbanized areas, both genders show low adoption, with males in nonurban areas slightly more likely to use rideshare services than females, although overall usage remains limited.

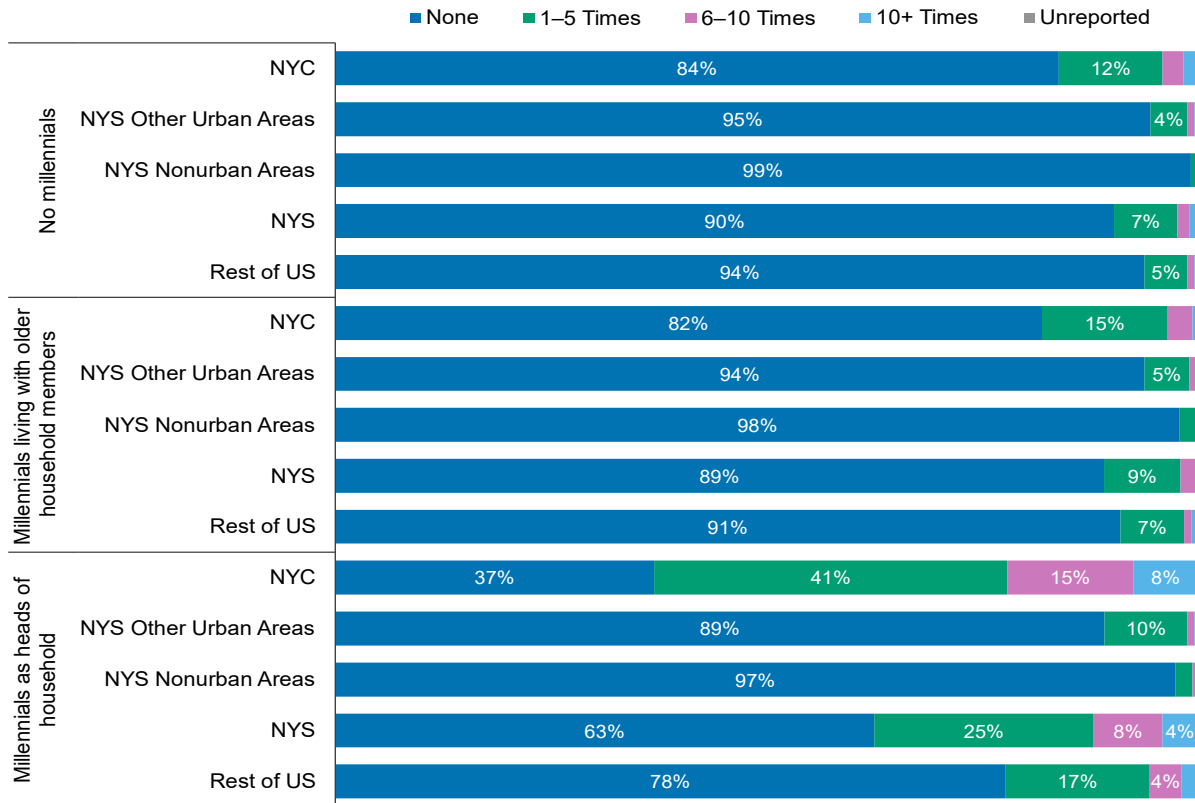


Figure 5-8. Distribution of rideshare app usage by millennial household status and region (2017 NHTS data).

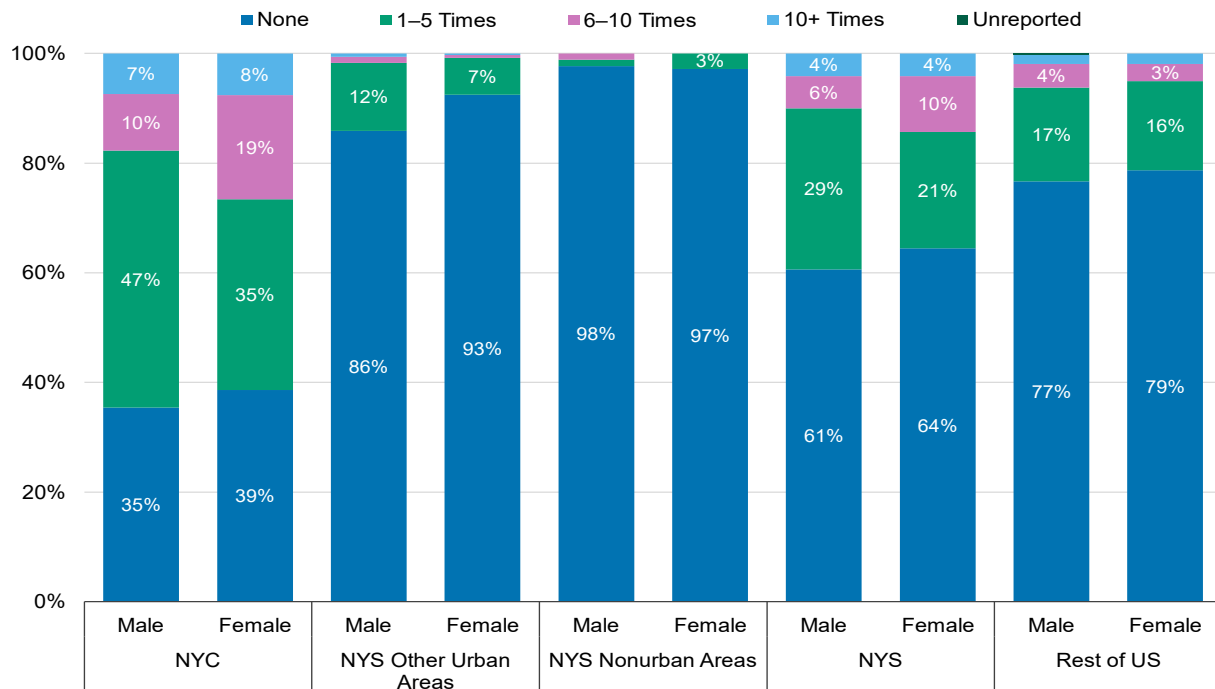


Figure 5-9. Distribution of rideshare app usage by region and the gender of millennials as heads of household (2017 NHTS data).

5.4 CARSHARE

Carsharing services allow users to rent vehicles on a short-term basis. The distribution of carshare usage from the 2017 NHTS data in Table 5-6 reveals low adoption rates among both millennials and baby boomers, though millennials show slightly higher engagement with carsharing services compared with that of baby boomers across most regions. For example, in NYC, 6% of millennials reported using carsharing services at least once, with 3% using it only one time and another 3% using it two or more times in the past 30 days. This represents the highest engagement level among millennials in any region. For millennials in other regions, carshare usage drops significantly. These findings suggest that carshare services are predominantly used by millennials in urban areas like NYC, where carshare infrastructure is likely more accessible and the density of users supports such services. Outside major urban centers, carshare services are rarely used by millennials, potentially owing to a lack of availability or reduced necessity given alternative transportation modes or personal vehicle ownership. Carshare usage among baby boomers is even lower than that of millennials in all regions. For instance, in NYC, 98% of baby boomers reported no usage, with 2% using carshare once and only 1% using it more than once.

Table 5-6. Distribution of carshare usage by millennials and baby boomers across regions (2017 NHTS data).

Generation	Region	None	One time	Two or more times
Millennials	NYC	94%	3%	3%
	NYS other urban areas	99%	1%	0%
	NYS nonurban areas	99%	1%	0%
	NYS	97%	2%	1%
	Rest of US	99%	1%	1%
Baby boomers	NYC	98%	2%	1%
	NYS other urban areas	99%	1%	0%
	NYS nonurban areas	100%	0%	0%
	NYS	99%	1%	0%
	Rest of US	100%	0%	0%

Note: Percentages may not total 100 due to rounding.

5.5 BIKESHARE

Bikeshare, a mode of transportation that has gained traction in urban areas, is especially popular among millennials, who show the second-highest bikeshare adoption rates in NYC. Figure 5-10 illustrates millennials' bikeshare behavior, revealing that in NYC, 24% of millennials used bikeshare two or more times, with an additional 5% using it once in the past 30 days. This 29% engagement rate is one of the highest across generations and regions, indicating that millennials in NYC are particularly inclined toward bikeshare. Outside of NYC, bikeshare usage drops significantly for millennials. In NYS other urban areas, only 2% of millennials reported using bikeshare two or more times, and 4% used it once, indicating limited adoption. Similarly, in NYS nonurban areas, 13% of millennials used bikeshare once, with no reports of multiple uses, which likely reflects restricted bikeshare access or lower demand in less densely populated regions.

Gen Z shows comparatively low bikeshare usage across all regions. In NYC, only 9% of Gen Zers used bikeshare two or more times, and 2% used it once, compared with millennials' 24% and 5%, respectively. This suggests that, whereas Gen Zers may be open to emerging technologies, their bikeshare adoption lags behind millennials. Gen Xers, however, demonstrate high bikeshare engagement in NYC, surpassing millennials with 38% reporting usage two or more times—the highest percentage across all generations in any region. Outside NYC, Gen X usage decreases significantly, with only 5% using bikeshare two or

more times in NYS other urban areas and 1% in NYS nonurban areas. Unlike millennials, who show slightly more engagement in suburban and rural areas, Gen X bikeshare usage is more polarized, peaking in NYC but tapering off elsewhere. Baby boomers report bikeshare usage similar to that of millennials in NYC. However, outside NYC, bikeshare adoption by baby boomers is minimal, with most regions showing over 95% reporting no usage.

In summary, NYC emerges as a hub for bikeshare usage across all generations. Millennials show the highest bikeshare adoption rates outside of Gen Xers in NYC, particularly in urban settings with accessible infrastructure. Compared with Gen Zers and baby boomers, millennials demonstrate a more consistent, albeit limited, adoption in suburban and nonurban areas. These trends suggest that millennials are relatively more inclined toward bikeshare across various settings, though usage could be heavily influenced by urban density and the availability of bikesharing services.

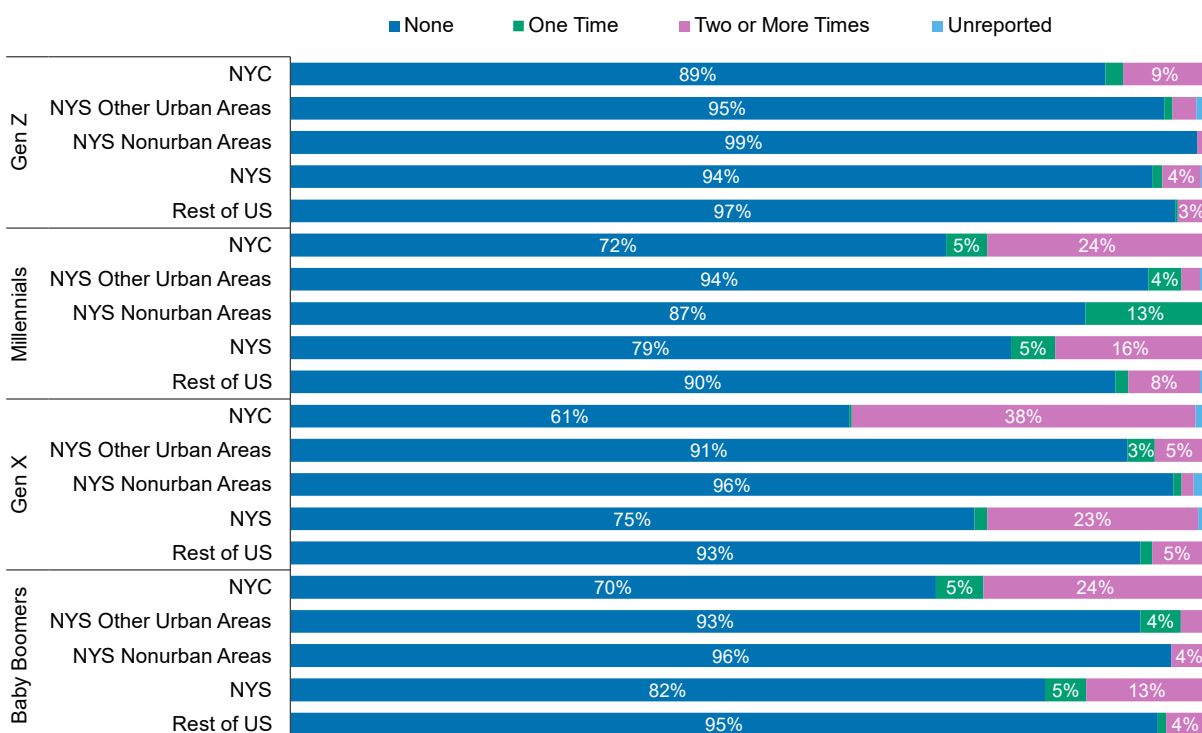


Figure 5-10. Distribution of bikeshare usage by generation and region (2017 NHTS data).

6. MILLENNIALS BEHAVIOR DURING COVID-19

The COVID-19 pandemic brought unprecedented shifts in daily routines, reshaping how people work, shop, and travel. Millennials, as one of the most adaptable generations in terms of technology use and lifestyle changes, demonstrated notable adjustments in their behavior during the pandemic. This chapter uses 2022 NHTS data to explore key aspects of millennials' behavior during COVID-19, including online shopping habits, public transit usage, and changes in commuting patterns to physical work locations. By examining these behaviors, we aim to understand the extent to which the pandemic influenced millennials' mobility choices and whether these shifts were temporary or have become lasting adaptations.

6.1 ONLINE SHOPPING

Figure 6-1 shows millennials' online shopping behavior during COVID-19 from the 2022 NHTS data. Millennials show notable patterns in the Mid-Atlantic region, where 39% reported that they ordered for delivery "more often than before," compared with pre-COVID levels. This is lower than the number of Gen Xers (45%) and baby boomers (43%) in the same region but higher than that of Gen Zers (28%), indicating that millennials in the Mid-Atlantic had a moderate increase in online shopping, though not as pronounced as Gen Xers. In the rest of the US, millennials' behavior was similar but slightly more active, with 41% indicating they ordered for delivery more frequently than before, which aligns with Gen Xers (41%) and baby boomers (43%). Gen Zers in the rest of the US also saw an increase, with 32% ordering for delivery more often than before, though they still showed lower engagement compared with the older generations.

In terms of maintaining the same delivery frequency, millennials in both regions report similar levels, with 49% in the Mid-Atlantic and 48% in the rest of the US saying their delivery habits had not changed from pre-COVID levels. This rate is consistent with other generations, showing a general stability in online shopping patterns among the majority of participants, regardless of region. When looking at those who reported ordering less often, millennials' rates were low across both regions (4% in the Mid-Atlantic and 5% in the rest of the US), suggesting minimal decline in delivery frequency.

Overall, whereas some regional and generational variations occurred in online shopping behavior during COVID-19, millennials in both the Mid-Atlantic and the rest of the US showed a trend toward increased or consistent online shopping and delivery usage, falling in line with Gen Xers and baby boomers in their usage patterns. Gen Zers, while increasing their online shopping, did so at a slightly lower rate. These patterns indicate that millennials adapted well to online shopping during COVID, but with a slightly lower surge compared with that of Gen Xers, especially in the Mid-Atlantic region, which includes NYS.

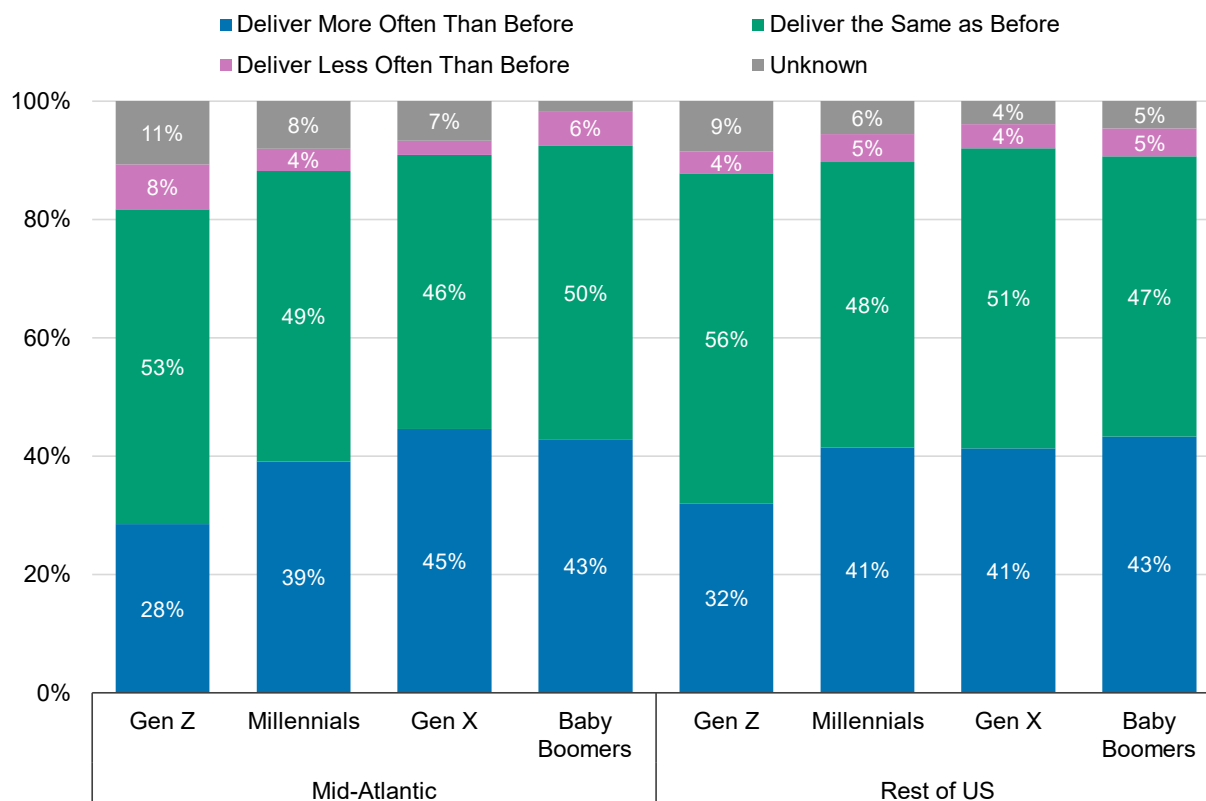


Figure 6-1. COVID effects on online purchases for home delivery by generation (2022 NHTS data).

Examining gender differences in millennials' online shopping behavior during COVID-19 reveals notable variations in delivery frequency (Table 6-1). In the Mid-Atlantic region, a significant 47% of millennial females reported ordering for delivery more often than before, compared with only 30% of millennial males. This trend suggests that female millennials in the Mid-Atlantic region were more inclined toward increased online shopping during the pandemic. Conversely, a larger proportion of millennial males (60%) reported maintaining the same delivery frequency, which indicates a more stable behavior pattern with less change from pre-COVID habits.

This gender-based difference is also evident, albeit to a lesser extent, in the rest of the US. Here, 47% of millennial females indicated they were ordering for delivery more frequently, whereas only 35% of millennial males reported the same. Again, a larger percentage of males (54%) stated that their online shopping frequency had not changed, compared with 43% of females. These findings suggest that, across regions, female millennials were more likely to increase their online shopping during COVID-19, whereas male millennials were more likely to keep their behaviors consistent with pre-COVID levels.

Comparing millennials to other generations reveals a similar pattern. Among Gen Xers in the Mid-Atlantic region, 49% of females reported ordering more frequently, compared with 41% of males. In the rest of the US, Gen X females (46%) also outpaced their male counterparts (36%) in increased online shopping frequency. Baby boomers followed this trend as well, with 49% of females in the Mid-Atlantic region reporting an increase, compared with 36% of males. This pattern suggests that across generations, female respondents were more likely than males to increase their delivery frequency during the pandemic.

Table 6-1. COVID effects on online purchases for home delivery by generation and gender (2022 NHTS data).

Region	Generation	Gender	Deliver more often than before	Deliver the same as before	Deliver less often than before	Unknown
Mid-Atlantic	Millennials	Female	47%	41%	3%	9%
		Male	30%	60%	5%	5%
	Gen X	Female	49%	38%	4%	9%
		Male	41%	54%	1%	4%
	Baby boomers	Female	49%	44%	5%	2%
		Male	36%	57%	7%	1%
Rest of US	Millennials	Female	47%	43%	6%	5%
		Male	35%	54%	4%	7%
	Gen X	Female	46%	47%	4%	4%
		Male	36%	55%	4%	4%
	Baby boomers	Female	47%	43%	5%	5%
		Male	40%	52%	4%	4%

Note: Percentages may not total 100 due to rounding.

Examining further the effect of COVID-19 on online purchases for home delivery, the 2022 NHTS data highlight generational and regional variations in the permanence of these behavioral changes (Figure 6-2). Millennials demonstrate a strong and likely lasting shift toward online shopping for delivery, with permanence levels slightly lower than those of Gen Xers. Compared with Gen Zers and baby boomers, whose changes appear more temporary, millennials' shift to online shopping shows a seemingly permanent trend.

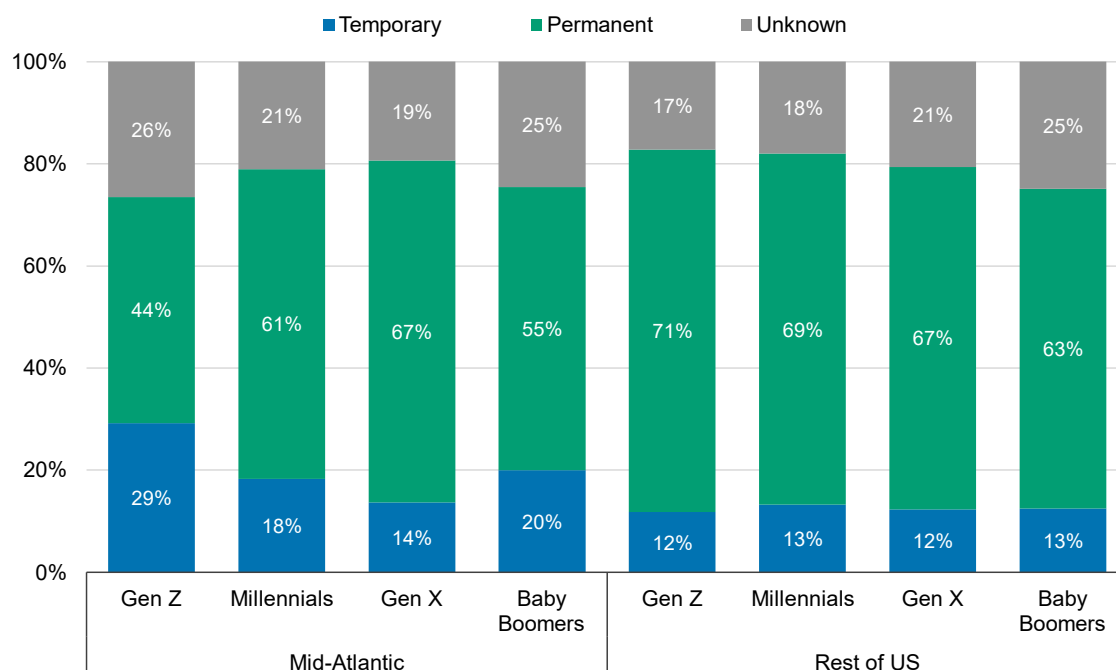


Figure 6-2. Permanent or temporary COVID effects on online purchases for home delivery by generation (2022 NHTS data).

6.2 PUBLIC TRANSIT USE

In the Mid-Atlantic region, millennials show a significant decrease in public transit use, with 46% reporting that they used it less often than they did before the pandemic (Figure 6-3). This is similar to the number of Gen Xers but lower than the number of baby boomers, who showed the highest decline at 54%. This greater decline among baby boomers aligns with their higher vulnerability to COVID-19 exposure, which likely influenced their cautious approach to public transit. Only 6% of millennials in this region used transit more frequently than before, a percentage comparable to that of Gen Xers (6%) but lower than that of Gen Zers, where 11% reported increased usage.

Millennials' transit usage in the rest of the US shows a slight variation compared with the Mid-Atlantic. In the rest of the US, a substantial portion of millennials (38%) also reported reduced transit use, though this decline was lower than that of baby boomers, who again showed the highest reduction in usage. Gen Xers in the rest of the US also exhibited a lower percentage compared with millennials. Notably, 47% of millennials in the rest of the US reported using transit the same as before, which is higher than the 40% who did so in the Mid-Atlantic region.

The data highlight that millennials' transit behavior has been less affected compared with that of baby boomers, who exhibited the highest reductions in both regions. Gen Z is the most likely to maintain or increase transit use, especially in the Mid-Atlantic region, where 11% of Gen Zers reported using public transit more often. This generational contrast suggests that millennials are more cautious than Gen Zers in returning to prepandemic transit levels but remain more adaptable than baby boomers, who appear most hesitant to resume transit use.

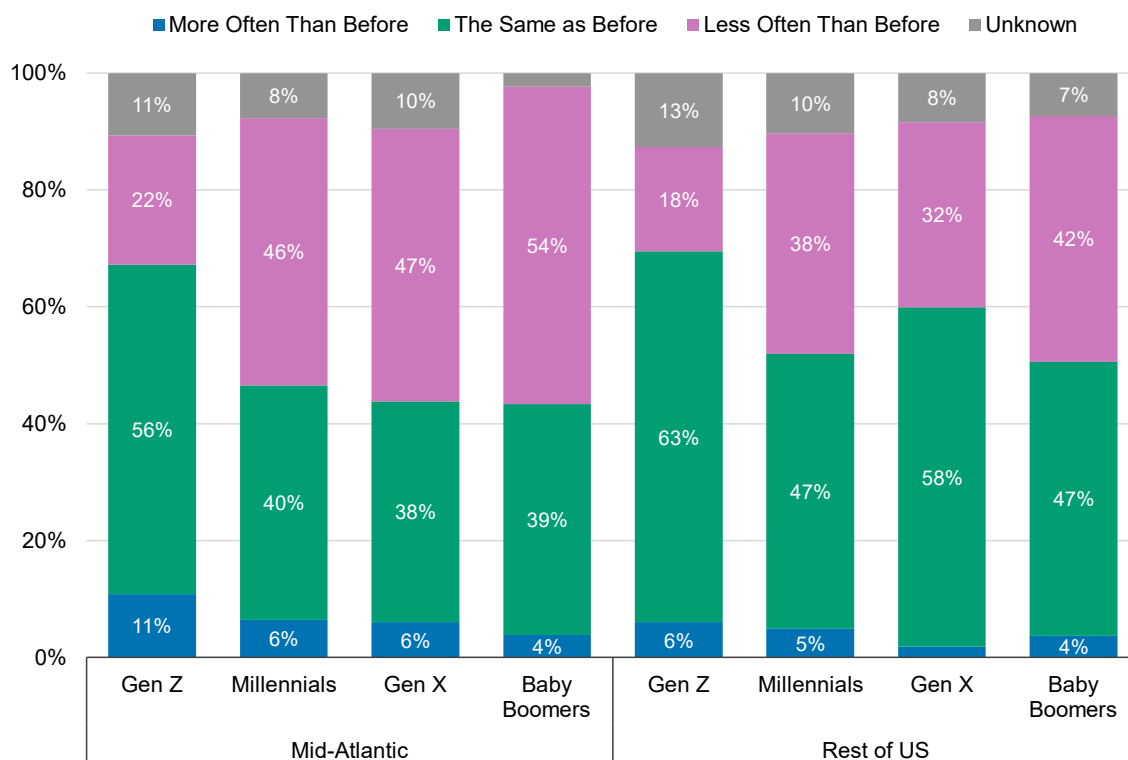


Figure 6-3. Use of public transit during COVID by generation (2022 NHTS data).

Additionally, millennials' transit usage during COVID-19 was explored by gender at the household status level (Table 6-2). In the Mid-Atlantic, millennials as heads of household show a stronger trend toward reduced transit usage. Among females in this category, 43% report using transit less often than before, whereas 6% report an increase. Males show a similar pattern, with 46% reducing their use and only 7% increasing it. In contrast, millennials as heads of household in the rest of the US demonstrate slightly more stable or even increased transit usage compared with their Mid-Atlantic counterparts. In this region, 7% of female millennials report using transit more often than before, with 48% maintaining the same level of usage, and only 33% reducing it. Male millennials in the rest of the US show similar stability, with 6% increasing transit use, 49% using it at the same level, and 34% reducing it.

Table 6-2. COVID effects on the use of public transit by millennial household status and gender (2022 NHTS data).

Region	Millennial household status	Gender	More often than before	Same as before	Less often than before	Unknown
Mid-Atlantic	No millennials	Female	10%	47%	35%	8%
		Male	2%	41%	53%	4%
	Millennials living with older household members	Female	9%	37%	36%	18%
		Male	3%	45%	46%	6%
	Millennials as heads of household	Female	6%	46%	43%	4%
		Male	7%	39%	46%	8%
Rest of US	No millennials	Female	4%	54%	35%	8%
		Male	4%	57%	32%	8%
	Millennials living with older household members	Female	3%	52%	34%	11%
		Male	4%	55%	27%	14%
	Millennials as heads of household	Female	7%	48%	33%	11%
		Male	6%	49%	34%	12%

Note: Percentages may not total 100 due to rounding.

Figure 6-4 on the COVID-19 impact on public transit use, classified as temporary or permanent changes by generation, reveals distinct generational and regional differences. In the Mid-Atlantic, millennials reported a higher proportion of temporary changes (44%) compared with permanent changes (25%), suggesting a cautious approach with a hope to resume previous levels of transit use. In contrast, Gen Xers and baby boomers in the same region have a more balanced distribution between temporary and permanent changes. For example, 33% of Gen Xers and 34% of baby boomers indicated permanent changes, suggesting that older generations may be more inclined to permanently reduce transit usage post-pandemic. Gen Z stands out in the Mid-Atlantic with the highest rate of temporary impact (53%), possibly indicating a greater likelihood of returning to prepandemic transit use habits.

In the rest of the US, the pattern shifts slightly. Here, millennials report a lower temporary impact rate (36%) and a higher permanent impact (31%) compared with the Mid-Atlantic region. Gen Z in the rest of the US shows a higher permanent impact rate (40%), diverging from their Mid-Atlantic counterparts, which might reflect less dependency on or availability of transit options in these regions. Overall, the data show that although millennials, especially in the Mid-Atlantic, are more inclined toward temporary changes, other generations indicate a more permanent reduction in transit use following COVID-19. It is important to note that a substantial portion of respondents marked “unknown” for this question.

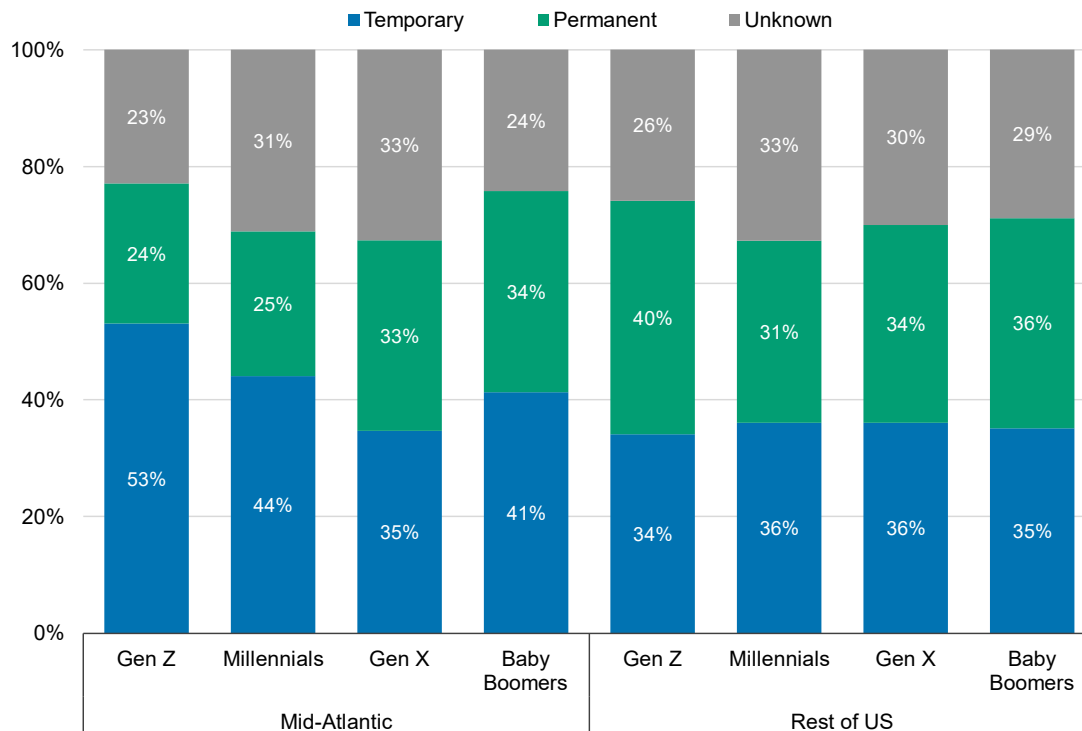


Figure 6-4. Permanent or temporary COVID effects on the use of public transit by generation (2022 NHTS data).

Further breakdown on gender in Table 6-3 reveals that among millennials, males are more likely to report a temporary reduction in transit use (52%) compared with females (35%). This suggests that millennial males in the Mid-Atlantic may have a greater inclination to eventually resume public transit use. Among baby boomers, males report the highest rate of temporary change (54%) across all groups, indicating that many male baby boomers may plan to return to public transit when conditions improve. Meanwhile, female baby boomers show a stronger tendency toward a permanent reduction in transit use.

Table 6-3. Permanent or temporary COVID effects on the use of public transit by generation and gender (2022 NHTS data).

Region	Generation	Gender	Temporary	Permanent	Unknown
Mid-Atlantic	Millennials	Female	35%	22%	42%
		Male	52%	27%	20%
	Gen X	Female	30%	42%	28%
		Male	34%	29%	38%
	Baby boomers	Female	32%	39%	29%
		Male	54%	28%	17%
Rest of US	Millennials	Female	37%	33%	30%
		Male	35%	30%	35%
	Gen X	Female	36%	35%	28%
		Male	36%	33%	32%
	Baby boomers	Female	35%	34%	31%
		Male	34%	39%	27%

Note: Percentages may not total 100 due to rounding.

6.3 COMMUTE BEHAVIOR

Figure 6-5 highlights generational and regional variations of COVID effects on travel to a physical work location. In the Mid-Atlantic region, 42% of millennials reported commuting less often than before the pandemic. This reduction is slightly higher than that of Gen Xers (36%) and baby boomers (32%). In the rest of the US, a similar trend is observed, though with some notable differences. Millennials in this region report slightly less reduction in commuting frequency, with 31% commuting less often, compared with 42% in the Mid-Atlantic. Gen Xers and baby boomers in the rest of the US also show lower rates of reduced commuting frequency compared with the Mid-Atlantic, with 32% of each generation commuting less often than before.

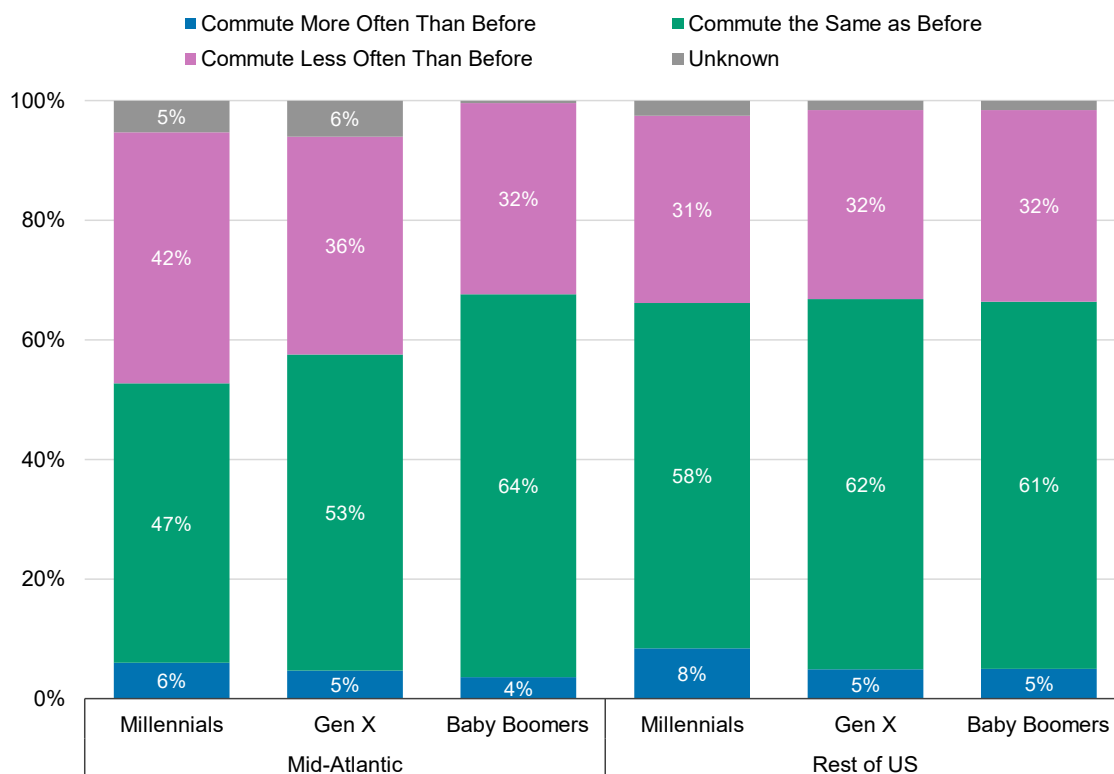


Figure 6-5. COVID effects on travel to a physical work location by generation (2022 NHTS data).

As illustrated in Figure 6-6, in the Mid-Atlantic region, millennials who are heads of household show a significant reduction in commuting frequency due to COVID-19, with 45% commuting less often than before. This is higher than both households without millennials (29%) and millennials living with older household members (23%), indicating that millennials as heads of household may be more likely to adopt remote or hybrid work arrangements. Only 5% of millennial-headed households report commuting more often than before, matching the rate for nonmillennial households. However, a higher rate of uncertainty is observed among millennial heads of household, with 9% marked as “unknown.”

We further dive into the state of the COVID-19 effect on commute changes, characterized as temporary or permanent in Figure 6-7. Millennials show a moderate rate of permanent change in the Mid-Atlantic region, with 51% indicating that their commuting habits have shifted permanently. This rate is similar to that of Gen Xers (50%) but lower than the baby boomers, who report only 35% permanent changes, suggesting that younger generations in this region may be more adaptable to changes in work routines.

Interestingly, baby boomers in the Mid-Atlantic show the highest temporary change rate at 39%, possibly reflecting a desire to return to prepandemic routines in the long term.

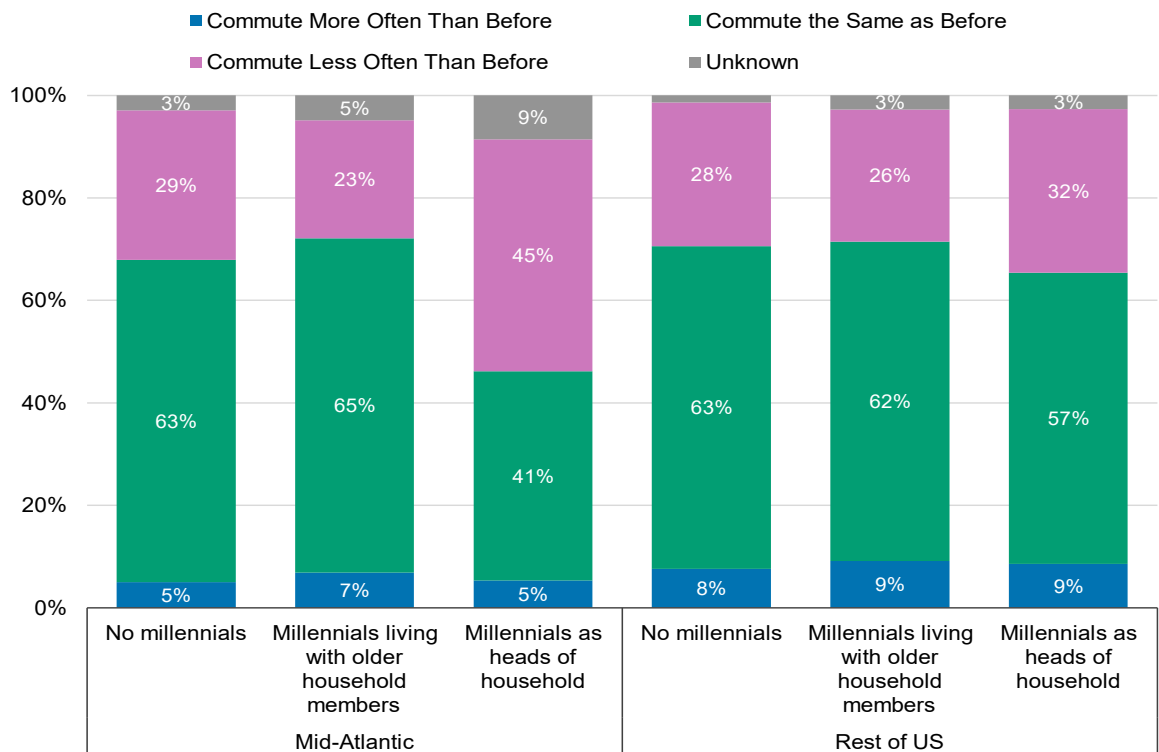


Figure 6-6. COVID effects on travel to a physical work location by millennial household status (2022 NHTS data).

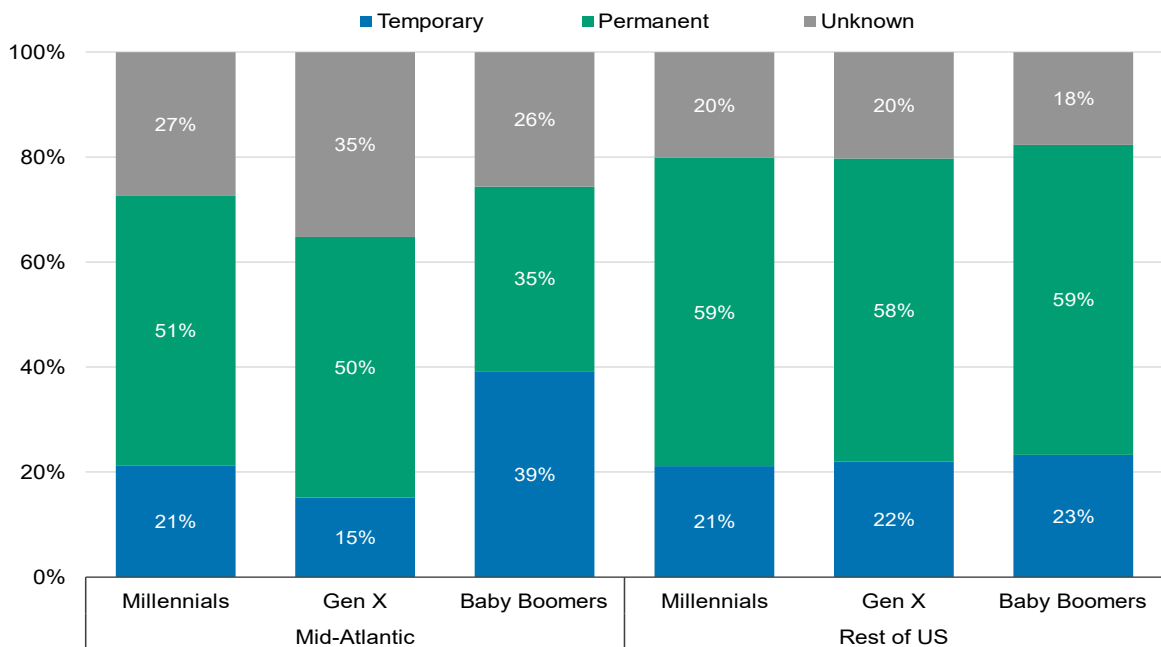


Figure 6-7. Permanent or temporary COVID effects on travel to a physical work location by generation (2022 NHTS data).

7. E-BIKE USAGE

The rise of e-bikes has revolutionized urban mobility, offering a sustainable and efficient alternative to traditional transportation. This section focuses on analyzing e-bike usage among millennials, examining their adoption patterns and behaviors in comparison to younger (Gen Z) and older (Gen X and baby boomer) generations. Using Citi Bike data in NYS from 2014 to 2019, the study investigates trip distributions, identifying yearly trends, seasonal variations, weekday versus weekend usage, and hourly activity patterns. By understanding these dynamics, we aim to uncover how millennials engage with bikesharing systems, particularly e-bikes, and how their usage evolves over time and across varying contexts.

7.1 E-BIKE TRIPS BY GENERATION OF THE USERS

The dataset analyzed here spans the years 2014 to 2019, comprising of 80,625,830 trips.² To focus on generational trends, the dataset was cleaned to include trips made only by users from four generational cohorts: Gen Zers, millennials, Gen Xers, and baby boomers. Figure 7-1 shows the monthly trend of e-bike trips in NYS from 2014 to 2019 for all four generations. Substantial growth in the number of e-bike trips taken by all generations can be observed. Millennials consistently led in trip numbers. Gen Xers followed closely, whereas baby boomers displayed a consistently lower participation. Gen Zers had the lowest trip numbers, likely reflecting their younger age and late exposure to this technology.

All generations, especially the millennials, demonstrate clear seasonality in their e-bike trips, with the highest usage occurring in the summer months (June through August) and the lowest during the winter months (December through February). This pattern is consistent across years, highlighting the influence of weather conditions on bikesharing. The peak months for millennials align with warmer weather and extended daylight hours, which are conducive to cycling. Although the overall seasonal pattern remains consistent, the absolute number of trips taken by millennials increases each year. The summer peak grows notably from 2016 onward, reflecting both the expansion of the Citi Bike system and millennials' growing adoption of e-bike usage. Winter usage, although lower, also shows slight year-over-year increases, suggesting that a growing number of millennials are incorporating bikesharing into their regular commuting patterns despite colder conditions.

Examining e-bike trips distribution by different generations and gender in Figure 7-2, e-bike usage has consistently been male-dominated across all groups over the years. For millennials, male users have consistently accounted for around 72%–74% of trips, whereas female participation has remained stable at approximately 27% in recent years. In contrast, Gen Z shows a noticeable increase in female participation over the years, rising from 18% in 2014 to 26% in 2019. This might indicate a growing interest and engagement from younger women in e-bike usage, which is supported by Uddin et al. (2023), who used the same data source. For older generations like Gen X and the baby boomers, male dominance remains more pronounced.

² <https://citibikenyc.com/system-data>

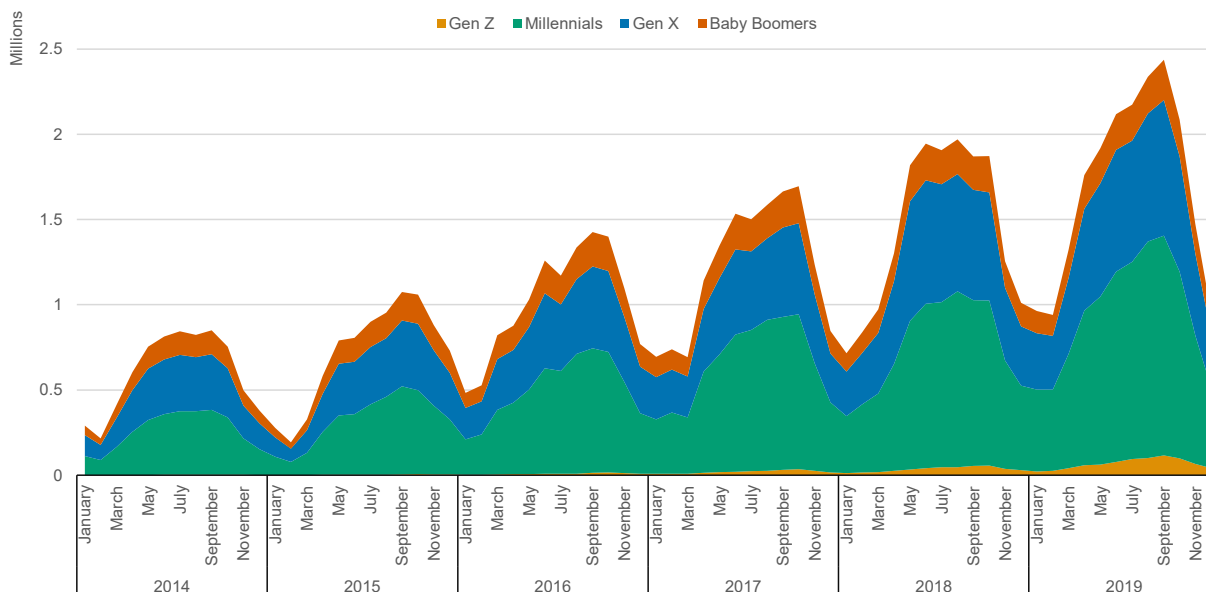


Figure 7-1. Area plot showing monthly Citi Bike trip trend from 2014 to 2019 by generation.

7.2 TRIP CHARACTERISTICS

The first metric of the e-bike trip characteristics considered here is the trip length. Figure 7-3 shows the distribution of monthly median trip lengths by generation. The trip lengths represent the great circle distance between the start and end stations of Citi Bike trips. This analysis aggregates data from all trips taken between 2014 and 2019 at monthly intervals, providing a comparative view of trip length patterns across generations.

The figure reveals notable differences in trip length across generations. Millennials have the highest monthly median trip length among the four generations, suggesting they are more likely to engage in longer-distance trips on average. In contrast, Gen Z shows the shortest monthly median trip length, indicating a tendency for shorter and more localized trips. Gen Xers and baby boomers have similar median trip lengths, slightly longer than those of Gen Zers but shorter than those of millennials. The distributions of trip lengths for millennials and Gen Zers appear more variable compared with other generations. This may reflect a wider range of trip purposes and destinations. Baby boomers and Gen Xers exhibit more uniform trip lengths, which might be indicative of more routine or purpose-specific travel.

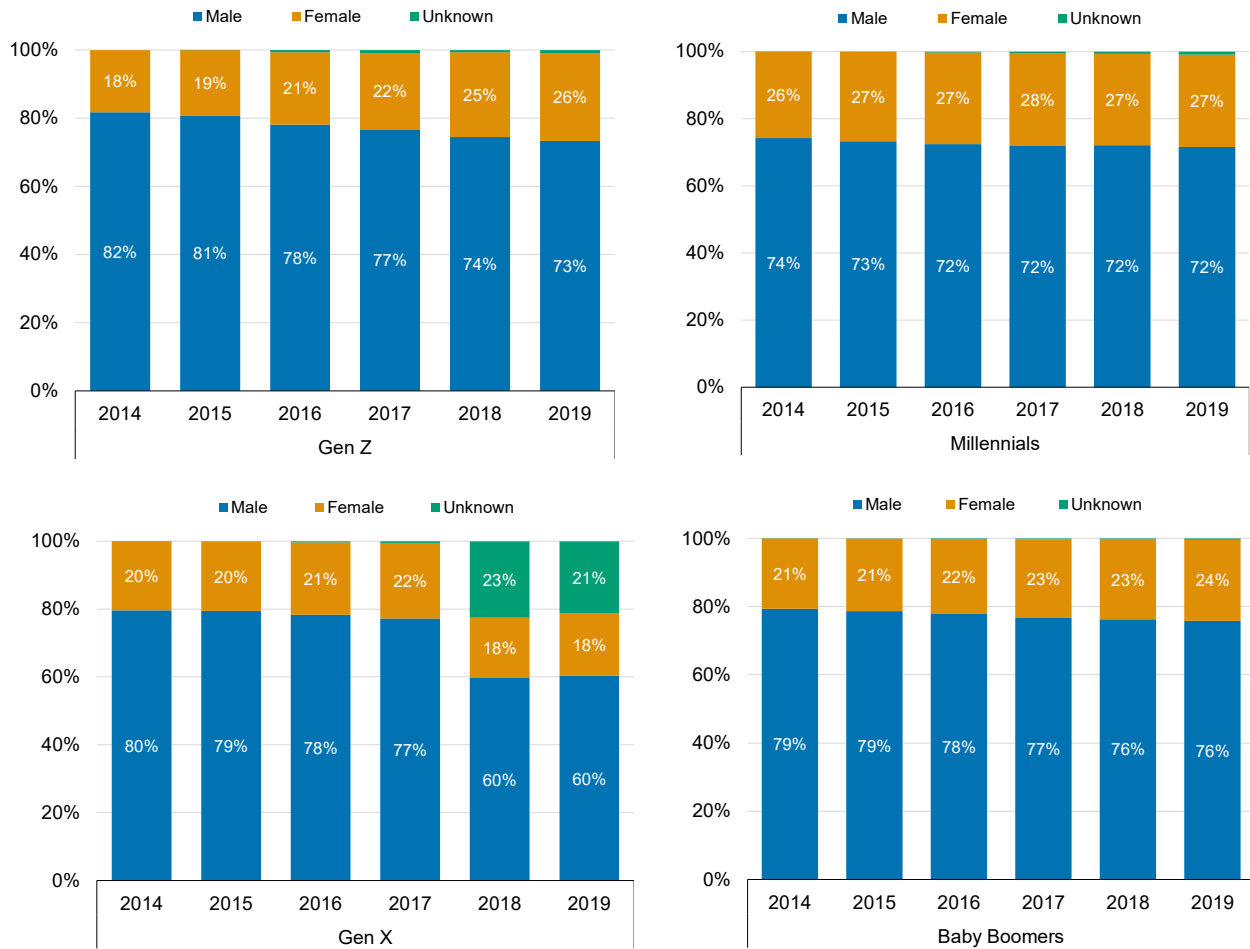


Figure 7-2. Share of Citi Bike trips by generation and gender.

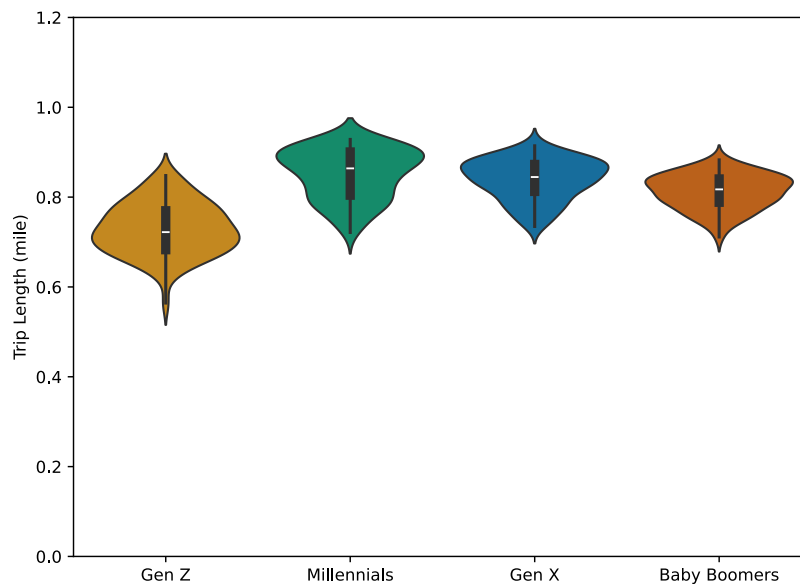


Figure 7-3. Distribution of monthly median trip length (miles) by generation.

Figure 7-4 illustrates the distribution of monthly median trip durations (in seconds) by generation. It highlights notable differences in how each generational group uses bikesharing services. Millennials exhibit higher median trip durations than those of Gen Zers, suggesting a more consistent use of bikesharing for longer rides. This trend might align with millennials' purposeful and utility-driven travel habits (usually not leisure- or recreation-based), where biking is often integrated into their daily routines, such as commuting or running errands, in big cities. The consistent higher median duration reflects their reliance on bikesharing as a practical mode of transportation rather than solely for recreational purposes. In contrast, Gen Z demonstrates a wider spread in trip durations, with the highest maximum trip duration observed across all generations. This broad variability indicates that although many of their trips may be shorter, a subset of trips skew toward longer durations. This could reflect occasional exploratory or leisurely rides that deviate from their regular usage patterns. When compared with Gen Xers and baby boomers, millennials' median trip durations are quite similar, with slight differences. Gen Xers and baby boomers exhibit slightly higher median durations.

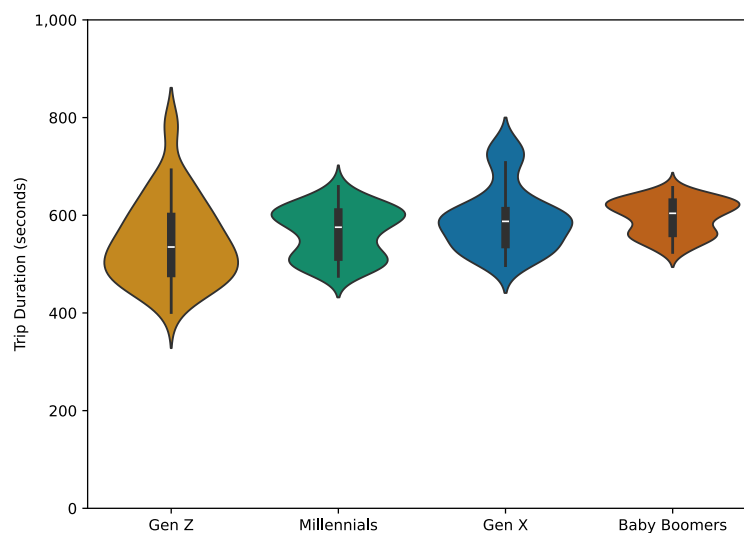


Figure 7-4. Distribution of monthly median trip duration (seconds) by generation.

The heatmaps in Figure 7-5 illustrate the distribution of e-bike trips by the day of the week and trip starting hour across generations, revealing distinct patterns in trip behavior and travel times. Gen Z demonstrates a unique distribution, with trips spread relatively evenly throughout weekdays. This could suggest a mix of activities such as commuting, attending classes, or recreational travel. Their heightened activity during weekends, particularly in the late mornings and afternoons, stands out compared with other generations. This active weekend pattern highlights Gen Z's flexible and diverse use of e-bikes, unbound by traditional weekday schedules. Millennials, on the other hand, exhibit a clear weekday pattern concentrated during morning (7–9 a.m.) and evening (5–7 p.m.) peaks, consistent with work-related travel, possibly for last-mile or first-mile connections. This structured usage aligns with commuting habits, indicating a reliance on e-bikes for daily transportation needs. On weekends, millennials' activity remains noticeable, though not as pronounced as during weekdays, yet still surpasses older generations in terms of weekend usage. Gen X and the baby boomers show weekday peak patterns similar to those of millennials, also concentrated in morning and evening rush hours, suggesting work-related trips or errands. However, their weekend activity is more subdued compared with those of millennials and Gen Zers, indicating a more limited use of e-bikes outside of weekdays.

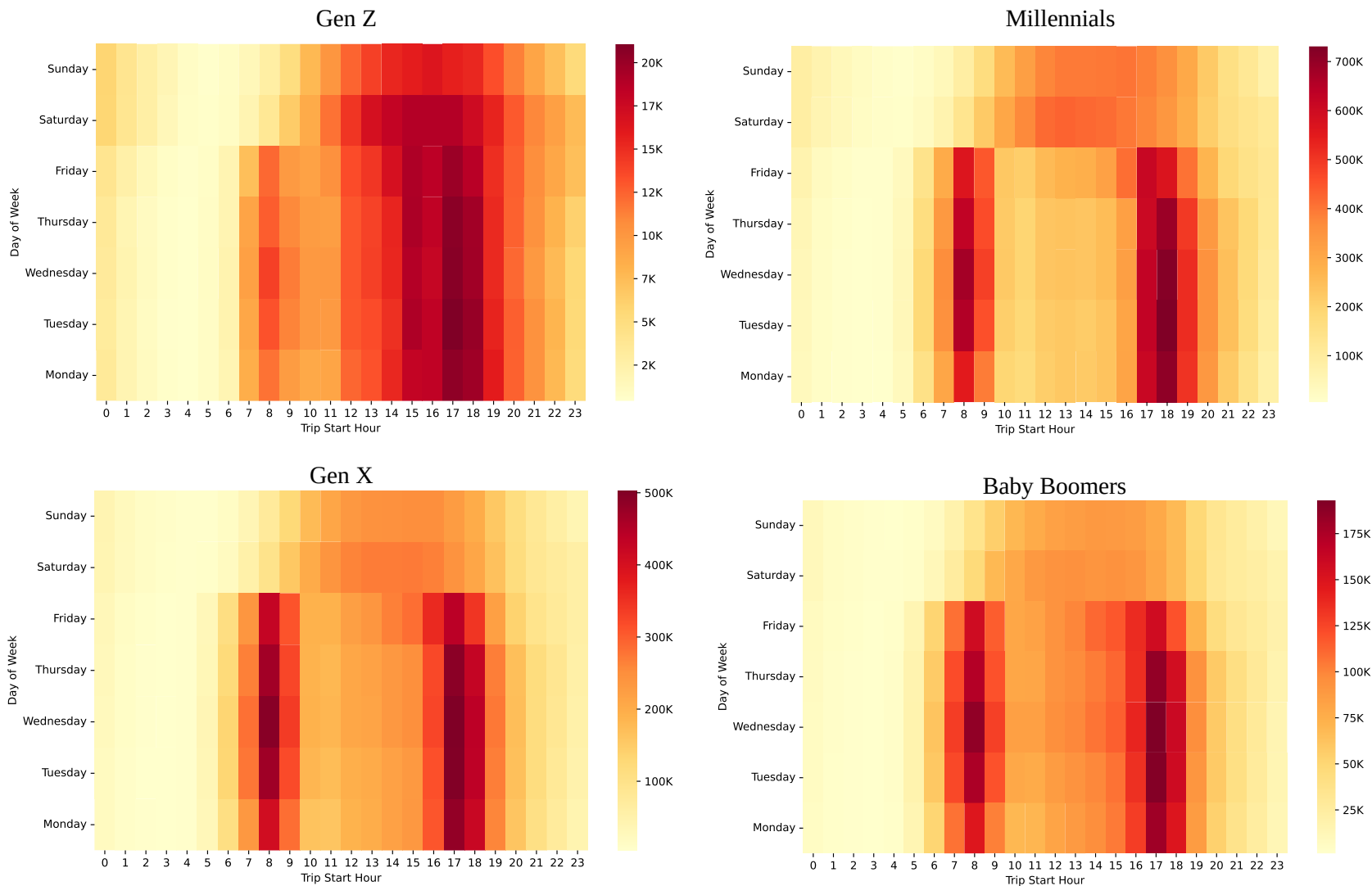


Figure 7-5. Distribution of trips by day of the week and trip start hour.

7.3 SPATIAL DISTRIBUTION OF TRIPS

The spatial distributions of trips by start and end stations, as shown in Figure 7-6 and Figure 7-7, provide important insights into generational differences in e-bike usage patterns and the purposes behind these trips. Gen Z demonstrates hotspots near educational institutions such as New York University and the Borough of Manhattan Community College, reflecting the high prevalence of e-bike use for academic and leisure purposes. The proximity to recreational areas like Rockefeller Park further supports this trend, indicating a mix of recreational and social trips.

Millennials, on the other hand, exhibit a broader and more diverse distribution of trip origins and destinations. High-density areas such as Grand Central Terminal and Union Square indicate a strong use of e-bikes for multimodal commutes, with bikes serving as a connection between other transit modes and final destinations. For example, millennials might connect from trains to bikes for the last-mile leg of their journey to work. The presence of commercial hubs among their hotspots suggests reverse travel patterns, where trips may begin at work or activity locations and transition to subways or buses for the remainder of the journey to home. This spatial diversity might align with millennials' structured commuting behaviors and their integration of e-bikes into daily routines, highlighting their utility-driven approach to bikesharing systems.

Gen X and the baby boomers exhibit overlapping patterns with millennials, particularly around major transit hubs and commercial areas. However, the intensity of these hotspots is significantly lower than it is for millennials, reflecting their comparatively lower adoption rates and less frequent use of e-bikes in such purposes. Although their trips are concentrated in similar key locations, the data suggest a narrower range of usage purposes and reduced reliance on bikesharing systems for daily transportation needs.

The alignment between start and end station hotspots across generations suggests consistent travel patterns within specific areas of New York City. Millennials' dominance in both spatial distributions highlights their key role in the city's bikesharing network. These findings reinforce the importance of targeted infrastructure investments to support the growing demand for e-bikes, particularly in high-density areas.

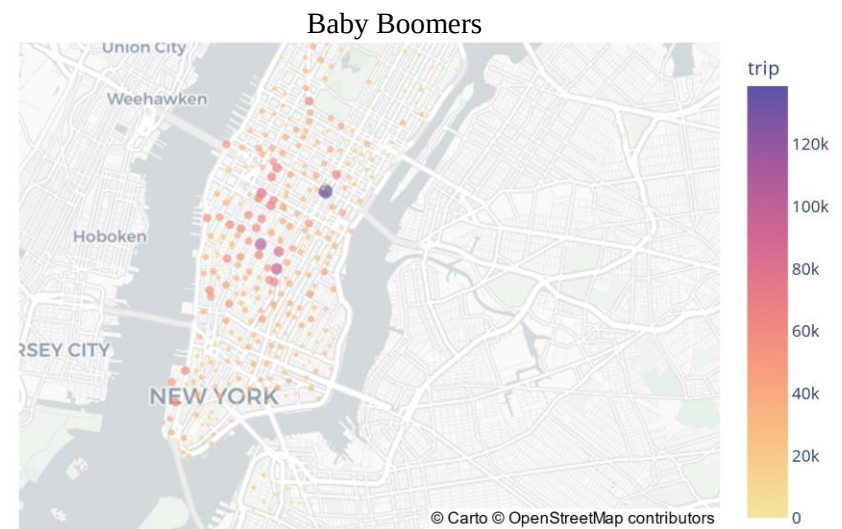
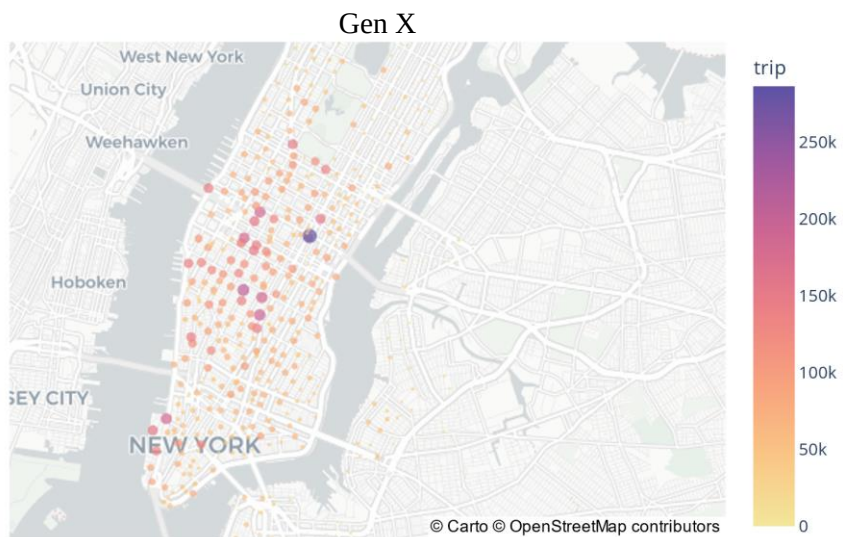
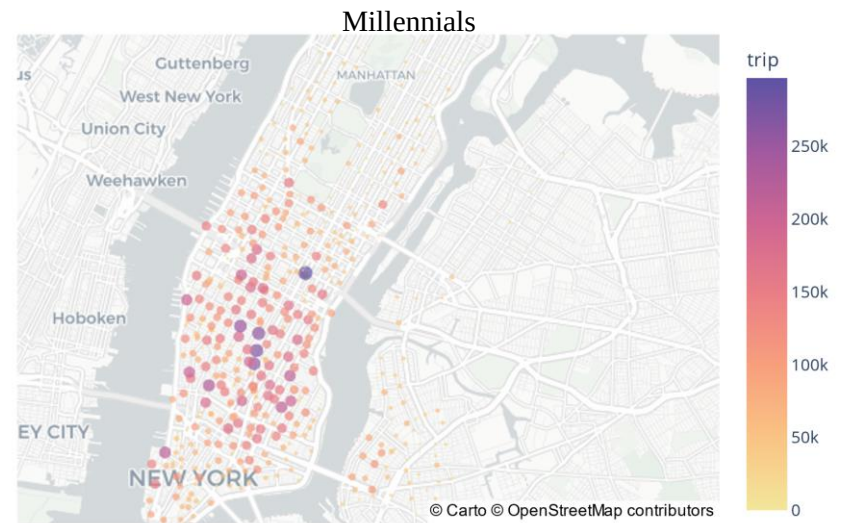
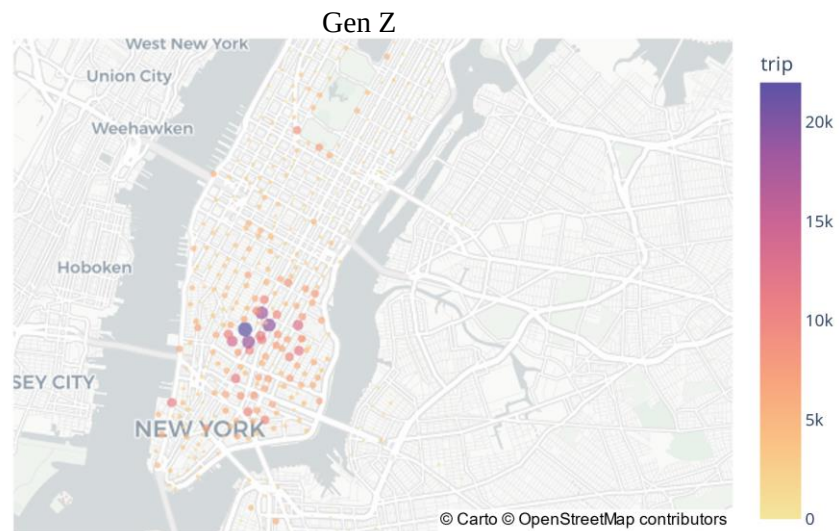


Figure 7-6. Spatial distribution of trips by start station.

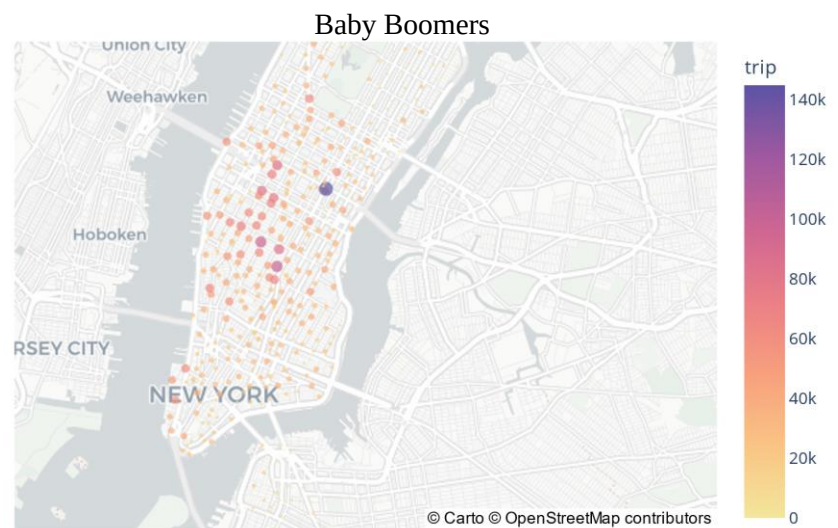
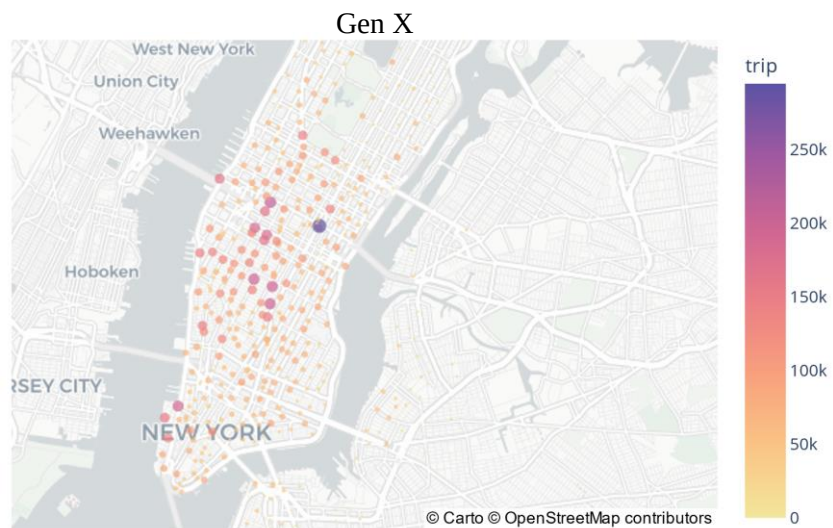
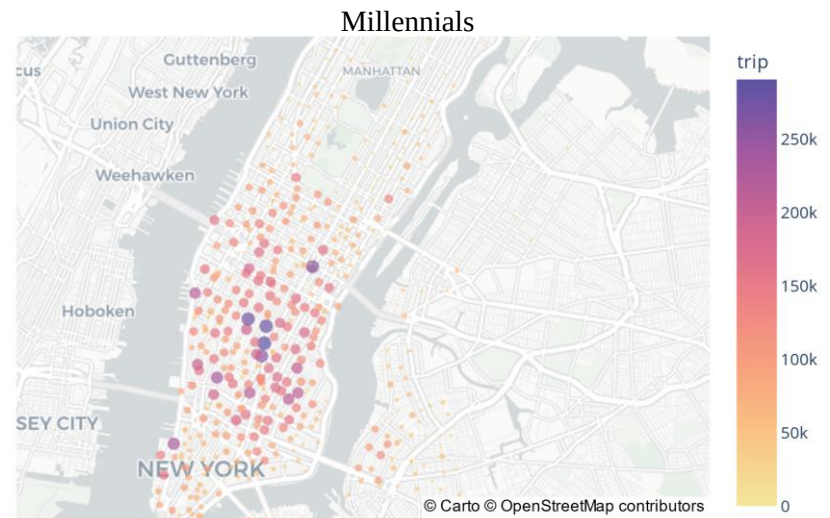
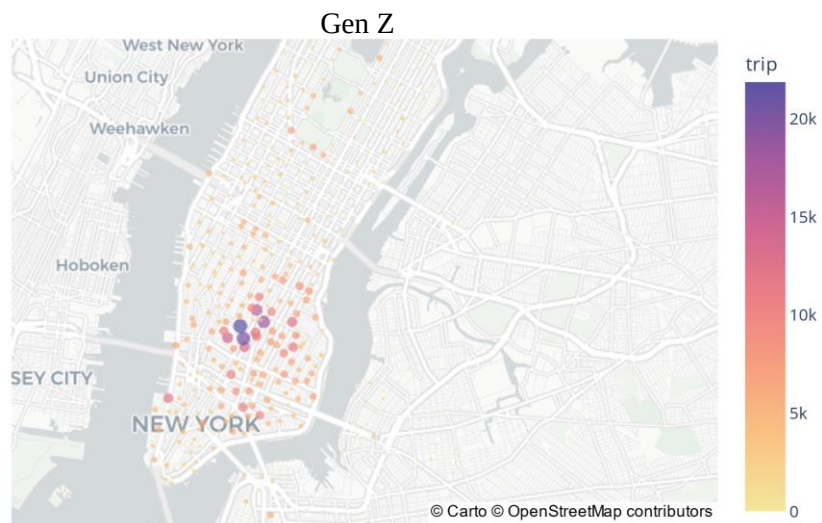


Figure 7-7. Spatial distribution of trips by end station.

8. CONCLUSIONS

This study was motivated by the need to understand the unique travel behaviors, sociodemographic characteristics, and transportation preferences of millennials, a generation that differs from earlier generations in their preferences for urban living, lower vehicle ownership, and delays in major life milestones. Focusing on NYS, the research aimed to analyze millennials' travel patterns in comparison with those of other generations, evaluate their adoption of transportation technologies, examine their views on transportation services, and assess the effects of the COVID-19 pandemic on their mobility. By using NHTS data (2009, 2017, and 2022) and other sources, this study provides insights to guide policy decisions, infrastructure planning, and service design that support the needs of this important demographic.

Key findings reveal that 64% of the millennial-headed households in NYC had no vehicles, a significant contrast to similar households in NYS other urban and nonurban areas, where multivehicle ownership was more common. Millennials averaged significantly fewer daily trips compared with Gen Xers and baby boomers in 2009 and 2017. Despite this, millennials consistently made the highest proportion of trips for earning a living (~28%) across all generations. In NYC, millennials relied on POVs for less than one-third of their daily trips, which was significantly lower than those of older generations. Notably, both male and female millennials in NYC had significantly fewer vehicle trips than those in the rest of the US. Across all regions, millennials showed significantly higher person miles traveled for solo trips, whereas Gen Zers with travel parties of 4+ traveled longer distances than millennials did in similar travel parties. Similar patterns were observed for vehicle miles traveled.

When examining millennials' views on transportation services, notable trends emerged. In NYC, 78% of millennial-headed households agree or strongly agree that they used public transit to save money, surpassing the 62% reported by nonmillennial-headed households. However, millennials were also more likely to disagree that gas prices affected their travel decisions when compared with other households in NYC. During the COVID-19 pandemic, millennials in the Mid-Atlantic region reported significant changes in travel behavior, with 46% reducing their public transit use, which was lower than that of older generations. Whereas 39% of millennials in this region increased use of online delivery orders, their adoption of delivery services lagged behind those of Gen Xers and baby boomers but exceeded Gen Zers' adoption of the services.

Millennials demonstrated a notable inclination toward adopting transportation technologies. They led in rideshare usage, with 25% reporting usage 1–5 times and 8% reporting usage 6–10 times in NYS. E-bike usage among millennials showed consistent growth, with male users accounting for over 70% of trips, indicating a gender imbalance in the usage of e-bike trips. Millennials recorded the highest monthly median trip lengths across generations and demonstrated structured weekday e-bike usage patterns during peak commuting hours. Spatial analysis of e-bike trips revealed hotspots near transit hubs like Grand Central Terminal and commercial areas near Union Square.

Future research could delve into the travel behavior of zero-vehicle millennials' households, focusing on their reliance on alternative transportation modes and the spatial characteristics of their trips. Furthermore, although this report examined millennials' COVID-19 behaviors at a broader regional level (i.e., Middle Atlantic Census Division), future studies could explore localized patterns, such as in NYC, to better understand how their travel behaviors change during a systemwide shock. Additionally, deeper investigations into sustainable transportation options such as e-scooters and e-bikes are necessary, especially to compare usage patterns and preferences between millennials and newer generations (i.e., Gen Z and Gen Alpha). This could provide insights to enhance transportation options, improve equity in access, and promote sustainable mobility solutions that support diverse demographic needs.

REFERENCES

- The Council of Economic Advisers. (2014). *15 economic facts about millennials*. Retrieved November 20 from https://obamawhitehouse.archives.gov/sites/default/files/docs/millennials_report.pdf
- Fry, R. (2020). *Millennials overtake Baby Boomers as America's largest generation*. Pew Research Center. Retrieved November 20 from <https://www.pewresearch.org/short-reads/2020/04/28/millennials-overtake-baby-boomers-as-americas-largest-generation/>
- Fry, R., Igielnik, R., & Patten, E. (2018). *How Millennials today compare with their grandparents 50 years ago*. Pew Research Center. Retrieved November 20 from <https://www.pewresearch.org/short-reads/2018/03/16/how-millennials-compare-with-their-grandparents/>
- Humi. (2024). *Humi Survey Finds Gen Z and Baby Boomers Are United on Remote Work Being Their Least Preferred Work Environment*. businesswire. Retrieved November 20 from <https://www.businesswire.com/news/home/20241024371385/en/Humi-Survey-Finds-Gen-Z-and-Baby-Boomers-Are-United-on-Remote-Work-Being-Their-Least-Preferred-Work-Environment>.
- Korhonen, V. (2024). *U.S. population by generation 2023*. Retrieved November 20 from <https://www.statista.com/statistics/296974/us-population-share-by-generation/>.
- Lee, Y., Mokhtarian, P. L., Guhathakurta, S., Circella, G., & Iogansen, X. (2021). Millennial Travelers Are More Multimodal than Older Travelers, but This Trend Might Change as They Age [Policy Brief].
- Liu, Y., Hwang, H.-L., Uddin, M., Reuscher, T., & Chin, S.-M. (2022). *Travel Patterns and Characteristics of Elderly Population in New York State: 2017 Update* (No. ORNL/TM-2022/2614). Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States).
- The Millennial Report. (2024). *How Millennials Are Leading the Charge Against Climate Change: Innovative Strategies and Actions*. Retrieved November 20 from <https://medium.com/@the.millennial.report.31/how-millennials-are-leading-the-charge-against-climate-change-innovative-strategies-and-actions-567de505fa92>.
- Moos, M., Pfeiffer, D., & Vinodrai, T. (2017). *The millennial city*. Taylor & Francis.
- NHTS. (2017). *National Household Travel Survey*. Federal Highway Administration, US Department of Transportation, Washington, DC. Retrieved May 20 from <https://nhts.ornl.gov/>
- Pan, M., Patwary, L., & Uddin, M. (2024). *Travel Patterns and Characteristics of Population in Rural Areas of New York State* (No. ORNL/SPR-2024/3216). Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States).
- Patwary, A. L., & Khattak, A. J. (2022). Interaction Between Information and Communication Technologies and Travel Behavior: Using Behavioral Data to Explore Correlates of the COVID-19 Pandemic. *Transportation Research Record*, 03611981221116626.
- Sadeghvaziri, E., & Tawfik, A. (2020). Using the 2017 National Household Travel Survey Data to Explore the Elderly's Travel Patterns. *International Conference on Transportation and Development 2020*.
- Uddin, M., Hwang, H.-L., & Hasnine, M. S. (2023). An interpretable machine learning framework to understand bikeshare demand before and during the COVID-19 pandemic in New York City. *Transportation Planning and Technology*, 46(4), 482-498.
- Uddin, M., Pan, M., Liu, Y., & Reuscher, T. (2024). *Travel Patterns and Characteristics of Low-Income Population in New York State: 2017 Update* (No. ORNL/TM-2023/2942). Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States).

APPENDIX A. GLOSSARY OF NHTS TERMS

APPENDIX A. GLOSSARY OF NHTS TERMS

This glossary provides the most commonly used terms in the National Household Travel Survey (NHTS) and the terms used in this report, along with definitions of those terms. These definitions are provided to assist the user in interpreting the NHTS data.

Adult	For NHTS, this is defined as a person 18 years or older.
Block Group	A subdivision of a Census Tract that contains 600–3,000 people. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Census Tract	A small subdivision of a county, generally having a population size between 1,200 and 8,000 people, with an optimum size of 4,000 people. The geographic size of the tract may vary considerably, depending on population density. Tracts were designed to be homogeneous in regard to population characteristics, economic status, and living conditions when they were first delineated. Because the first tracts were delineated for the 1890 Census, today's tracts may be far from homogeneous. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Destination	For travel day trips, the destination is the point at which there is a break in travel, except if the break is only to change vehicles or means of transport.
Driver	A driver is a person who operates a motorized vehicle. If more than one person drives on a single trip, the person who drives the most miles is classified as the principal driver.
Elderly	A person age 65 years or older.
Elderly Household	Households with one or more persons age 65 years or older.
Employed	A person is considered employed if that person worked for pay, either full time or part time, during the week before the interview. This includes persons who work at home and/or have more than one job.
Household	A group of persons whose usual place of residence is a specific housing unit; these persons may or may not be related to each other. The total of all US households represents the total civilian noninstitutionalized population. A household does not include group quarters (i.e., 10 or more persons living together, none of whom are related).
Household Income	Household income is the money earned by all family members in a household, including those temporarily absent. Annual income consisted of the income earned 12 months preceding the interview. Household income includes monies from all sources, such as wages and salary, commissions, tips, cash bonuses, income from a business or farm, pensions, dividends, interest, unemployment or workmen's compensation, social security, veterans' payments, rent received from owned property (minus the operating costs), public assistance payments, regular gifts of money from friends or relatives not living in the household, alimony, child support, and other kinds of periodic money income other than earnings. Household income excludes in-kind

	income such as room and board, insurance payments, lump-sum inheritances, occasional gifts of money from persons not living in the same household, withdrawal of savings from banks, tax refunds, and the proceeds of the sale of one's house, car, or other personal property.
Household Members	Household members include all people, whether present or temporarily absent, whose usual place of residence is in the sample unit. Household members also include people staying in the sample unit who have no other usual place of residence elsewhere.
Household Vehicle	A household vehicle is a motorized vehicle that is owned, leased, rented, or company-owned and available to be used regularly by household members during the 2 week travel period. Household vehicles include vehicles used solely for business purposes or business-owned vehicles, as long as they are driven home and can be used for the home-to-work trip (e.g., taxicabs and police cars). Household vehicles include all vehicles that were owned or available for use by members of the household during the travel period, even though a vehicle may have been sold before the interview. Vehicles excluded from household vehicles are (1) those which were not working and were not expected to be working within 60 days and (2) those that were purchased or received after the designated travel day.
Journey-to-Work Trips (Commute Trips)	Includes travel to and from a place where one reports for work. Does not include any other work-related travel. Does not include any trips for persons who work at home.
Means of Transportation	A mode of travel used for going from one place (origin) to another (destination). A means of transportation includes private and public transit modes, as well as walking. The following transportation modes, grouped by major mode, are included in the NHTS data. Active modes—include modes where a person must actively move from one place to the next. 1. Walk: This category includes walking and jogging. 2. Bicycle: This category includes bicycles of all speeds and sizes that do not have a motor. Private vehicle—a stipulation for being a private vehicle is that the vehicle is privately operated, including rental cars. 3. Car: Includes cars and station wagons. Leased and rented cars are included if privately operated and not used for picking up passengers in return for fare. 4. Sport utility vehicle: Includes vehicles that are a hybrid of design elements from a van, a pickup truck, and a station wagon. Examples include a Ford Explorer, Jeep Cherokee, or Nissan Pathfinder. 5. Van: Includes vans or minivans designed to carry 5–13 passengers or to haul cargo. 6. Pickup truck: Includes vehicles with an enclosed cab that usually accommodates two to three passengers and has an open cargo area in the rear. Late-model pickups often have a back seat that allows for total seating of four to six passengers. Pickup trucks usually have the same size of wheel-base as a full-size station wagon. This category also includes pickups with campers. 7. Motorcycle/moped: This category includes large, medium, and small motorcycles and mopeds. 8. RV (motor home, ATV, snowmobile): An RV or motor home includes a self-powered recreational vehicle that is operated as a unit without being towed by

another vehicle (e.g., a Winnebago motor home). This category also includes ATVs and snowmobiles.

Rental car—Includes Zipcar and Car2Go, in addition to commercially rented cars for private use.

9. Golf cart/Segway: This includes all electric or gas-operated vehicles designed for use on a golf course, but whose use has recently extended to within smaller, often gated, communities. NOTE: Travel taken via golf cart is NOT included in vehicle trips and vehicle miles of travel.

Public transportation—as used in Federal Highway Administration publications and analysis of NHTS data, typically includes the following (indicated in italics in the following text): public or commuter bus, commuter rail, and subway/elevated rail/light rail/streetcar.

Bus—This category includes:

11. *Public or commuter buses*: local public transit buses available to the general public and buses used for short-distance public transport purposes (e.g., city bus or public bus), school buses
13. Private/charter/tour/shuttle buses: private buses operating on a fixed schedule between population centers and buses that shuttle passengers from one fixed place to another (e.g., airport shuttles)
14. City-to-city buses: buses that run from one urban center to another (e.g., Greyhound)

Train—This category includes:

15. *Amtrak/commuter rail*: trains that run from one urban center to another
16. *Subway/elevated rail/light rail/streetcar* (also known as rail rapid transit): a high-capacity system operated on a fixed-rail or guideway system on a private right of way, and vehicles that run on a fixed-rail system powered by electricity obtained from an overhead power distribution system

Other modes—This category includes:

10. School buses
12. Paratransit/dial-a-ride
17. Taxi/limo: Use of a taxicab by a passenger for fare, including limousines. In 2017, this category also includes ridesharing such as Uber and Lyft.
19. Airplane: includes commercial airplanes and smaller planes available for use by the general public in exchange for a fare. Private and corporate planes and helicopters are also included.
20. Boat/ferry/water taxi: includes travel by passenger line ferries
97. Something else: includes any type of transportation not previously listed (e.g., skateboards, roller blades, sailboats, cruise ships)

Metropolitan Statistical Area (MSA)

Except in the New England states, a metropolitan statistical area is a county or group of contiguous counties which contains at least one city of 50,000 inhabitants or more, or “twin cities” with a combined population of at least 50,000. In addition, contiguous counties are included in an MSA if, according to certain criteria, they are socially and economically integrated with the central city. In the New England states, MSAs consist of towns and cities instead of counties. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).

Motorized Vehicle

Motorized vehicles are all vehicles that are licensed for highway driving. Snow mobiles and minibikes are specifically excluded.

New York City (NYC)	New York City is defined in this report as the five-county area: Bronx, Kings, Queens, New York (Manhattan), and Richmond.
Occupancy	Occupancy is the number of persons, including driver and passenger(s) in a vehicle.
Occupancy Rate	NHTS occupancy rates are generally defined as the mileage-weighted averages of the number of persons on a vehicle trip.
Origin	Origin is the starting point of a trip.
Passenger	For a specific trip, a passenger is any occupant of a motorized vehicle, other than the driver.
Person Miles of Travel (PMT)	PMT is a primary measure of person travel. When one person travels 1 mile, 1 person mile of travel results. Where two or more persons travel together in the same vehicle, each person makes the same number of person miles as the vehicle miles. Therefore, four persons traveling 5 miles in the same vehicle results in 20 person miles ($4 \times 5 = 20$).
Person Trip	A person trip is a trip by one or more persons in any mode of transportation. Each person is considered as making one person trip. For example, four persons traveling together in one auto are counted as four person trips.
Population Density	Category of population density (persons per square mile) in the Census Block Group of the household's home location: 0–99, 100–499, 500–999, 1,000–1,999, 2,000–3,999, 4,000–9,999, 10,000–24,999, and 25,000–999,999.
Privately Owned Vehicle (POV)	A privately-owned or privately-operated vehicle. In either case, the vehicle is not available to the public for a fee (i.e., it is not a bus, subway, taxi, or other vehicle for which a fee is charged for use).
Travel Day	A travel day is a 24 hour period from 4:00 a.m. to 3:59 a.m. designated as the reference period for studying trips and travel by members of a sampled household.
Travel Day Trip	A travel day trip is defined as any time the respondent went from one location to another by private motor vehicle, public transportation, bicycle, walking, or other means during the NHTS-assigned reporting travel day. However, a separate trip is not counted in two instances: 1. When the sole purpose for the trip is to get to another vehicle or mode of transportation to continue to the destination. 2. Travel within a shopping center, mall, or shopping areas of four to five blocks is to be considered as travel to one destination.
Vehicle	In the 2017 NHTS, the term vehicle includes autos, passenger vans, sport utility vehicles, pickups and other light trucks, RVs, motorcycles, and mopeds owned or available to the household.
Vehicle Miles of Travel (VMT)	VMT is a unit to measure vehicle travel made by a private vehicle, such as an automobile, van, pickup truck, or motorcycle. Each mile traveled is counted as one vehicle mile regardless of the number of persons in the vehicle.
Vehicle Occupancy	Vehicle occupancy is the number of persons, including driver and passenger(s) in a vehicle; it also includes persons who did not complete a whole trip.

Vehicle Occupancy Rate	NHTS vehicle occupancy rates are generally calculated as person miles divided by vehicle miles.
Vehicle Trip	A trip by a single POV regardless of the number of persons in the vehicle.
Work-Related Travel	Trips related to business activities except travel to the place of work; for example, a plumber drives to a wholesale dealer to purchase supplies for his business or a company executive travels from his office to another building to attend a business meeting. Business, out-of-town trips, and professional conventions are also included.
Worker	See “Employed.”

